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Affidavit

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Preamble

Abstract

Superconductivity is a classic example of a many-body phenomenon, where electronic correlations lead to macroscopic quantum coherence. While conventional superconductivity, as described by BCS theory, can arise even in weakly correlated materials, unconventional superconductors — such as the famous example of the cuprates or the recently discovered nickelates — require a more sophisticated theoretical treatment. In particular, the infinite-layer nickelates are believed to exhibit d-wave superconductivity, which crucially depends on the spatial structure of electronic correlations. Standard dynamical mean-field theory (DMFT), which treats correlations locally, cannot capture such nontrivial pairing symmetries, since the local irreducible pairing vertex Γ_{pp} obtained from DMFT alone would never lead to a d-wave instability. To overcome this limitation, diagrammatic extensions of DMFT are needed for a better description. One such method, the dynamical vertex approximation (DΓA) incorporates non-local correlations by using the irreducible vertex as a building block, thereby granting access to momentum- and frequency-dependent self-energies, susceptibilities, and pairing interactions. This makes the DΓA particularly well-suited for studying phase transitions such as unconventional superconductivity, where non-local fluctuations and spatially extended correlations are essential.

In this thesis, we developed and implemented a code for the self-consistent ladder dynamical vertex approximation applied to the multi-orbital Hubbard model. The implementation exploits vertex asymptotics, enabling calculations down to very low temperatures and thus the determination of superconducting transition temperatures from two-particle correlations and their feedback into single-particle properties. Compared to DMFT, which neglects all spatial correlations, this framework systematically incorporates them and provides a natural extension to the local theory.

A central motivation for this work is the multi-orbital nature of correlated superconductors. While single-orbital models capture essential aspects of certain materials, such an effective description breaks down in many real systems. For instance, cuprates or the infinite-layer nickelates (such as $NdNiO_2$) can, to a good approximation, be described within a single-orbital framework and indeed the dynamical vertex approximation has already been successfully applied to predict their superconducting critical temperature T_c. In contrast, the recently discovered bilayer nickelate $La_3Ni_2O_7$, which hosts two inequivalent Ni sites per unit cell and exhibits the highest known T_c among the nickelates, is believed to require a multi-orbital description to account for interlayer interactions and orbital interplay. Our numerical implementation of a multi-orbital DΓA framework provides the means to capture these effects, including orbital selectivity, Hund's coupling and intertwined instabilities, which are necessary components for understanding the superconducting pairing symmetry and critical temperature in such systems.

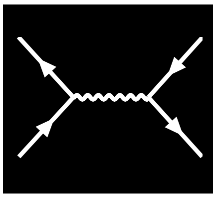
PREAMBLE

DEUTSCHE KURZFASSUNG

Die Supraleitung stellt ein klassisches Beispiel für ein Vielteilchenphänomen dar, bei dem elektronische Korrelationen zu makroskopischer quantenmechanischer Kohärenz führen. Während konventionelle Supraleitung, wie sie durch die BCS-Theorie beschrieben wird, bereits in schwach korrelierten Materialien auftreten kann, erfordern unkonventionelle Supraleiter — etwa die bekannten Kuprate oder die jüngst entdeckten Nickelate — eine deutlich anspruchsvollere theoretische Beschreibung. Besonders bei den Nickelaten mit unendlicher Schicht-Kristallstruktur geht man davon aus, dass sie d-Wellen-Supraleitung aufweisen, die entscheidend von der räumlichen Struktur elektronischer Korrelationen abhängt. Die Dynamische Molekularfeldtheorie (DMFT), die Korrelationen ausschließlich lokal behandelt, ist hierfür nicht ausreichend, da der lokale irreduzible Paarungsvertex Γ_{pp} allein keine d-Wellen-Instabilität erzeugen kann. Um diese Einschränkung zu überwinden, werden diagrammatische Erweiterungen der DMFT benötigt. Eine prominente Methode ist die Dynamische Vertex Approximation (DΓA), die nichtlokale Korrelationen durch die Verwendung des irreduziblen Vertex als Baustein einbezieht. Auf diese Weise eröffnet die DΓA den Zugang zu impuls- und frequenzabhängigen Selbstenergien, Suszeptibilitäten und Paarungswechselwirkungen und eignet sich damit besonders gut zur Untersuchung von Phasenübergängen wie unkonventioneller Supraleitung, bei denen nichtlokale Fluktuationen und räumlich erweiterte Korrelationen eine zentrale Rolle spielen.

In dieser Arbeit präsentieren wir einen Code für die selbstkonsistente Leiter-Dynamische Vertex Approximation, angewandt auf das multi-orbitale Hubbard-Modell. Durch die Nutzung von Vertex-Asymptotiken ermöglicht die Implementierung Rechnungen bis in den Bereich sehr tiefer Temperaturen und damit die Bestimmung supraleitender Übergangstemperaturen aus Zweiteilchenkorrelationen sowie deren Rückwirkung auf Einteilcheneigenschaften. Im Gegensatz zur DMFT, die sämtliche räumlichen Korrelationen vernachlässigt, werden diese hier systematisch berücksichtigt, sodass sich eine natürliche Erweiterung der lokalen Theorie ergibt.

Eine zentrale Motivation dieser Arbeit ist die multi-orbitale Natur korrelierter Supraleiter. Ein-Orbital-Modelle erfassen zwar wesentliche Eigenschaften bestimmter Materialien, stoßen jedoch bei vielen realen Systemen an ihre Grenzen. So lassen sich Kuprate oder unendliche Schichtnickelate (wie $NdNiO_2$) in guter Näherung durch ein Ein-Orbital-Modell beschreiben, und tatsächlich wurde die Dynamische Vertex Approximation bereits erfolgreich eingesetzt, um ihre kritische Temperatur T_c vorherzusagen. Im Gegensatz dazu erfordert das kürzlich entdeckte Doppelschicht-Nickelat $La_3Ni_2O_7$, das pro Einheitszelle zwei nichtäquivalente Ni-Atome enthält und die bislang höchste bekannte T_c unter den Nickelaten aufweist, eine multi-orbitale Beschreibung, um Schichtkopplungen und das Zusammenspiel der Orbitale adäquat zu erfassen. Mit der hier vorgestellten numerischen Implementierung eines multi-orbitalen DΓA-Verfahrens lassen sich diese Effekte nun berücksichtigen — einschließlich Orbitalselektivität, Hund'scher Kopplung und verschränkter Instabilitäten. Diese Aspekte sind unerlässlich, um die Symmetrie der supraleitenden Paarung sowie die kritische Temperatur in solchen Systemen zu verstehen.



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Chapter 1 Introduction

The theoretical description of strongly correlated electron systems remains one of the most challenging and fascinating problems in condensed matter physics. Unlike weakly interacting systems, where electrons behave as nearly independent quasiparticles, strong Coulomb interactions lead to the emergence of collective phenomena such as the Mott metal–insulator transition [1], non-Fermi-liquid behavior, unconventional magnetism [2] and high-temperature unconventional superconductivity [3]. Understanding these effects requires theoretical frameworks that go beyond single-particle approximations and simple mean-field descriptions. The treatment of strongly correlated electron systems is very challenging, even with modern computational approaches due to the exponential growth of the Hilbert space with system size and the presence of multiple competing energy scales.

Among modern approaches, dynamical mean-field theory (DMFT) [4] has established itself as a cornerstone for modeling local electronic correlations. By mapping a lattice problem onto a quantum impurity model [5], DMFT provides non-perturbative access to single-particle observables, including spectral functions and quasiparticle renormalizations. An important property of DMFT is that it becomes exact in the limit of infinite spatial dimensions, where non-local correlations vanish. In finite dimensions, however — and particularly in low-dimensional systems — spatial correlations can play a crucial role. This is especially relevant for unconventional superconductors, which are often effectively described as quasi-two-dimensional materials. It should be noted, though, that spatial correlations are not equally important for all collective phenomena: for example, the Mott metal-insulator transition is a collective effect that can still be accurately captured within DMFT. In contrast, d-wave superconductivity represents the opposite extreme, as its pairing symmetry fundamentally relies on the spatial structure of electronic correlations, which cannot be obtained from DMFT alone.

To overcome this limitation, so-called cluster extensions or diagrammatic extensions of DMFT have been developed (more on that in Ch. 4), among these the dynamical vertex approximation (DΓA) [6, 7] is a state-of-the-art method. DΓA incorporates non-local correlations by taking two-particle irreducible quantities as building blocks and constructing ladder or parquet diagrammatic expansions to compute momentum- and frequency-resolved quantities. In this way, DΓA grants access to susceptibilities, self-energies and pairing instabilities beyond the local approximation, enabling the study of competing phases on equal footing [8].

While DΓA has been successfully applied to single-orbital systems [9, 10], many strongly correlated materials exhibit a pronounced multi-orbital character. The importance of this for a realistic description of materials is highlighted by the recent discovery of high-temperature superconductivity in compounds such as the bilayer nickelates $La_3Ni_2O_7$ [11], where orbital selectivity, Hund's coupling and intertwined instabilities strongly influence low-energy physics and superconducting pairing mechanisms [12]. Accurately describing these systems requires a multi-orbital generalization of the dynamical vertex approximation, which significantly increases computational and algorithmic complexity due to the higher dimensionality of the involved vertex functions. Such a formalism has been developed previously in Ref. [13]. In that work however, they did not consider superconductivity and the implications of multi-orbital correlations on it.

This thesis presents the implementation of a self-consistent multi-orbital ladder-DΓA framework applied to the multi-orbital Hubbard model, which is based on previous works, see Refs. [13–15]. The newly developed implementation is designed for numerical efficiency, scalability, and flexibility, enabling calculations for realistic multi-orbital systems. By exploiting vertex asymptotics, it allows us to access the correlated electronic structure at low temperatures which can provide us insights into the mechanisms relevant for superconductivity.

This work is structured as follows: Ch. 2 introduces a general theoretical background of condensed matter physics and how we model realistic materials. We mainly focus on the multi-orbital Hubbard Hamiltonian and its various simplifications, including the well-known Kanamori parametrization. We then introduce and develop the specific tools and concepts required for DΓA, including one- and two-particle Green's functions, susceptibilities, the Bethe-Salpeter equation and the Schwinger-Dyson equation in Ch. 3. Furthermore, we also cover vertex asymptotics and explain their role in reducing numerical complexity while simultaneously significantly enhancing the capabilities of low-temperature calculations. Next, the ladder DΓA formalism and its connection to superconducting pairing instabilities via the Eliashberg equation are derived in Ch. 4, and in Ch. 5 we describe the design and implementation of the code which was developed in the course of this thesis. We explain the parallelization strategy using MPI, detail the data structures and APIs and discuss optimization techniques such as symmetry exploitation and vertex asymptotics. This chapter also presents benchmark results, scaling analyses and a validation of the implementation against known reference cases. Lastly, we summarize the main contributions and outline possible future extensions in Ch. 6.

Chapter 2 Foundations and Models

2.1 Solid-state Hamiltonian

In the realm of solid state physics, the many-body Schrödinger equation plays a crucial role in understanding the behavior of electrons and ions in crystalline materials. This equation provides a fundamental framework for describing the quantum mechanical interactions between multiple electrons and ions within a solid. The Schrödinger Hamiltonian for a system consisting of N_{el} electrons and N_{ion} ions — in its full beauty — reads

$$\begin{aligned}\mathcal{H} = &-\frac{\hbar^2}{2}\sum_{i}^{N_{ion}}\frac{1}{M_i}\nabla^2_{\mathbf{R}_i} - \frac{\hbar^2}{2m}\sum_{i}^{N_{el}}\nabla^2_{\mathbf{r}_i} + \frac{1}{2}\sum_{i\neq j}^{N_{ion}}\frac{Z_i Z_j e^2}{|\mathbf{R}_i - \mathbf{R}_j|} + \\ &+ \frac{1}{2}\sum_{i\neq j}^{N_{el}}\frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{i}^{N_{el}}\sum_{j}^{N_{ion}}\frac{Z_j e^2}{|\mathbf{r}_i - \mathbf{R}_j|} + \mathcal{V}_R,\end{aligned} \tag{2.1}$$

where $\mathbf{r}_i$ ($\mathbf{R}_i$) are the positions of the electrons (ions), m (M_i) their respective masses and Z_i the atomic number of the atoms from which the crystal is constructed. The Hamiltonian above can be described in a very simple manner: the first two terms describe the kinetic energy of the ions and electrons, whereas the third and fourth terms denote the mutual Coulomb interaction between all ions (third term) and all electrons (fourth term). The second to last term corresponds to the coupling between the ions and electrons and finally $\mathcal{V}_R$ expresses any relativistic effects[1]. The "only" thing left to do is solve the (time-independent) Schrödinger equation using the Hamiltonian of Eq. (2.1)

$$\mathcal{H}\psi(\{\mathbf{r}_i\},\{\mathbf{R}_i\}) = E\psi(\{\mathbf{r}_i\},\{\mathbf{R}_i\}), \tag{2.2}$$

which is easier said than done. The primary problem one faces by trying to do exactly that is the sheer number of particles present in a crystalline material. To be more precise, the wavefunction needed to fulfil the eigenvalue equation above would have to depend on all the degrees of freedom of $\mathcal{O}(10^{23})$ electrons and atoms present in solids. The exponential growth in the number of configurations makes it computationally infeasible to directly calculate or even store this quantity, and already a very small number of electrons and atoms $\mathcal{O}(10)$ exceeds the capability of modern supercomputers.

[1] e.g., spin-orbit coupling, relativistic mass correction, anomalous magnetic moments, relativistic Doppler shifts, Zitterbewegung, etc. An instructive example are the relativistic effects that take place in mercury: The binding orbitals of mercury are Lorentz-contracted due to the high speed of the electrons inside these orbitals. This results in a reduction in bonding strength, where ultimately mercury is forming a liquid at room temperatures [16]. In most real solids however, relativistic effects are mostly negligible and this term will not play a significant role in the calculations we will be concerned with.

Thus, one goal of condensed matter physicists is not to solve the fundamental Hamiltonian but instead to construct approximate models that contain all the physics necessary while still being controllable and numerically feasible.
Usually, approaches to approximate Eq. (2.1) fall into two main categories: (i) using approximate methods to solve the full Hamiltonian, or (ii) applying (more) exact methods to solve an approximate, mathematically simpler model Hamiltonian. Density functional theory (DFT), for example, falls under the first category, whereas dynamical mean-field theory (DMFT) or the dynamical vertex approximation (DΓA) reside in category two. The latter build upon a model Hamiltonian which usually incorporates only a subset of all degrees of freedom, e.g. specific orbitals of atomic shells. In modern calculations, a combination of different tools is used to calculate physical properties of materials. For example, a DFT calculation provides a band structure, from which the most relevant orbitals are identified, whereafter more sophisticated methods, such as DMFT or DΓA can be applied. Within this thesis, we mainly focus on the dynamical vertex approximation, however, we will give a short introduction to DFT and DMFT in the following subsections and Sec. 4.1, respectively, to complete the picture.

2.2 Born-Oppenheimer approximation

The timescales of electronic and ionic motion differ tremendously due to the large mass disparity between them: An electron is approximately 1800 times lighter than a proton, and atomic nuclei in real solids — especially those containing transition-metal elements — are typically about five orders of magnitude heavier than electrons, which was already recognized almost a century ago. This allows us, to a very good approximation, to solve the electronic degrees of freedom for static ionic coordinates. This approximation, called the Born-Oppenheimer approximation, is highly useful in condensed matter physics since it enables the study of electronic band structures, energy levels, and transport properties without the need to explicitly account for the complex interactions between electrons and lattice vibrations induced by the motion of ionic cores. However, there are certain phenomena, e.g. conventional phonon-mediated superconductivity, where the coupling of lattice vibrations with electrons can not be neglected. Luckily, there exist methods, like perturbation theory, which allow circumventing this apparent issue. Within the regime of the Born-Oppenheimer approximation, the Hamiltonian of Eq. (2.1) for the electrons reduces to

$$\mathcal{H}_{\text{B-O}} = -\frac{\hbar^2}{2m}\sum_{i}^{N_{el}} \nabla_{\mathbf{r}_i}^2 + \frac{1}{2}\sum_{i\neq j}^{N_{el}} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \sum_{i}^{N_{el}} \mathcal{V}_{ext}(\mathbf{r}_i), \tag{2.3}$$

where $\mathcal{V}_{ext}$ describes any external potential, e.g. from the ions in the lattice. The other quantities appearing in Eq. (2.3) have already been introduced in Eq. (2.1). This approximation will serve as the basis for all upcoming models if not stated otherwise.
Despite this large simplification, the eigenvalues and the wavefunctions of the Hamiltonian in Eq. (2.3) are still impossible to compute for real-world solids. Most of the difficulties lie in the second term of Eq. (2.3), which describes the Coulomb interaction between pairs of electrons. Due to this mutual electromagnetic interaction, the evolution of every electron is influenced by all other electrons present. Hence, all electrons are correlated with each other. In general, these effects are of course not negligible, however in some materials, e.g. in simple metals, where the electrons are very mobile, screening effects take place which weaken the Coulomb interaction between the electrons. In this context, a very successful ansatz to make Eq. (2.3) treatable is to model the system as electrons

interacting with an effective mean-field generated by all other electrons in its proximity. Such a mean field approach are the Hartree-Fock approximation or the Kohn-Sham equation of density functional theory.

2.3 Density functional theory

The basic idea of density functional theory (DFT) is to work with a simple quantity, the electron density $\rho(\mathbf{r})$, instead of trying to solve the ab-initio Hamiltonian of Eq. (2.3) — or other, mathematically simpler variants — through complicated many-body wavefunctions. This is possible, at least for the ground state energy and its derivatives, thanks to the two Hohenberg-Kohn theorems [17], which in essence state that (i) the ground state energy is a unique functional of the electron density $E[\rho(\mathbf{r})]$ which (ii) is minimized at the ground state density $\rho(\mathbf{r}) = \rho_0(\mathbf{r})$. This looks already quite promising, however, it lacks in some aspects: firstly, the two H-K theorems do not suggest any constructive/mathematical way to obtain the energy functional; secondly, this only holds for the ground state density. There is no way to extract any information from excited states; and lastly, the $3N$-dimensional object $\psi(\{\mathbf{r}_i\})$ is compressed into a three-dimensional object $\rho(\mathbf{r})$.
The big question is: "How does one obtain the energy functional $E[\rho(\mathbf{r})]$?". To answer this, we start by splitting up the energy functional into different contributions,

$$E[\rho(\mathbf{r})] = T[\rho(\mathbf{r})] + E_H[\rho(\mathbf{r})] + \int d^3r\, \rho(\mathbf{r})\mathcal{V}_{\text{ext}}(\mathbf{r}) + E_{XC}[\rho(\mathbf{r})]. \tag{2.4}$$

Here, $T[\rho(\mathbf{r})]$ denotes the kinetic energy of the electrons. $E_H[\rho(\mathbf{r})]$ is the electrostatic energy, given by the Hartree-term

$$E_H[\rho(\mathbf{r})] = \frac{1}{2}\int d^3r \int d^3r'\, \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}. \tag{2.5}$$

The potential term $\mathcal{V}_{\text{ext}}$ includes energies of electrons in external potentials (e.g. nuclei) and lastly, $E_{XC}[\rho(\mathbf{r})]$ includes everything else, i.e. the exchange and correlation energies. Dealing with the potential and the Hartree-term is easier compared to the treatment of the kinetic and exchange/correlation functional. Within Kohn-Sham DFT for example, XC functionals are often obtained by the local density approximation (LDA) or higher-order variants, such as different flavors of generalized gradient approximations (GGA). Stepping higher on Jacob's ladder[2], more sophisticated methods are hyper-GGA or several variants of (range-separated) hybrid functionals [18].
This already sounds very promising, why should we not stop here and solely describe our materials with DFT? This question can be answered straightforwardly: If we had the exact energy functional, it should pose no issue. However, in realistic settings one can never acquire the exact energy functional and has to employ, e.g. Kohn-Sham DFT. Since those effective mean-field approaches assume that interactions are relatively weak and treat electrons as largely independent, they become less accurate when correlations are strong. Orbitals with higher angular momentum (e.g. d or f orbitals) are

[2] This refers to a hierarchical classification of density functional approximations (DFAs) used in Kohn-Sham density functional theory (DFT) [18]. The ladder represents a sequence of increasingly accurate and complex approximations for the exchange-correlation energy, which is a key component of the total energy in DFT. Jacob's Ladder begins with the simplest and least accurate approximations at the lower rungs and progresses towards more sophisticated and accurate methods at the higher rungs. Each rung represents a level of approximation that includes additional physical effects and improves upon the previous level.

generally more localized in real space for a fixed principal quantum number, which leads to higher electronic densities and thus stronger local Coulomb interactions. However, the strength of these correlations depends not only on the orbital localization but also on the occupancy: correlations are particularly strong when these orbitals are partially filled, as is the case for transition-metal d shells or rare-earth f shells. In such situations, standard screening mechanisms are less effective, and mean-field theories tend to break down. Therefore, since e.g. K-S DFT is such a mean-field approach, it is not very successful in describing those systems. For example, DFT in LDA/GGA generically predicts a metallic behavior in partially filled 3d transition metal or 4f rare-earth elements, whereas experiments often find them to be insulating. That is, DFT fails for Mott insulators because it treats electrons as too delocalized, missing strong local Coulomb correlations that drive the insulating state. Thus, for strongly correlated materials, sophisticated many-body methods are needed. These do not approximate the full Born-Oppenheimer Hamiltonian of Eq. (2.3) directly — since many-body techniques tend to be numerically expensive — and instead rely on model Hamiltonians. The most important example is the Hubbard Hamiltonian, the standard Hamiltonian of strongly correlated condensed matter systems, which is described in more detail in Sec. 2.5.

2.4 Wannier projection

We now want to define an effective (low-energy) subspace, where we can use more advanced many-body methods. One commonly used approach is to obtain localized wavefunctions, also called Wannier functions, from extended Bloch wavefunctions used in DFT by Wannier projections [19]. These localized Wannier functions are constructed for the bands near the Fermi energy to filter out irrelevant bands and obtain a simplified model for the low-energy physics. They can then be interpreted as basis orbitals localized on lattice sites, forming the foundation for effective model descriptions. In the extreme case, when only a single band is retained, this reduces to the one-band Hubbard model, see Sec. 2.6. The eigenstates of the Schrödinger equation in a periodic lattice are Bloch functions of the form

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\mathbf{r}} u_n(\mathbf{r}), \tag{2.6}$$

where $u_n(\mathbf{r}+\mathbf{R}) = u_n(\mathbf{r})$ incorporates the periodicity of the lattice and in the simplest case, n denotes the index of a single isolated band. To obtain a local set of basis functions, one applies a Fourier transform to the Bloch functions,

$$w_{n\mathbf{R}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}\in BZ} e^{-i\mathbf{k}\mathbf{R}} \psi_{n\mathbf{k}}(\mathbf{r}), \tag{2.7}$$

where $\mathbf{R}$ is a lattice vector and $w_{n\mathbf{R}}(\mathbf{r})$ is a Wannier function centered around the lattice site $\mathbf{R}$. Wannier functions are an equivalent basis compared to the Bloch functions, however they are not eigenstates of the lattice Hamiltonian. Additionally, the choice for Eq. (2.7) is not unique, since the Bloch functions have a gauge freedom of a unitary transformation, which reads for a single isolated orbital n

$$\tilde{\psi}_{n\mathbf{k}}(\mathbf{r}) = e^{i\varphi_n(\mathbf{k})} \psi_{n\mathbf{k}}(\mathbf{r}). \tag{2.8}$$

Consequently, there is some degree of freedom to Wannier function construction. To reduce this arbitrariness one can exploit it and optimize the phases of Eq. (2.8), such that one obtains *maximally localized* Wannier functions.

2.4.1 Multi-orbital Wannier functions

So far, we assumed a single isolated band with band index n. This does not capture real-world materials, where one usually has a "spaghetti" of entangled bands. Let us assume that the number of Bloch bands, which are encompassed in the Wannier projection, is L and these bands are well separated from the other bands in the material. In this case, the transformation of Eq. (2.8) can be generalized to

$$\tilde{\psi}_{m\mathbf{k}}(\mathbf{r}) = \sum_{n=1}^{L} U^{\mathbf{k}}_{mn} \psi_{n\mathbf{k}}(\mathbf{r}), \tag{2.9}$$

where $U^{\mathbf{k}}_{mn} \in M \times L$. The construction of the Wannier functions follows

$$w_{m\mathbf{R}}(\mathbf{r}) = \sum_{\mathbf{k}\in BZ} e^{-i\mathbf{k}\mathbf{R}} \sum_{n=1}^{L} U^{\mathbf{k}}_{mn} \psi_{n\mathbf{k}}(\mathbf{r}), \tag{2.10}$$

which allows for an additional mixing of the Bloch states. In this case, the unitary matrix $U^{\mathbf{k}}_{mn}$ encodes the gauge freedom of the Wannier projection. A popular unique "gauge fixing" of $U^{\mathbf{k}}_{mn}$ is again through maximally localized Wannier functions.

2.4.2 Maximally localized Wannier functions

The idea here is to choose $U^{\mathbf{k}}_{mn}$ such that the sum of quadratic spreads of the Wannier functions is minimized [19]. This spread of the set of L Wannier functions in real space is quantified by the localization functional Ω and reads

$$\Omega = \sum_{m} \left[\left\langle \mathbf{r}^2 \right\rangle_m - \langle \mathbf{r} \rangle^2_m \right] = \sum_{m} \left[\langle w_{m\mathbf{0}} | \, \mathbf{r}^2 \, | w_{m\mathbf{0}} \rangle - | \langle w_{m\mathbf{0}} | \, \mathbf{r} \, | w_{m\mathbf{0}} \rangle |^2 \right]. \tag{2.11}$$

Applying a minimization procedure to the spread of Wannier functions results in a "gauge-fixed" $\tilde{U}^{\mathbf{k}}_{mn}$, with a unique unitary transformation. The resulting Wannier functions are then called maximally localized.

2.5 Multi-orbital Hubbard model

Now we have acquired enough tools to describe one of the most important models of strongly correlated electron systems. As already discussed at the end of Sec. 2.3, for elements with partially filled d or f orbitals, a mean-field description, as in DFT, is not enough to successfully describe microscopic and macroscopic phenomena due to strong electronic correlation.

A sophisticated and accurate treatment on the level of the Born-Oppenheimer approximation (Eq. (2.3)) is currently not feasible. Hence, the method of choice is the introduction of a mathematically simpler, but conceptually still very relevant model Hamiltonian that is able to capture the essential low-energy physics, and to derive the bandstructure for this model from DFT and the interaction from the constrained random phase approximation (cRPA). The Hamiltonian is named after J. Hubbard.

As a starting point we choose the Hamiltonian of Eq. (2.3) in second quantization

$$\begin{aligned}\mathcal{H} = \sum_{\sigma} \int d^3r \, \hat{\psi}^{\dagger}_{\sigma}(\mathbf{r}) \left[-\frac{\hbar^2}{2m} \nabla^2_{\mathbf{r}} + \mathcal{V}(\mathbf{r}) \right] \hat{\psi}_{\sigma}(\mathbf{r}) \\ + \frac{1}{2} \sum_{\sigma\sigma'} \int d^3r \int d^3r' \, \hat{\psi}^{\dagger}_{\sigma}(\mathbf{r}) \hat{\psi}^{\dagger}_{\sigma'}(\mathbf{r}') \frac{e^2}{|\mathbf{r}-\mathbf{r}'|} \hat{\psi}_{\sigma'}(\mathbf{r}') \hat{\psi}_{\sigma}(\mathbf{r}),\end{aligned} \tag{2.12}$$

where $\hat{\psi}^{(\dagger)}_{\sigma}(\mathbf{r})$ are field operators that annihilate (create) an electron with spin σ at position $\mathbf{r}$. We can expand these field operators in terms of maximally localized Wannier functions introduced in Sec. 2.4.2, yielding

$$\hat{\psi}^{(\dagger)}_{\sigma}(\mathbf{r}) = \sum_{\mathbf{R},m} w^{(*)}_{m\mathbf{R}}(\mathbf{r}) \, \hat{c}^{(\dagger)}_{\mathbf{R}m\sigma}, \tag{2.13}$$

where $w_{m\mathbf{R}}$ is the (maximally localized) Wannier function of orbital m centered around $\mathbf{R}$, and $\hat{c}_{\mathbf{R}m\sigma}$ the corresponding annihilation operator. Note that the orbitals m are usually the localized 3d or 4f orbitals in transition metals or rare earth elements. With the help of Eq. (2.13), the Hamiltonian of Eq. (2.12) can be rewritten as

$$\begin{aligned}\mathcal{H} = -\sum_{\substack{\mathbf{R}\mathbf{R}' \\ mm',\sigma}} t_{mm'}(\mathbf{R},\mathbf{R}') \hat{c}^{\dagger}_{\mathbf{R}m\sigma} \hat{c}_{\mathbf{R}'m'\sigma} \\ + \frac{1}{2} \sum_{\substack{\mathbf{R}_1\mathbf{R}_2\mathbf{R}_3\mathbf{R}_4 \\ ll'mm' \\ \sigma\sigma'}} U_{lmm'l'}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3,\mathbf{R}_4) \hat{c}^{\dagger}_{\mathbf{R}_1 l\sigma} \hat{c}^{\dagger}_{\mathbf{R}_3 m'\sigma'} \hat{c}_{\mathbf{R}_4 l'\sigma'} \hat{c}_{\mathbf{R}_2 m\sigma},\end{aligned} \tag{2.14}$$

where $ll'mm'$ denote different Wannier orbitals, while $\mathbf{R}_i$ corresponds to a lattice site. Hence, the first term in the Hubbard Hamiltonian represents the kinetic energy and describes electrons hopping from orbital m' at site $\mathbf{R}'$ to orbital m at site $\mathbf{R}$. The hopping matrix element $t_{mm'}(\mathbf{R},\mathbf{R}')$ (in units of energy) quantifies the amplitude for such a process and thus determines how easily an electron can move between the two orbitals. The second term in Eq. (2.14) encodes the Coulomb interaction between two electrons. Some of the terms of the multiorbital Hubbard Hamiltonian are visually depicted in Fig. 2.1, where a subset of effects of $t_{mm'}(\mathbf{R},\mathbf{R}')$ and $U_{lmm'l'}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3,\mathbf{R}_4)$ are shown for two lattice sites. Both the hopping matrix element $t_{mm'}(\mathbf{R},\mathbf{R}')$ and Coulomb interaction

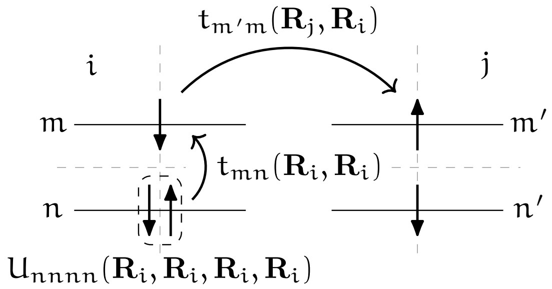


FIGURE 2.1 – Visual representation of a subset of interaction and hopping terms of Eq. (2.14). We show two lattice sites i and j with orbitals m, n and m', n', respectively.

$U_{lmm'l'}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3,\mathbf{R}_4)$ are given by

$$t_{mm'}(\mathbf{R},\mathbf{R}') = -\int d^3r\, w^*_{m\mathbf{R}}(\mathbf{r})\left[-\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}} + \mathcal{V}(\mathbf{r})\right] w_{m'\mathbf{R}'} \quad \text{and} \tag{2.15a}$$

$$U_{lmm'l'}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3,\mathbf{R}_4) = \int d^3r \int d^3r'\, w^*_{l\mathbf{R}_1}(\mathbf{r}) w^*_{m'\mathbf{R}_3}(\mathbf{r}') \frac{e^2}{|\mathbf{r}-\mathbf{r}'|} w_{l'\mathbf{R}_4}(\mathbf{r}') w_{m\mathbf{R}_2}(\mathbf{r}). \tag{2.15b}$$

While for Eq. (2.14) we have already restricted ourselves to a subspace of orbitals, spanned by the Wannier function we picked previously, it is still too complex to be solved exactly in the general case. The complexity can be further reduced by either placing further assumptions and restrictions or by using symmetries of the Hamiltonian, which are present in the hopping and interaction terms. Note, that Eq. (2.15b) overestimates the repulsive Coulomb interaction since screening effects from electrons outside the orbital subspace are not considered. One way to partly overcome this issue is offered by the constrained Random Phase Approximation (cRPA). For details on how cRPA works, we refer the reader to, e.g. Ref. [20].

2.5.1 Fourier transform of the Coulomb term

For reasons that become apparent later, it is more convenient to express the interaction term in Fourier space for diagrammatic calculations. For this we follow Ref. [13], where we simplify the notation by employing a compound index $i = \{\mathbf{R}_i, n_i\}$, denoting a combination of an orbital index and a real-space vector or a momentum vector, depending on the context if they are not explicitly written down. If some parameters contained in a compound index are explicitly written down, the compound index is often additionally placed to denote the remaining quantities. The real-space interaction term in Eq. (2.14) inherently depends on four lattice vectors. However, to simplify the notation using translational symmetry, one of these arguments (e.g. $\mathbf{R}_4$) can be eliminated by expressing the distances relative to the lattice site 4 rather than using their absolute positions. Therefore, [13],

$$\hat{U} = \frac{1}{2}\sum_{\substack{\mathbf{R}_1\mathbf{R}_2\mathbf{R}_3\\1234\\\sigma\sigma'}} U_{1234}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3)\hat{c}^\dagger_{1\sigma}\hat{c}^\dagger_{3\sigma'}\hat{c}_{\mathbf{R}_4=\mathbf{0};4\sigma'}\hat{c}_{2\sigma}. \tag{2.16}$$

This quantity naturally fulfills a swapping symmetry, where the simultaneous exchange of both incoming and outgoing particles keeps it invariant, i.e.

$$U_{1234}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3) = U_{2143}(\mathbf{R}_3-\mathbf{R}_2,-\mathbf{R}_2,\mathbf{R}_1-\mathbf{R}_2). \tag{2.17}$$

The general Fourier transform of the tensor in Eq. (2.16) with respect to the lattice positions $\mathbf{R}_i$ is defined as

$$U^{\mathbf{qkk}'}_{1234} := \sum_{\mathbf{R}_1\mathbf{R}_2\mathbf{R}_3} e^{i\mathbf{k}\mathbf{R}_1} e^{-i(\mathbf{k}-\mathbf{q})\mathbf{R}_2} e^{i(\mathbf{k}'-\mathbf{q})\mathbf{R}_3} U_{1234}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3), \tag{2.18}$$

or for the full interaction operator in Eq. (2.14)

$$\hat{U} = \frac{1}{2}\sum_{\substack{\mathbf{qkk}'\\1234\\\sigma\sigma'}} U^{\mathbf{qkk}'}_{1234}\hat{c}^\dagger_{\mathbf{k}1\sigma}\hat{c}^\dagger_{\mathbf{k}'-\mathbf{q}3\sigma'}\hat{c}_{\mathbf{k}'4\sigma'}\hat{c}_{\mathbf{k}-\mathbf{q}2\sigma}, \tag{2.19}$$

where the Fourier transforms of the creation and annihilation operators are given by

$$\hat{c}^{(\dagger)}_{\mathbf{k}1\sigma} = \sum_{\mathbf{R}} e^{(-)i\mathbf{k}\mathbf{R}} \hat{c}^{(\dagger)}_{\mathbf{R}1\sigma}. \tag{2.20}$$

In the case where Wannier orbitals outside the same unit cell have negligible overlap (effectively pairing up the creation and annihilation operators at sites $\mathbf{0}$ and $\mathbf{R}$), the interaction becomes diagonal in real space. As a result, the Coulomb matrix elements depend only on the relative positions of the sites and their Fourier transform depends at most on the momentum transfer $\mathbf{q}$, while the explicit dependence on $\mathbf{k}$ and $\mathbf{k}'$ vanishes,

$$U_{1234} := U_{1234}(\mathbf{0},\mathbf{0},\mathbf{0}), \quad \text{and} \tag{2.21a}$$

$$V^{\mathbf{q}}_{1234} := \sum_{\mathbf{R}\neq\mathbf{0}} e^{i\mathbf{q}\mathbf{R}} U_{1234}(\mathbf{R},\mathbf{R},\mathbf{0}), \tag{2.21b}$$

corresponding to a possible split into a purely local interaction U_{1234} and a purely non-local interaction $V^{\mathbf{q}}_{1234}$. In this case the exchange symmetry of the incoming and outgoing electron reduces to $U_{1234} = U_{2143}$ and $V^{\mathbf{q}}_{1234} = V^{-\mathbf{q}}_{2143}$.

2.5.2 More symmetries, assumptions and the Kanamori-Hamiltonian

Despite the simplifications of the interaction tensor due to the translational symmetry, the object still has a complicated structure for a multi-orbital system. While one can in principle calculate all of these elements using methods such as cRPA, there are certain symmetries and assumptions that can help reduce the amount of individual elements of $U_{1234}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3,\mathbf{R}_4)$ drastically. Similar simplifications can be made for the kinetic term in Eq. (2.14). Due to the localized nature of Wannier orbitals, the Hopping amplitudes reduce (rapidly) in magnitude the further apart the orbitals are located in real space due to reduced orbital overlap. The decay of the hopping elements $t_{12}(\mathbf{R}_1,\mathbf{R}_2)$ behaves as at least $|\mathbf{R}_1 - \mathbf{R}_2|^{-1}$[3]. Hence, for practical calculations, a very good approximation — at least in the one-band case — can often be achieved by considering only the three closest (in distance) hopping terms: nearest (t), next-nearest (t'); and next-next-nearest (t'') neighbor, see Fig. 2.3 for an illustration. Elements of $U_{1234}(\mathbf{R}_1,\mathbf{R}_2,\mathbf{R}_3,\mathbf{R}_4)$ usually decay even faster than the hopping terms due to the overlap integral between four orbitals in Eq. (2.15b) compared to only two.

To further reduce the degrees of freedom of Eq. (2.14) and make the model more tractable, additional assumptions about the interaction can be made: (i) the Wannier functions respect SU(2) symmetry; (ii) the Wannier functions of the subset of orbitals preserve orbital symmetry; and (iii) the Coulomb interaction of electrons on different lattice sites is small compared to the on-site Coulomb interaction, e.g. $U_{1234}(\mathbf{R},\mathbf{R},\mathbf{R}) \gg U_{1234}(\mathbf{R},\mathbf{R},\mathbf{R}') \gg U_{1234}(\mathbf{R},\mathbf{R}',\mathbf{R}'')$. Let us briefly mention why these assumptions hold for systems we want to consider: (i) holds when spin-orbit coupling is ignored, the material is in the paramagnetic phase, and no external electromagnetic fields are applied [24]; (ii) applies to degenerate orbital subsets, like the t_{2g} and e_g orbitals in the 3d shell; and (iii) is valid as long as the Wannier functions remain localized around their respective lattice sites, which is generally true for d or f shells which do not (strongly) hybridize with ligand orbitals. This leads to

[3] More precisely, it was found that for one-dimensional and centrosymmetric three-dimensional systems the decay behaves purely exponential, i.e. $w(\mathbf{r}) \propto e^{-|\mathbf{r}|/\xi}$ [21]. For two- or three-dimensional insulating systems, $w(\mathbf{r}) \propto |\mathbf{r}|^{-\alpha} e^{-|\mathbf{r}|/\xi}$ [22] and for metals $w(\mathbf{r}) \propto |\mathbf{r}|^{-\alpha}$, where in this case they cannot decay exponentially [23], were found.

the assumption of the interaction in Eq. (2.21) to be purely local, i.e. $V^{q}_{1234} = 0$, since we are only keeping creation and annihilation operators in the same unit cell[4].
Under the assumptions (i)-(iii) and for a two-band model $U_{1234}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3)$ is constrained to a single unit cell and has only three independent entries, commonly denoted by the intra-orbital and inter-orbital interactions U and U', respectively, and the Hund's exchange J [24, 25]. A visual representation of the effects of U, U' and J can be found in Fig. 2.2. If the model consists of additional orbitals and/or sites, the above assumptions are not enough to reduce $U_{1234}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3)$ to just three terms. These quantities are given by

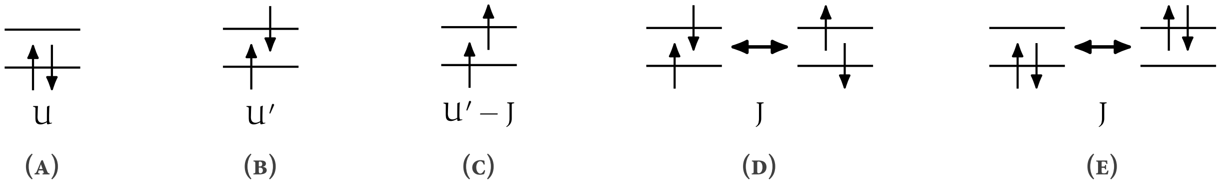


FIGURE 2.2 – Kanamori parameters U, U' and J as described in the text: (A) intra-orbital Coulomb interaction, (B)-(C) inter-orbital Coulomb interaction, (D) spin-flip of an electron pair and (E) pair hopping. Adapted from Ref. [26] to match the style of this thesis.

$$U = U_{1111} = \int d^3r \int d^3r' \, |w_1(\mathbf{r})|^2 \mathcal{V}_c(\mathbf{r} - \mathbf{r}') |w_1(\mathbf{r}')|^2, \tag{2.22a}$$

$$U' = U_{1122} = \int d^3r \int d^3r' \, |w_1(\mathbf{r})|^2 \mathcal{V}_c(\mathbf{r} - \mathbf{r}') |w_2(\mathbf{r}')|^2 \quad \text{and} \tag{2.22b}$$

$$J = U_{1221} = \int d^3r \int d^3r' \, w_1^*(\mathbf{r}) w_2^*(\mathbf{r}') \mathcal{V}_c(\mathbf{r} - \mathbf{r}') w_1(\mathbf{r}') w_2(\mathbf{r}), \tag{2.22c}$$

where $\mathcal{V}_c(\mathbf{r} - \mathbf{r}')$ denotes the partially screened Coulomb interaction. The terms in Eq. (2.22) correspond to the interactions of the Kanamori Hamiltonian. Since this Hamiltonian is an effective low-energy model, the high-energy electronic degrees of freedom have been integrated out, and their effect is incorporated as an effective screening of the local electron-electron repulsion. For cubic symmetry — due to constraint (ii) — there are only two independent quantities in the interaction, resulting in $U' = U - 2J$. With the help of these three parameters we can simplify the Hamiltonian of Eq. (2.14) and for the general case the so-called Kanamori-Hamiltonian [24, 25] reads

$$\begin{aligned} \mathcal{H}_{\text{Kanamori}} = &- \sum_{ab,\sigma} t_{ab} \hat{c}^{\dagger}_{a\sigma} \hat{c}_{b\sigma} + U \sum_{a} \hat{n}_{a\uparrow} \hat{n}_{a\downarrow} + \sum_{\substack{\mathbf{R}, a \neq b \\ \sigma\sigma'}} (U' - J\delta_{\sigma\sigma'}) \hat{n}_{\mathbf{R}a\sigma} \hat{n}_{\mathbf{R}b\sigma'} \\ &- J \sum_{\mathbf{R}, a \neq b} \hat{c}^{\dagger}_{\mathbf{R}a\uparrow} \hat{c}_{\mathbf{R}a\downarrow} \hat{c}^{\dagger}_{\mathbf{R}b\downarrow} \hat{c}_{\mathbf{R}b\uparrow} + J \sum_{\mathbf{R}, a \neq b} \hat{c}^{\dagger}_{\mathbf{R}a\uparrow} \hat{c}^{\dagger}_{\mathbf{R}a\downarrow} \hat{c}_{\mathbf{R}b\downarrow} \hat{c}_{\mathbf{R}b\uparrow}, \end{aligned} \tag{2.23}$$

where $\hat{n}_{a\sigma}$ is the number operator ($\hat{c}^{\dagger}_{a\sigma} \hat{c}_{a\sigma}$) that counts the number of electrons in orbital a with spin σ. The Kanamori Hamiltonian is a widely used model for strongly correlated systems and it can capture a lot of low-energy physics.

[4]While it is generally true that the local interaction is stronger than $U(|\mathbf{R}_i| > 0)$, the non-local part is often not negligible. For the case of $La_3Ni_2O_7$, the non-local contributions of the interaction are believed to be long-range in-plane.

2.6 Single-band Hubbard Hamiltonian

To benchmark the code developed in the course of this thesis, we consider the well-studied case of the single-band Hubbard model, which has been extensively analyzed within the dynamical vertex approximation. We therefore briefly introduce its Hamiltonian in the following. The single-band version of Eq. (2.14) is one of the most widely used model Hamiltonians to describe strongly correlated materials and reads

$$\mathcal{H}_{\text{Hubbard}} = -\sum_{i\neq j,\sigma} t_{ij}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} - \mu\sum_{i}(\hat{n}_{j\uparrow}+\hat{n}_{j\downarrow}), \tag{2.24}$$

where i and j denote the lattice site in the N-dimensional lattice and the hopping is spin-independent, and μ denotes the chemical potential. Since the model is reduced to a single orbital, the inter-orbital interaction terms U' and J no longer appear, as they are only defined between different orbitals.

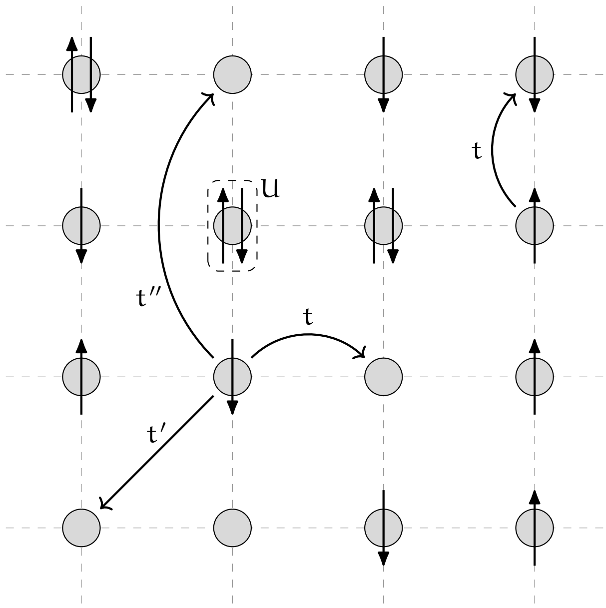


FIGURE 2.3 – Visual representation of the two-dimensional square lattice Hubbard model. Each lattice site either contains zero, one or two electrons. Every circle represents a local orbital at any site. Here, t, t' and t'' denote nearest-neighbor hopping, next-nearest-neighbor hopping and next-next-nearest-neighbor hopping, respectively. Realistic values for t, t', t'' and U are obtained from first principles or *ab-initio* methods, e.g. density functional theory (DFT) [27], ARPES [28] or cRPA [29]. For e.g. cuprates, one finds that $t \approx 0.3$-0.5 eV, $t' \approx -0.2t$, $t'' \approx 0.1t$ and $U \approx 8t$.

This is one of the simplest forms of the general Hubbard Hamiltonian of Eq. (2.14), however it can still not be solved exactly since the kinetic term is diagonal in momentum space whereas the interaction term is diagonal in real space. A general analytical solution to the model only exists for one-dimension [30] or two sites [31], however recent advances allowed numerical calculations to exactly calculate the eigen decomposition of the Hubbard Hamiltonian for a few lattice sites [32]. To visually illustrate the model's description, a representation of the hopping and interaction terms of Eq. (2.24) can be seen in Fig. 2.3.

In the previous sections, we looked at the fundamental Hamiltonians and a few specific models that create the theoretical foundation for modelling strongly correlated electron systems, such as transition metal oxides (e.g. manganites, cuprates or nickelates), heavy fermion systems (e.g. compounds containing rare-earth or actinide elements), Mott insulators (e.g. V_2O_3) and many more. We have developed the fundamental ideas required to explain the distinctive behavior of particles in systems where interactions play an important role.
With the fundamentals established, in the following chapters we will introduce the mathematical framework for performing realistic material calculations within these models. The next chapter presents a comprehensive toolkit, including many-body Green's functions and diagrammatic methods such as the dynamical mean-field theory (DMFT) and the dynamical vertex approximation (DΓA), which allow us to compute physically relevant observables. The equations that follow form the core of the program developed as part of this thesis, bridging the theoretical models with their numerical implementation.

Chapter 3 Quantum many-body framework

This chapter is dedicated to introducing the necessary tools needed to perform realistic material calculations using dynamical mean-field theory (DMFT) or the dynamical vertex approximation (DΓA). We start with the most central objects — the Green's functions — and subsequently assemble a complete framework which enables us to calculate a number of physical properties, e.g. the susceptibility, optical conductivity, superconductivity, pseudogap formation and more, that stem from the correlated interplay between electrons or holes.

3.1 One-particle Green's functions

Green's functions are fundamental tools in many-body physics, used to study the properties and behavior of interacting quantum systems. These mathematical objects encode information about the propagation of particles or excitations within a system and serve as the cornerstone for analyzing both equilibrium and nonequilibrium states. In the many-body regime, Green's functions extend beyond single-particle descriptions to include complex interactions among multiple particles. They provide insights into phenomena such as quasiparticle lifetimes, collective excitations and response functions. Specifically, one- and two-particle Green's functions describe the propagation of a single particle and the correlated motion of particle pairs, respectively. Alongside of the mathematical description of Green's function-based expressions, we also provide a pictorial description of the equations using Feynman diagrams. These Feynman diagrams allow us to easily describe interaction processes visually. Lines and vertices within the diagrams indicate particle propagators and interaction vertices, respectively. External legs will be colored gray while inner legs and interaction lines will be colored black.

In condensed matter physics, we are usually interested in finite-temperature effects. Hence, it is practical to use the so-called Matsubara formalism [33] in imaginary time by performing a Wick rotation $t \to -i\tau$ [34].

Furthermore, most many-body quantities described in the following are n-point (correlation) functions[1]. These functions typically include a subset of the following parameters for each external leg: (Matsubara) frequency (ν), spin index (σ), orbital index (o), lattice position ($\mathbf{R}$), momentum ($\mathbf{k}$) or imaginary time (τ). This introduces a huge amount of parameters appended to a variable and can be very cumbersome to read through if explicitly written down. Therefore, to increase readability, we follow Ref. [35] and group all indices that are not explicitly written down into a compound index, e.g. $i = \{o_i, \sigma_i, \mathbf{R}_i\}$. If an equation containing spatial or momentum-dependent quantities is written down with a compound index, it applies equally in both real and Fourier

[1] Here, n denotes the number of "connection points" of the corresponding Feynman diagram.

space. Furthermore, summing over these compound indices means summing over all individual components they include, with a normalization of $\frac{1}{\beta}$ for frequency sums and $\frac{1}{N_{\mathbf{k}}}$ for momentum sums, where $\beta = \frac{1}{k_B T}$ is the inverse temperature and $N_{\mathbf{k}}$ is the total number of reciprocal lattice points. After this short introduction, let us dive in by defining the most central quantity, the two-point (one-particle) Green's function, in a system in thermal equilibrium as

$$G^{\mathbf{k}}_{12} = -\left\langle \mathcal{T}\left[\hat{c}_{\mathbf{k};1}\hat{c}^{\dagger}_{\mathbf{k};2}\right]\right\rangle, \tag{3.1}$$

where $\hat{c}_{\mathbf{k};i}$ ($\hat{c}^{\dagger}_{\mathbf{k};i}$) are fermionic annihilation (creation) operators which annihilate (create) an electron with parameters $i = \{\sigma_i, o_i, \tau_i\}$ and momentum $\mathbf{k}$ in the system. $\langle\cdot\rangle = \frac{1}{Z}\mathrm{Tr}\{e^{-\beta\hat{\mathcal{H}}}\ \cdot\}$ with $Z = \mathrm{Tr}\{e^{-\beta\hat{\mathcal{H}}}\}$ describing the thermal expectation value. Note, that the time evolution operator $e^{i\hat{\mathcal{H}}t} = e^{\hat{\mathcal{H}}\tau}$ is real after performing the Wick rotation to imaginary time and the Boltzmann weight $e^{\beta\hat{\mathcal{H}}}$ describes nothing more than an additional evolution in imaginary time. Lastly, $\mathcal{T}[\cdot]$ is the imaginary time ordering operator, where

$$\mathcal{T}\left[\hat{c}^{(\dagger)}_1(\tau_1)\hat{c}^{(\dagger)}_2(\tau_2)\right] = \Theta(\tau_1-\tau_2)\hat{c}^{(\dagger)}_1(\tau_1)\hat{c}^{(\dagger)}_2(\tau_2) - \Theta(\tau_2-\tau_1)\hat{c}^{(\dagger)}_2(\tau_2)\hat{c}^{(\dagger)}_1(\tau_1). \tag{3.2}$$

In simple quantum-mechanical terms, Green's functions are often introduced as probability amplitudes describing the propagation of a particle from one point in space and time to another. While this picture provides useful intuition, it becomes too restrictive in our context. Here, Green's functions are defined as thermal averages, i.e. they involve a summation over many states and their physical significance is more naturally understood in terms of response functions: Green's functions are directly related to measurable quantities such as spectral functions, densities of states, susceptibilities and conductivities. They are the most central objects in our theory and enter all Feynman diagrams; hence we start with their Feynman diagram in Fig. 3.1.

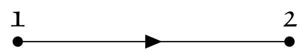
$G_{12} =$ 

Figure 3.1 – Diagrammatic representation of the one-particle Green's function G_{12}. The electron is annihilated at 1, propagates in imaginary time from τ_1 to τ_2 and is created at 2.

In homogeneous, i.e. translationally invariant, systems the Green's function reflects this symmetry and becomes invariant under spatial translations. This means that $G_{12}(\mathbf{R}_1, \mathbf{R}_2) = G_{12}(\mathbf{R}_1 - \mathbf{R}_2, \mathbf{0}) = G_{12}(\mathbf{R})$. In Fourier space, due to momentum conservation, the Green's function is only dependent on a single momentum parameter, $G_{12}(\mathbf{k}_1, \mathbf{k}_2) = G_{12}(\mathbf{k}_1 - \mathbf{k}_2, \mathbf{0}) = G_{12}(\mathbf{k})$. Furthermore, if the system possesses time-translational invariance or is stationary, i.e. the Hamiltonian is not explicitly dependent on imaginary time, then $G_{12}(\tau_1, \tau_2) = G_{12}(\tau_1 - \tau_2, 0) = G_{12}(\tau)$. If the system additionally possesses SU(2) symmetry, the Green's functions are spin-diagonal ($G_{\uparrow\downarrow} = G_{\downarrow\uparrow} = 0$) and the diagonal components are identical $G_{\uparrow\uparrow} = G_{\downarrow\downarrow} = G$. This also holds for the self-energy, which will be discussed later. Additionally, the *fermionic* Green's function can be shown to be anti-periodic in imaginary time with a period of β[2],

$$G_{12}(\tau) = -G_{12}(\beta-\tau) \quad \text{for } \tau > 0 \qquad \text{and} \tag{3.3a}$$

[2] This originates from the Boltzmann term $e^{\beta\hat{\mathcal{H}}}$ in the thermal expectation value and restricts the imaginary time domain to $\tau \in [-\beta, \beta)$.

$$G_{12}(\tau) = -G_{12}(\beta+\tau) \quad \text{for } \tau < 0. \tag{3.3b}$$

This anti-periodicity in time leads to the Fourier transform to be defined over a discrete set of imaginary frequencies, the so-called Matsubara frequencies, which for fermionic quantities are given by $\nu_n = (2n+1)\frac{\pi}{\beta}$. In contrast, *bosonic* Green's functions are periodic in imaginary time with period β and therefore the Fourier transform includes the set of bosonic Matsubara frequencies $\omega_n = (2n)\frac{\pi}{\beta}$. In the following, we will always denote fermionic Matsubara frequencies with ν_n and bosonic ones with ω_n. The fermionic- and bosonic-like Fourier transforms of $G_{12}(\tau)$ become, respectively,

$$G_{12}(i\nu_n) = \int_0^\beta d\tau\, G_{12}(\tau)e^{i\nu_n\tau} \qquad \text{and} \qquad G_{12}(i\omega_n) = \int_0^\beta d\tau\, G_{12}(\tau)e^{i\omega_n\tau} \tag{3.4}$$

and only require the knowledge of Green's functions for positive imaginary times τ. The inverse Fourier transforms are given by

$$G_{12}(\tau) = \frac{1}{\beta}\sum_n G_{12}(i\nu_n)e^{-i\nu_n\tau} \qquad \text{and} \qquad G_{12}(\tau) = \frac{1}{\beta}\sum_n G_{12}(i\omega_n)e^{-i\omega_n\tau}. \tag{3.5}$$

By expressing the fermionic Green's function in Eq. (3.1) in its spectral representation, performing the discrete Fourier transform and subsequently transforming to Matsubara frequencies, we obtain

$$G_{12}(i\nu_n) = \frac{1}{Z}\sum_{mn}\left(e^{-\beta E_n} + e^{-\beta E_m}\right)\frac{\langle n|\,\hat{c}_1\,|m\rangle\,\langle m|\,\hat{c}_2^\dagger\,|n\rangle}{i\nu - (E_m - E_n)}, \tag{3.6}$$

where $|n\rangle$ and $|m\rangle$ are eigenstates of the Hamiltonian. If we take $\mathbf{R}_1 = \mathbf{R}_2$, which corresponds to the *local* Green's function, we can introduce the $\mathbf{k}$-resolved spectral function

$$A(\mathbf{k},\nu) = \frac{1}{Z}\sum_{mn} e^{-\beta E_n}\left(1 + e^{-\beta\nu}\right)|\,\langle n|\,\hat{c}_\mathbf{k}\,|m\rangle\,|^2\delta(\nu - E_m - E_n), \tag{3.7}$$

where

$$G(\mathbf{k}, i\nu_n) = \int_\mathbb{R} d\nu\,\frac{A(\mathbf{k},\nu)}{i\nu_n - \nu}. \tag{3.8}$$

The spectral function $A(\mathbf{k},\nu)$ characterizes the probability of adding or removing a particle with momentum $\mathbf{k}$ and spin σ at energy ν. In this sense, it directly reflects the processes measured in angle-resolved photoemission spectroscopy (ARPES) and inverse photoemission, where an electron is removed from or injected into the system. For this reason, $A(\mathbf{k},\nu)$ is referred to as the single-particle ($\mathbf{k}$-resolved) spectral function. Moreover, since it represents a normalized probability distribution in energy, the integral of $A(\mathbf{k},\nu)$ over all frequencies equals unity. It is directly measurable using techniques like angle-resolved photoemission spectroscopy (ARPES) [36, 37], which probes electronic states in the material and reveals electronic structure properties, such as band gaps and quasiparticle dispersions. Peaks in $A(\mathbf{k},\nu)$ correspond to quasiparticle states, with their position indicating the energy of these states and their width related to the lifetime or decay rate of the quasiparticles. A narrow peak indicates a well-defined quasiparticle with a long lifetime and weak decay rate, while a broad peak suggests the contrary. The spectral function is a key quantity in many-body theory as it connects real-world experimental results to the Green's function formalism, as is shown in Refs. [36, 37] and the following. By performing analytic continuation in the upper complex plane ($i\nu \to \nu + i0^+$), one can define the retarded Green's function as

$$G^R(\mathbf{k}, \nu) = \int_{\mathbb{R}} d\nu' \, \frac{A(\mathbf{k}, \nu')}{\nu - \nu' + i0^+}, \tag{3.9}$$

from where one obtains through the Kramers-Kronig relations

$$A(\mathbf{k}, \nu) = -\frac{1}{\pi} \Im G^R(\mathbf{k}, \nu). \tag{3.10}$$

Thus, the spectral function is directly connected to the imaginary part of the retarded Green's function, connecting many-body theory with physical experiments.

As an example, which will prove itself useful for the future, let us consider non-interacting electrons in a system with translational symmetry that are described by the Hamiltonian

$$\hat{\mathcal{H}}_0 = \sum_{\mathbf{k};12} \varepsilon_{12}(\mathbf{k}) \hat{c}^\dagger_{\mathbf{k};1} \hat{c}_{\mathbf{k};2}, \tag{3.11}$$

where $\varepsilon_{12}(\mathbf{k}) = -\sum_{\mathbf{R};12} t_{12}(\mathbf{R}) e^{i\mathbf{k}\mathbf{R}}$ is the momentum dispersion of the band and is measured with respect to the chemical potential μ. In this case, it is easy to show, that

$$G_{0;12}(\nu, \mathbf{k}) = [i\nu\delta_{12} - \varepsilon_{12}(\mathbf{k})]^{-1}. \tag{3.12}$$

For the one-particle spectral function, we find in this case

$$A_{12}(\varepsilon, \mathbf{k}) = \delta(\varepsilon - \varepsilon_{\mathbf{k};12})_{12}, \tag{3.13}$$

corresponding to stable quasiparticles with infinite lifetime, due to zero width of the quasiparticle peak of $A_{12}(\varepsilon, \mathbf{k})$.

A direct computation of the Green's function as defined in Eq. (3.1) generally incurs an exponential increase in computational cost with lattice size. While this approach can be carried out exactly for a limited class of Hamiltonians, it becomes infeasible for the Hubbard Hamiltonian, which we employ throughout this work. A standard approach, since the non-interacting situation can be treated straightforwardly, is to begin with the non-interacting case and construct a perturbative expansion in terms of the surrounding interaction. In this context, we will start with the non-interacting case described above. The perturbation expansion in the interaction U will not be derived in detail in this thesis, as there are many wonderful resources that explain it very thoroughly [38, 39], thus we will only sketch it here. In essence, the expansion is constructed using the so-called S-matrix and uses Wick's theorem [34, 40, 41] and the linked cluster theorem [42]. The expansion begins by specifying the term to be expanded, for instance $\hat{\mathcal{H}}_I$ of Eq. (2.14) in the case of an interaction expansion:

$$\hat{\mathcal{H}}_I = \frac{1}{2} \sum_{1234} U_{1234} \hat{c}^\dagger_1 \hat{c}^\dagger_3 \hat{c}_4 \hat{c}_2. \tag{3.14}$$

The expansion of the Green's function then reads

$$G_{12}(\tau) = -\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \cdots \int_0^\beta d\tau_n \left\langle \mathcal{T} \left[\hat{c}_1 \hat{c}^\dagger_2 \hat{\mathcal{H}}_I(\tau_1) \cdots \hat{\mathcal{H}}_I(\tau_n) \right] \right\rangle_0^{\text{conn}}, \tag{3.15}$$

where $\langle\cdot\rangle_0^{\text{conn}}$ now indicates that only connected diagrams are considered in the expansion[3]. We can transform this into a diagrammatic representation by applying Wick contractions, which allow us to collect pairs of creation and annihilation operators to form independent expectation values. In the case of the $n = 1$ term in the perturbation expansion of the Green's function,

$$\begin{aligned} G_{1;12}(\tau) = \sum_{abcd} \int_0^\beta d\tau_1\, U_{abcd} \times \\ \times \Big[- \left\langle \mathcal{T}\left[\hat{c}_1(\tau)\hat{c}_a^\dagger(\tau_1)\right]\right\rangle_0 \left\langle \mathcal{T}\left[\hat{c}_d(\tau_1)\hat{c}_c^\dagger(\tau_1)\right]\right\rangle_0 \left\langle \mathcal{T}\left[\hat{c}_b(\tau_1)\hat{c}_2^\dagger(0)\right]\right\rangle_0 \\ + \left\langle \mathcal{T}\left[\hat{c}_1(\tau)\hat{c}_a^\dagger(\tau_1)\right]\right\rangle_0 \left\langle \mathcal{T}\left[\hat{c}_b(\tau_1)\hat{c}_c^\dagger(\tau_1)\right]\right\rangle_0 \left\langle \mathcal{T}\left[\hat{c}_d(\tau_1)\hat{c}_2^\dagger(0)\right]\right\rangle_0 \Big]. \end{aligned} \tag{3.16}$$

The remaining expectation values are nothing more than minus the non-interacting Green's function as seen from Eq. (3.12). The diagrammatic representation of the first order term of Eq. (3.15), i.e. Eq. (3.16), is shown in Fig. 3.2.

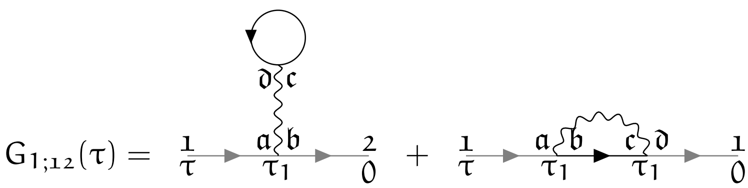


FIGURE 3.2 – First order diagrams for the single-particle Green's function. The corresponding diagrams are coined the Hartree- and the Fock-term, respectively. As the name suggests, taking only these two diagrams as the whole perturbation expansion leads to the Hartree-Fock approximation, formulated as a diagrammatic theory.

3.2 Self-energy

The one-particle Green's function perturbative expansion is already much simplified in the graphical portrayal. However, the number of diagrams increases exponentially with order n[4]. Therefore, the Feynman diagram approach would be of little practical use if the only way to progress was to calculate each diagram one at a time. However, in reality the technique is actually quite useful, since it hints on how to sum up the perturbative series. Let us therefore review how it applies to single-particle Green's functions. First, let us define another compound index which will further levitate readability. We will set $k = \{\nu, \mathbf{k}\}$ and $q = \{\omega, \mathbf{q}\}$ as a compound index including momentum and frequency in one index for fermionic (k) and bosonic (q) frequencies and momenta, respectively. Note, that if we write ν instead of k, we refer to the *local* part of the quantity with no momentum dependence instead of the full *non-local* one.
Single-particle diagrams in general can be split up into two classes: (i) diagrams that, by cutting a single internal Green's function line, transform into two lower-order diagrams, which are called

[3] Disconnected diagrams are those that consist of two or more separate, unlinked parts.
[4] For $n = 2$ there exist 10 diagrams, for $n = 7$ the number of diagrams is well over a million.

one-particle reducible and (ii) diagrams that are not one-particle reducible, which are called one-particle irreducible (1PI). Let us define as the one-particle self-energy Σ^k_{12} the sum of all irreducible diagrams without external legs. For example, the black part of the diagrams in Fig. 3.2 corresponds to the first-order diagrams of the self-energy. It is then straightforward to rewrite Eq. (3.15) in terms of the self-energy, yielding

$$G^k_{12} = G^k_{0,12} + \sum_{ab} G^k_{0,1a} \Sigma^k_{ab} G^k_{b2}, \tag{3.17}$$

which results in — when taking Eq. (3.12) as the non-interacting Green's function and inverting in frequency and momentum space — a way to express the full (interacting) single-particle Green's function in terms of the self-energy

$$G^k_{12} = \left[i\nu_n \delta_{12} - \varepsilon_{12}(\mathbf{k}) - \Sigma^k_{12} \right]^{-1}. \tag{3.18}$$

Eq. (3.17) is commonly known as the Dyson equation for the single-particle Green's function and for completeness, its diagrammatic representation can be found in Fig. 3.3. It is a geometric series, where each term in the sum subsequently adds irreducible diagrams that are pieced together by non-interacting Green's functions to generate all possible diagrams, including reducible ones.

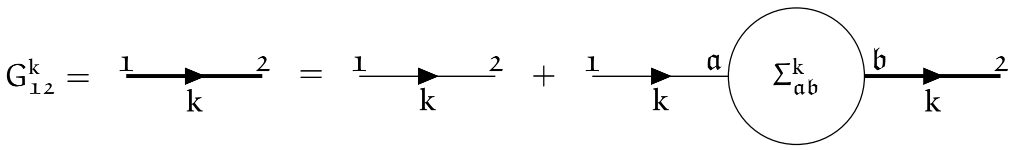


FIGURE 3.3 – Diagrammatic representation of Eq. (3.17). The full Green's function G^k_{12} is drawn as a thick black line.

The self-energy can be obtained by simply inverting Eq. (3.17),

$$\Sigma^k_{12} = (G^k_{0;12})^{-1} - (G^k_{12})^{-1}. \tag{3.19}$$

Thus, perturbation theory can be done in two distinct ways: the simpler approach is to determine the Green's function directly, for example, up to order n, the second method is to compute the self-energy up to order n and insert its expression in the Dyson equation Eq. (3.17). This gives an approximate Green's function containing each order of perturbation theory with only a subset of all possible diagrams. As a result, Eq. (3.17) represents a way to sum up perturbation theory more compactly, and is actually more relevant from a physical standpoint. Indeed, many methods leverage the usefulness of this approach, for example the dynamical mean-field theory or the dynamical vertex approximation, as we will see later.
Let us add some physical background to the self-energy in the following. Σ represents the effects of the interaction on the single electron propagators. The real and imaginary part of the self-energy have significant impact on the physical properties of the system. The real part of the self-energy contributes to an energy shift of the electronic states of the single-particle energy levels $\varepsilon(\mathbf{k})$ due to interactions. In contrast, the imaginary part of the self-energy is related to the decay or lifetime of quasiparticles and introduces a spectral broadening or damping of the electronic states, indicating how long an excitation or quasiparticle persists before interacting with other excitations or scattering events. A larger imaginary part implies a shorter lifetime and stronger interactions. The first order derivative in frequency, $\frac{\partial \Sigma^k}{\partial \nu}$, contributes to an effective renormalization of the bare electron mass. It

also leads to a redistribution of spectral weight in the one-particle spectral function, which provides information about the distribution of energy levels available for excitations. A broader spectral function, whose broadening is controlled by the imaginary part of the self-energy, typically indicates stronger interactions and a more correlated system.

3.3 Two-particle Green's functions

In analogy to the one-particle Green's function, the two-particle Green's function describes the amplitude of the propagation of two particles, two holes or a particle and a hole. A similar expression to Eq. (3.1) can be formulated for the two-particle Green's function

$$G^{qkk'}_{1234} = -\left\langle \mathcal{T}\left[\hat{c}_{\mathbf{k};1}\hat{c}^{\dagger}_{\mathbf{k}-\mathbf{q};2}\hat{c}_{\mathbf{k}'-\mathbf{q};3}\hat{c}^{\dagger}_{\mathbf{k}';4}\right]\right\rangle . \tag{3.20}$$

It describes two particles being created in the system at times τ_2 and τ_4, propagating in the system until they are removed again at τ_1 and τ_3, respectively. Assuming a stationary Hamiltonian that satisfies time-translation invariance, it is possible to eliminate one imaginary time variable by setting it zero. Then, the Green's function only depends on three time variables and through Fourier transform, also only on three frequency variables. Furthermore, since momentum is conserved, the two-particle Green's function is parameterized only by three independent momenta instead of four. In addition, similar to the case for the one-particle Green's function, the absence of spin-orbit coupling reduces the number of spin degrees of freedom to two:

$$G^{qkk'}_{\sigma\sigma';1234} = G^{qkk'}_{\sigma\sigma\sigma'\sigma';1234} \qquad \text{and} \tag{3.21a}$$

$$G^{qkk'}_{\overline{\sigma\sigma'};1234} = G^{qkk'}_{\sigma\sigma'\sigma'\sigma;1234}. \tag{3.21b}$$

In this case, the total incoming and outgoing spin must be equal and this puts a heavy constraint on the possible spin combinations,

$$\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4, \qquad (\sigma_1 = \sigma_2) \neq (\sigma_3 = \sigma_4) \qquad \text{and} \qquad (\sigma_1 = \sigma_4) \neq (\sigma_2 = \sigma_3), \tag{3.22}$$

totaling to only six unique spin combinations, each of which can be seen in Fig. 3.4. Furthermore, if

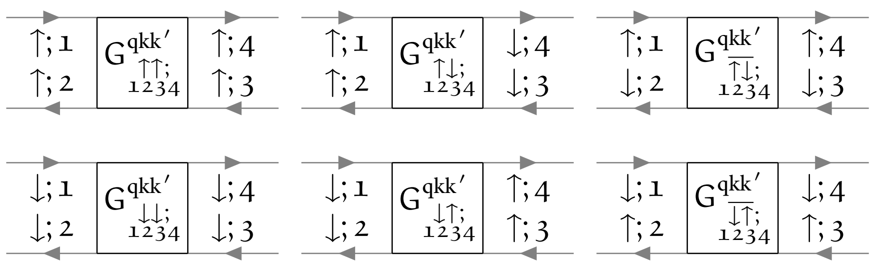


FIGURE 3.4 – In the absence of spin-orbit coupling, these are the only non-vanishing spin combinations of the two-particle Green's function.

one describes quantities in the paramagnetic phase, where the system is SU(2)-symmetric, there are only two independent spin configurations,

$$G^{qkk'}_{\sigma\sigma';1234} = G^{qkk'}_{(-\sigma)(-\sigma');1234} = G^{qkk'}_{\sigma'\sigma;1234} \qquad \text{and} \tag{3.23a}$$

$$G^{qkk'}_{\sigma\sigma;1234} = G^{qkk'}_{\sigma(-\sigma);1234} + G^{qkk'}_{\overline{\sigma(-\sigma)};1234}. \tag{3.23b}$$

For these two spin combinations, the (d)ensity, (m)agnetic, (s)inglet and (t)riplet channel[5] specified in the following are a particularly helpful choice of notation, as we shall see in the upcoming sections:

$$G^{qkk'}_{d;1234} = G^{qkk'}_{\uparrow\uparrow;1234} + G^{qkk'}_{\uparrow\downarrow;1234}, \tag{3.24a}$$
$$G^{qkk'}_{m;1234} = G^{qkk'}_{\uparrow\uparrow;1234} - G^{qkk'}_{\uparrow\downarrow;1234} = G^{qkk'}_{\overline{\uparrow\downarrow};1234}, \tag{3.24b}$$
$$G^{qkk'}_{s;1234} = G^{qkk'}_{\uparrow\downarrow;1234} - G^{qkk'}_{\overline{\uparrow\downarrow};1234} \quad \text{and} \tag{3.24c}$$
$$G^{qkk'}_{t;1234} = G^{qkk'}_{\uparrow\downarrow;1234} + G^{qkk'}_{\overline{\uparrow\downarrow};1234}. \tag{3.24d}$$

Besides the spin-symmetries discussed above, the Green's function also inherits other properties of the system, such as time-reversal symmetry, which manifests itself by

$$G^{qkk'}_{1234} = G^{q\bar{k}'\bar{k}}_{4321}, \tag{3.25}$$

where $\bar{k} = \{\nu, -\mathbf{k}\}$ is the time-reversed compound momentum variable. Furthermore, the two-particle Green's function satisfies the crossing symmetry, which is a direct manifestation of Pauli's principle. Crossing symmetry refers to the antisymmetric property of the Green's function with respect to the exchange of the incoming and outgoing fermions

$$G^{qkk'}_{\sigma\sigma';1234} = -G^{(k'-k)(k'-q)k'}_{\overline{\sigma'\sigma};3214} \tag{3.26a}$$
$$= -G^{(k-k')k(k-q)}_{\overline{\sigma\sigma'};1432} \tag{3.26b}$$
$$= G^{(-q)(k'-q)(k-q)}_{\sigma'\sigma;3412}. \tag{3.26c}$$

The symmetries Eq. (3.26a) - Eq. (3.26c) describe the frequency changes that occur by exchanging the positions of the gray fermion lines, leaving the vertex unchanged. The last line corresponds to a full swap of the incoming and outgoing particle labels. The crossing symmetry is visually shown in Fig. 3.5. Lastly, the Green's function also possesses symmetry with respect to complex conjugation,

$$\left(G^{qkk'}_{\sigma\sigma';1234}\right)^* = G^{(-q)(-k)(-k')}_{\sigma'\sigma;4321}. \tag{3.27}$$

At the beginning of the section it was stated that the (four-point) two-particle Green's function only requires three frequency arguments to be fixed since the last one is automatically given by momentum conservation. There are three equivalent choices (coined channels) of independent frequencies (with ν, ν' being a fermionic and ω a bosonic Matsubara frequency)

$$\text{ph-notation: } \{\nu_1 = \nu, \quad \nu_2 = \nu - \omega, \quad \nu_3 = \nu' - \omega, \quad \nu_4 = \nu'\}, \tag{3.28a}$$
$$\overline{\text{ph}}\text{-notation: } \{\nu_1 = \nu, \quad \nu_2 = \nu', \quad \nu_3 = \nu' - \omega, \quad \nu_4 = \nu - \omega\} \quad \text{and} \tag{3.28b}$$
$$\text{pp-notation: } \{\nu_1 = \nu, \quad \nu_2 = \omega - \nu', \quad \nu_3 = \omega - \nu, \quad \nu_4 = \nu'\}. \tag{3.28c}$$

To show the effects of this frequency shift and how they affect the labels of the diagrams, we show the two-particle Green's function in the three notations of Eq. (3.28a) - Eq. (3.28c) in Fig. 3.6. Obviously,

[5] As we will see later, in ph/$\overline{\text{ph}}$-notation, the Bethe-Salpeter equation will become diagonal for the d/m spin combination as well as for pp-notation in the s/t spin combination.

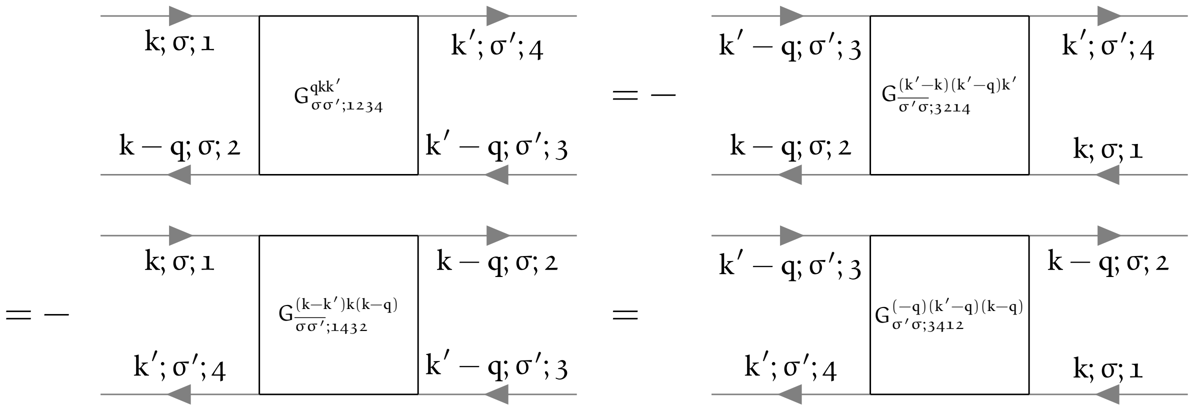


Figure 3.5 – Diagrammatic representation of the crossing symmetry relations in order of Eq. (3.26a) - Eq. (3.26c).

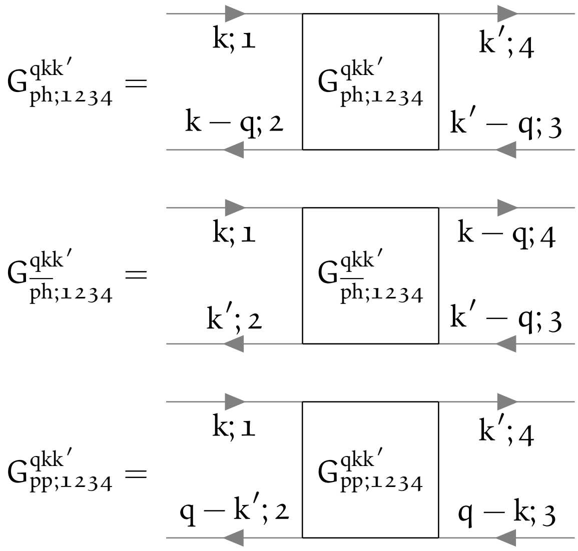


Figure 3.6 – Two-particle Green's functions in the different frequency notations of Eq. (3.28a) - Eq. (3.28c) expressed in Feynman diagrams. The frequency notation is encoded in an additional subscript, $\{\mathrm{ph}, \overline{\mathrm{ph}}, \mathrm{pp}\}$.

the choice of frequency convention should not have any impact on the physical content of the Green's function. Hence, it is possible to switch between these channels by applying channel-specific frequency shifts [43],

$$G^{qkk'}_{\overline{\mathrm{ph}};1234} = G^{(k-k')k(k-q)}_{\mathrm{ph};1234}, \tag{3.29a}$$

$$G^{qkk'}_{\mathrm{ph};1234} = G^{(k-k')k(k-q)}_{\overline{\mathrm{ph}};1234}, \tag{3.29b}$$

$$G^{qkk'}_{\mathrm{ph};1234} = G^{(k+k'-q)kk'}_{\mathrm{pp};1234} \quad \text{and} \tag{3.29c}$$

$$G^{qkk'}_{\mathrm{pp};1234} = G^{(k+k'-q)kk'}_{\mathrm{ph};1234}. \tag{3.29d}$$

Let us finally note that the two-particle Green's function represents all diagrams that are possible that involve two electrons, two holes or an electron and a hole. It can therefore be split into two parts: (i) a part, where the electrons do not interact with each other and propagate independently; and

(ii) a part, where the electrons do interact with each other through an infinite number of processes. Part (ii) is commonly called the connected two-particle Green's function and can be obtained by removing the disconnected parts (i) from the full Green's function

$$G^{qkk'}_{\sigma\sigma';1234} = G^{conn;qkk'}_{\sigma\sigma';1234} + \delta_{q0} G^{k}_{\sigma;12} G^{k'}_{\sigma';34} - \delta_{\sigma\sigma'}\delta_{kk'} G^{k}_{\sigma;14} G^{k-q}_{\sigma;32}. \tag{3.30}$$

Diagrammatically, the connected two-particle Green's function contains all diagrams, where two propagating electrons interact with each other. The first terms up to interaction order two are given in Fig. 3.7. The other terms in Eq. (3.30) are more straightforward, they merely describe the electrons

$G^{conn;qkk'}_{\sigma\sigma';1234} =$

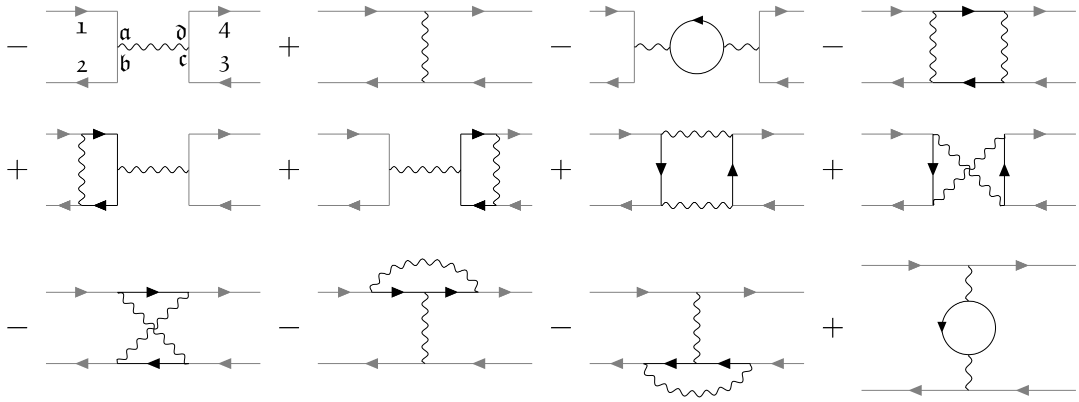


FIGURE 3.7 – Diagrammatic representation of the two-particle connected Green's function up to order two. It includes the possible interaction processes that occur between two propagating electrons. The particle labels are only written down for the first diagram, the other diagrams are labeled analogously.

propagating in the system without ever interacting with the other. The diagrammatic content of the second and third term of Eq. (3.30) can be seen in Fig. 3.8. The connected two-particle Green's

$\delta_{q0} G^{k}_{\sigma;12} G^{k'}_{\sigma';34} - \delta_{\sigma\sigma'}\delta_{kk'} G^{k}_{\sigma;14} G^{k-q}_{\sigma;32} =$

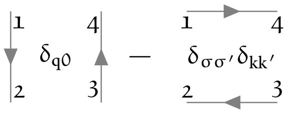


FIGURE 3.8 – Disconnected part (second and third term on the right-hand side of Eq. (3.30)) of the two-particle Green's function.

function without the external legs is commonly called the full vertex F, whereby convention an additional minus sign is introduced,

$$G^{conn;qkk'}_{1234} = -\frac{1}{\beta}\sum_{abcd} G^{k}_{1a} G^{k-q}_{b2} F^{qkk'}_{abcd} G^{k'-q}_{3c} G^{k'}_{d4}. \tag{3.31}$$

By the previous definition, the vertex function $F^{qkk'}_{1234}$ contains all diagrams that connect two incoming and two outgoing lines. All diagrams contained in the full vertex $F^{qkk'}_{1234}$ can be classified by their two-particle reducibility (2PR). A diagram is called two-particle reducible, if it can be split into two separate parts by cutting two internal Green's function lines. Furthermore, this classification is described by three channels, $\{\mathrm{ph}, \overline{\mathrm{ph}}, \mathrm{pp}\}$[6], depending on which incoming and outgoing lines are being separated by this "cutting"-process[7]. The ph-channel corresponds to those diagrams, where the sub-parts connect 12 and 34, for the $\overline{\mathrm{ph}}$-channel 14 and 23 and for the pp-channel 13 and 24 are connected. These reducible diagrams are categorized in $\Phi^{qkk'}_{\mathrm{ph};1234}$, $\Phi^{qkk'}_{\overline{\mathrm{ph}};1234}$ and $\Phi^{qkk'}_{\mathrm{pp};1234}$, respectively. All diagrams that are not two-particle reducible are grouped together in the quantity $\Lambda^{qkk'}_{1234}$. This procedure is called the parquet decomposition and reads mathematically

$$F^{qkk'}_{1234} = \Lambda^{qkk'}_{1234} + \Phi^{qkk'}_{\mathrm{ph};1234} + \Phi^{qkk'}_{\overline{\mathrm{ph}};1234} + \Phi^{qkk'}_{\mathrm{pp};1234}. \tag{3.32}$$

Another possibility is to split up the full vertex F in reducible and irreducible diagrams in a specific channel $r \in \{\mathrm{ph}, \overline{\mathrm{ph}}, \mathrm{pp}\}$,

$$F^{qkk'}_{1234} = \Gamma^{qkk'}_{r;1234} + \Phi^{qkk'}_{r;1234}, \tag{3.33}$$

where $\Gamma^{qkk'}_{r;1234}$ is the irreducible vertex in channel r and $\Phi^{qkk'}_{r;1234}$ the corresponding reducible one. Note, that in channel r irreducible diagrams might still be reducible in another channel $r' \neq r$. For example, $\Gamma^{qkk'}_{\mathrm{ph};1234}$ contains all fully irreducible diagrams, but also all diagrams which are reducible in channels $\overline{\mathrm{ph}}$ and pp,

$$\Gamma^{qkk'}_{\mathrm{ph};1234} = \Lambda^{qkk'}_{1234} + \Phi^{qkk'}_{\overline{\mathrm{ph}};1234} + \Phi^{qkk'}_{\mathrm{pp};1234}. \tag{3.34}$$

This will be important later in Sec. 3.5, where we will discuss the Bethe-Salpeter equations. Furthermore, the following relations for the ph and $\overline{\mathrm{ph}}$ channel for $\Phi^{qkk'}_{r;1234}$ hold:

$$\begin{aligned} \Phi^{qkk'}_{\overline{\mathrm{ph}};1234;\sigma\sigma'} &= -\Phi^{(k'-k)(k'-q)k'}_{\mathrm{ph};3214;\overline{\sigma'\sigma}}, &\text{(3.35a)}\\ &= -\Phi^{(k-k')k(k-q)}_{\mathrm{ph};1432;\overline{\sigma'\sigma}}. &\text{(3.35b)} \end{aligned}$$

From this one can construct the density and magnetic contributions of $\Phi^{qkk'}_{\overline{\mathrm{ph}};1234}$ in terms of $\Phi^{qkk'}_{\mathrm{ph};1234}$

$$\begin{aligned} \Phi^{qkk'}_{d;\overline{\mathrm{ph}};1234} &= -\frac{1}{2}\Phi^{(k'-k)(k'-q)k'}_{d;\mathrm{ph};3214} - \frac{3}{2}\Phi^{(k'-k)(k'-q)k'}_{m;\mathrm{ph};3214} &\text{(3.36a)}\\ \Phi^{qkk'}_{m;\overline{\mathrm{ph}};1234} &= -\frac{1}{2}\Phi^{(k'-k)(k'-q)k'}_{d;\mathrm{ph};3214} + \frac{1}{2}\Phi^{(k'-k)(k'-q)k'}_{m;\mathrm{ph};3214}. &\text{(3.36b)} \end{aligned}$$

The relations Eq. (3.35) and Eq. (3.36) also apply to the full- and irreducible vertices $F^{qkk'}_{\overline{\mathrm{ph}};1234;\sigma\sigma'}$ and $\Gamma^{qkk'}_{\overline{\mathrm{ph}};1234;\sigma\sigma'}$ similarly. Note that there is no connection between the pp channel and the other two

[6]The attentive reader remembers that there are also three (identically-named) frequency notations that were introduced previously. The channel reducibility and frequency notation are — in general — detached. From now on, if only one subscript $r \in \{\mathrm{ph}, \overline{\mathrm{ph}}, \mathrm{pp}\}$ is given per vertex quantity, it means that the channel reducibility will belong to r and the frequency notation will be ph-like. Otherwise, the channel reducibility and frequency notation will be denoted in the sub- and superscript of the variable, respectively. This will be relevant later when discussing the Bethe-Salpeter equation (BSE), see Sec. 3.5. The BSE is only diagonal in the bosonic transfer momentum and frequency q if the channel reducibility is the same as the frequency notation.

[7]A given diagram is reducible in *at most one* of these three channels, hence making it characterized fully by the two-particle reducibility and we obtain Eq. (3.32).

channels. Hence, it is not possible to transform the channel reducibility from either ph or $\overline{\text{ph}}$ to pp. The pp-channel fulfills a crossing symmetry on its own,

$$\Phi^{qkk'}_{pp;1234;\sigma\sigma'} = -\Phi^{(k-k')k(k-q)}_{pp;1432;\overline{\sigma\sigma'}}. \tag{3.37}$$

In the next section, we will briefly introduce the concept of linear response and the central physical property describing the response of a system with respect to an external perturbation[8], the susceptibility.

3.4 Susceptibility

Experimentally, it is not possible to directly extract information (just by "looking") about the interactions between electrons in a system. Instead, in spectroscopic experiments, a perturbation is applied to the system and one measures the system's response. In particular, one measures how expectation values $\langle\cdot\rangle$ of operators (take the multi-dimensional operator $\hat{O}_i(\tau)$ as an example) changes when an external field $h_j(\tau)$ is applied. In linear response theory, this response is — as the name suggests — taken to be linear in the perturbing field, requiring this field to be sufficiently small in order for this description to be a good approximation [44, 45],

$$\left\langle \hat{O}_i(\tau) \right\rangle_{\mathbf{h}} - \left\langle \hat{O}_i(\tau) \right\rangle_{\mathbf{h}=0} = \int_0^\beta d\tau' \, \chi_{ij}(\tau-\tau') h_j(\tau') + \mathcal{O}(\mathbf{h}^2), \tag{3.38}$$

where $\chi_{ij}(\tau-\tau')$ is coined physical susceptibility and does not depend on the field $h_j(\tau)$. $\chi(\tau-\tau')$ fulfills the Kramers-Kronig relations, is causal ($\chi_{ij}(\tau-\tau') = \Theta(\tau-\tau')\chi_{ij}(\tau-\tau')$) and is finite ($\forall\, \tau,\tau' :\ \exists\, C \in \mathbb{R} :\ |\chi_{ij}(\tau-\tau')| < C$). The susceptibility can be computed straightforwardly by taking the functional derivative of $\hat{O}_i(\tau)$ with respect to the field $h_j(\tau)$,

$$\chi_{ij}(\tau-\tau') = \left.\frac{\delta \left\langle \hat{O}_i(\tau') \right\rangle}{\delta h_j(\tau)}\right|_{\mathbf{h}=0} = \left\langle \mathcal{T} \hat{O}_i(\tau) \hat{O}_j(\tau') \right\rangle - \left\langle \hat{O}_i \right\rangle \left\langle \hat{O}_j \right\rangle. \tag{3.39}$$

On the two-particle level, the density's response to variations in the one-particle energy, χ_d, and the magnetization's response to variation in the electromagnetic field in the same direction, χ_m, can be shown to be non-zero [44],

$$\chi_d(\tau) = \chi_{nn}(\tau) = -\left.\frac{\delta \langle \hat{n} \rangle}{\delta \mu(\tau)}\right|_{\mu=0} \qquad \text{and} \tag{3.40a}$$

$$\chi_m(\tau) = \chi_{ii}(\tau) = \left.\frac{\delta \langle \hat{\sigma}_i \rangle}{\delta h_i(\tau)}\right|_{\mathbf{h}=0}, \tag{3.40b}$$

where $i \in \{x, y, z\}$. The physical susceptibilities are included in the two-particle Green's function, as we will see shortly. In fact, the sum of terms corresponding to the two disconnected, horizontal Green's function lines and the vertex part is denoted as the generalized susceptibility and can be

[8] For example an external electromagnetic field.

expressed as

$$
\begin{aligned}
\chi^{ph;qkk'}_{1234} &= \beta\left(G^{ph;qkk'}_{1234} - \delta_{q0} G^{k}_{14} G^{k'}_{32}\right) \\
&= -\beta \delta_{kk'} G^{k}_{14} G^{k-q}_{32} - \sum_{\mathfrak{abcd}} G^{k}_{1\mathfrak{a}} G^{k-q}_{\mathfrak{b}2} F^{ph;qkk'}_{\mathfrak{abcd}} G^{k'-q}_{3\mathfrak{c}} G^{k'}_{\mathfrak{d}4} \\
&= \chi^{qkk'}_{0;1234} - \frac{1}{\beta^2} \sum_{\mathfrak{abcd}} \chi^{qkk}_{0;12\mathfrak{ba}} F^{ph;qkk'}_{\mathfrak{abcd}} \chi^{qk'k'}_{0;\mathfrak{dc}34},
\end{aligned}
\tag{3.41}
$$

where

$$
\chi^{qkk'}_{0;1234} = -\beta \delta_{kk'} G^{k}_{14} G^{k-q}_{32} \tag{3.42}
$$

is commonly denoted as the generalized bubble susceptibility of Fig. 3.8[9]. From this, one can obtain physical susceptibilities by contracting the inner legs, i.e.

$$
\chi^{q}_{14} = \sum_{kk';23} \chi^{qkk'}_{1234}. \tag{3.44}
$$

The diagrammatic content of the physical susceptibility is shown in Fig. 3.9 below. Similarly, the

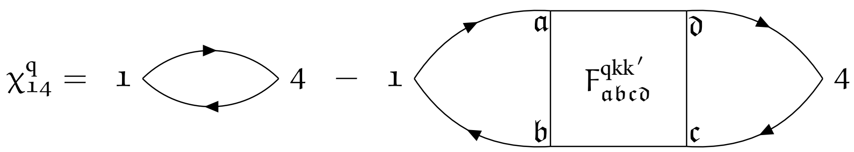


FIGURE 3.9 – The physical susceptibility is defined as the generalized susceptibility with contracted legs, see Eq. (3.44).

contracted bubble susceptibility can be written as

$$
\chi^{q}_{0;14} = \sum_{kk';23} \chi^{qkk'}_{0;1234}. \tag{3.45}
$$

Analogously to Eq. (3.24a) and Eq. (3.24b), one can write down density and magnetic spin combinations of the physical susceptibility,

$$
\chi^{q}_{d} = \chi^{q}_{\uparrow\uparrow} + \chi^{q}_{\uparrow\downarrow} \quad \text{and} \tag{3.46a}
$$

$$
\chi^{q}_{m} = \chi^{q}_{\uparrow\uparrow} - \chi^{q}_{\uparrow\downarrow} = \chi^{q}_{\overline{\uparrow\downarrow}}. \tag{3.46b}
$$

Note that expressions Eq. (3.46a) and Eq. (3.46b) are nothing more than the Fourier-transformed versions of Eq. (3.40a) and Eq. (3.40b), respectively.

[9] The generalized susceptibility in the particle-particle channel reads

$$
\chi^{pp;qkk'}_{1234;\sigma\sigma'} = \beta\left(1 - \delta_{\sigma\sigma'}\right)\left[G^{pp;qkk'}_{1234;\sigma\sigma'} - \delta_{q0} G^{-k}_{14} G^{k'}_{32}\right]. \tag{3.43}
$$

3.5 Bethe-Salpeter equation

In the section above we have specified $\Gamma^{qkk'}_{r;1234}$ as the fully irreducible vertex in channel r and $\Phi^{qkk'}_{r;1234}$ as the reducible diagrams in channel r. We can now construct the full set of $\Phi^{qkk'}_{r;1234}$ from the irreducible vertex by chaining them together with pairs of Green's functions connecting two irreducible vertices. This results in so-called ladders,

$$\begin{aligned}\Phi^{qkk'}_{r;1234} &= \sum_{k_1;abcd} \Gamma^{qkk_1}_{r;12ab} G^{k_1}_{bc} G^{k_1-q}_{da} \Gamma^{qk_1k'}_{r;cd34} \\ &\quad + \sum_{\substack{k_1k_2;abcd\\efgh}} \Gamma^{qkk_1}_{r;12ab} G^{k_1}_{bc} G^{k_1-q}_{da} \Gamma^{qk_1k_2}_{r;cdef} G^{k_2}_{fg} G^{k_2-q}_{he} \Gamma^{qk_2k'}_{r;gh34} + \dots \qquad (3.47a) \\ &= -\frac{1}{\beta} \sum_{k_1;abcd} \Gamma^{qkk_1}_{r;12ab} \chi^{qk_1k_1}_{0;badc} \Gamma^{qk_1k'}_{r;cd34} \\ &\quad + \frac{1}{\beta^2} \sum_{\substack{k_1k_2;abcd\\efgh}} \Gamma^{qkk_1}_{r;12ab} \chi^{qk_1k_1}_{0;badc} \Gamma^{qk_1k_2}_{r;cdef} \chi^{qk_2k_2}_{0;fehg} \Gamma^{qk_2k'}_{r;gh34} + \dots \qquad (3.47b)\end{aligned}$$

By combining Eq. (3.33) and Eq. (3.47), one can obtain the Bethe-Salpeter equation (BSE) in all three channels [46–48], cf. Fig. 3.10:

$$\begin{aligned}F^{qkk'}_{1234} &= \Gamma^{qkk'}_{ph;1234} - \frac{1}{\beta} \sum_{k_1k_2;abcd} \Gamma^{qkk_1}_{ph;12ba} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{ph;dc34} \qquad (3.48a) \\ &= \Gamma^{qkk'}_{\overline{ph};1234} + \frac{1}{\beta} \sum_{k_1k_2;abcd} \Gamma^{qkk_1}_{\overline{ph};1ad4} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{\overline{ph};b23c} \qquad (3.48b) \\ &= \Gamma^{qkk'}_{pp;1234} - \frac{1}{2\beta} \sum_{k_1k_2;abcd} \Gamma^{qkk_1}_{pp;1b3a} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{pp;d2c4}, \qquad (3.48c)\end{aligned}$$

where we have expressed the full vertex $F^{qkk'}_{1234}$ in terms of the irreducible vertex $\Gamma^{qkk'}_{r;1234}$ and the generalized (bubble) susceptibility as a Dyson-like equation. Note that for Eq. (3.48) we used the same channel reducibility and frequency notation for each separate line. This results in a key property of the BSE: the equation becomes diagonal in the bosonic transfer momentum and frequency q. This can be taken advantage of in numerical computations as it makes it especially easy to parallelize over q as it is independent of all other quantum numbers. If we had not chosen the channel reducibility equal to the frequency notation, then the new q' would depend on the fermionic frequencies (the precise dependency is channel-specific, but in general $q' = f(q, k, k')$) and the transformed BSE would not be diagonal in q'.
Eq. (3.48) can be rewritten in terms of susceptibilities with the help of Eq. (3.41),

$$\begin{aligned}\chi^{qkk'}_{1234} &= \chi^{qkk'}_{0;1234} - \frac{1}{\beta^2} \sum_{k_1k_2;abcd} \chi^{qkk_1}_{0;12ba} \Gamma^{qk_1k_2}_{ph;abcd} \chi^{qk_2k'}_{ph;dc34} \qquad (3.49a) \\ &= \chi^{qkk'}_{0;1234} + \frac{1}{\beta^2} \sum_{k_1k_2;abcd} \chi^{qkk_1}_{0;1ad4} \Gamma^{qk_1k_2}_{\overline{ph};abcd} \chi^{qk_2k'}_{\overline{ph};b23c} \qquad (3.49b) \\ &= \chi^{qkk'}_{0;1234} - \frac{1}{2\beta^2} \sum_{k_1k_2;abcd} \chi^{qkk_1}_{0;1b3a} \Gamma^{qk_1k_2}_{pp;abcd} \left(\chi^{qk_2k'}_{pp;d2c4} + \chi^{qk_2k'}_{0;d2c4} \right). \qquad (3.49c)\end{aligned}$$

This is the most general form of the Bethe-Salpeter equations, where energy and momentum is conserved. If the system under consideration is additionally SU(2)-symmetric, then we can formulate the BSE in terms of density and magnetic spin components,

$$F^{qkk'}_{d/m;ph;1234} = \Gamma^{qkk'}_{d/m;ph;1234} - \frac{1}{\beta} \sum_{k_1 k_2;abcd} \Gamma^{qkk_1}_{d/m;ph;12ba} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{d/m;ph;dc34} \quad \text{and} \tag{3.50a}$$

$$F^{qkk'}_{d/m;\overline{ph};1234} = \Gamma^{qkk'}_{d/m;\overline{ph};1234} - \frac{1}{\beta} \sum_{k_1 k_2;abcd} \Gamma^{qkk_1}_{d/m;\overline{ph};1ad4} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{d/m;\overline{ph};b23c}. \tag{3.50b}$$

Likewise, the BSE decouples for other spin-symmetric combinations in for the pp-channel. These are the (s)inglet and (t)riplet channel, where

$$F^{qkk'}_{s;pp;1234} = \Gamma^{qkk'}_{s;pp;1234} + \frac{1}{2\beta} \sum_{k_1 k_2;abcd} \Gamma^{qkk_1}_{s;pp;1b3a} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{s;pp;d2c4} \quad \text{and} \tag{3.51}$$

$$F^{qkk'}_{t;pp;1234} = \Gamma^{qkk'}_{t;pp;1234} - \frac{1}{2\beta} \sum_{k_1 k_2;abcd} \Gamma^{qkk_1}_{t;pp;1b3a} \chi^{qk_1k_2}_{0;abcd} F^{qk_2k'}_{t;pp;d2c4}. \tag{3.52}$$

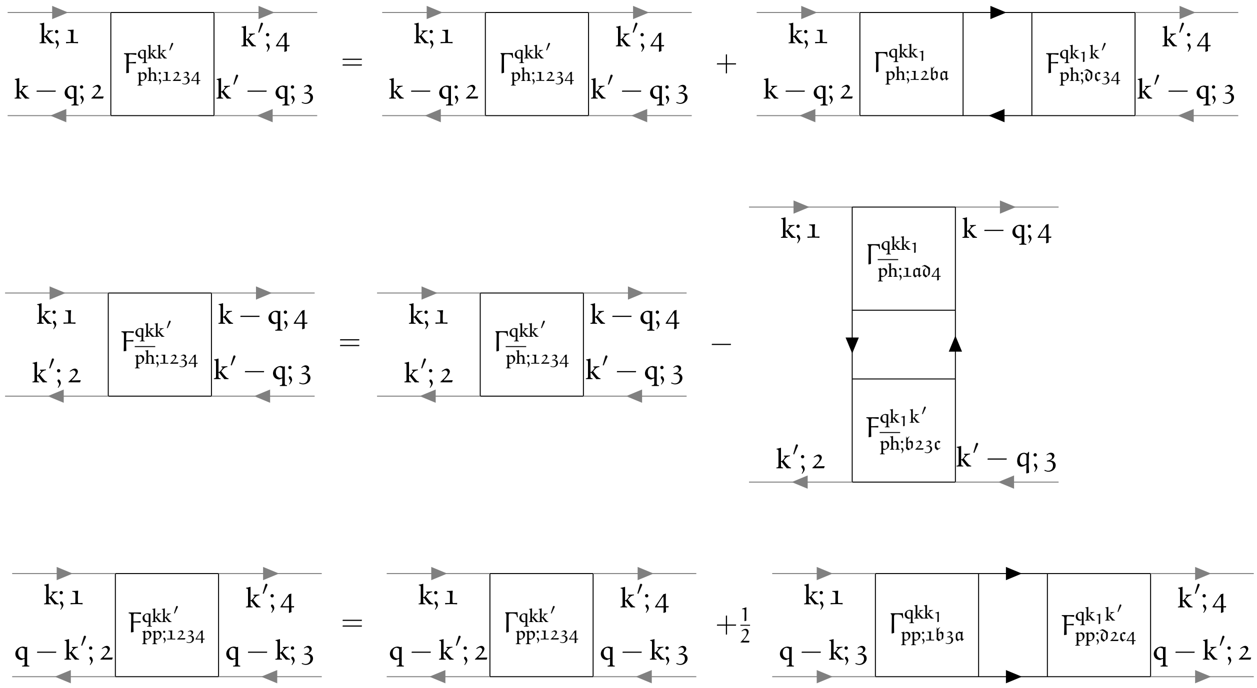


FIGURE 3.10 – Diagrammatic version of the BSE in all three channels for arbitrary spin components, see Eq. (3.48). The lower legs are exchanged in pp-notation, hence their direction is reversed.

The BSE of Eq. (3.48) needs to be inverted in order to be able to yield the full vertex $F^{qkk'}_{1234}$. This can be done in a smart way by viewing the multiplications of the vertex functions in the BSE as matrix multiplications written down in grouped indices. This means that for each transfer momentum q, we can generate a matrix from a vertex by grouping the left (right) two orbital indices and fermionic frequencies into a new grouped index $n = \{1, 2, k\}$ ($m = \{4, 3, k'\}$), such that [13]

$$\boldsymbol{F}^{q}_{d/m;nm} = F^{qkk'}_{d/m;1234}. \tag{3.53}$$

That allows for a straightforward numerical implementation of vertex products and inversions with respect to these grouped indices. Hence, this yields for the full vertex $\boldsymbol{F}^{q}_{d/m;nm}$ obtained from the BSE,

$$\boldsymbol{F}^{q}_{d/m;nm} = \boldsymbol{\Gamma}^{q}_{d/m;nl}\left(\mathbb{1}_{ac} + \frac{1}{\beta}\boldsymbol{\Gamma}^{q}_{d/m;ab}\chi^{q}_{0;bc}\right)^{-1}_{lm}. \tag{3.54}$$

3.6 Schwinger-Dyson equation

To fill the missing gap in our (almost) complete theory of correlated electrons on a lattice, we have to consider the calculation of the self-energy which enters the many-body Green's function through Eq. (3.17). This is done by the Schwinger-Dyson equation, which is derived as an equation of motion for the self-energy [48] and connects the self-energy to the full vertex via[10] [13, 49]

$$\begin{aligned}\Sigma^{k}_{12} &= \Sigma^{k}_{HF;12} + \Sigma^{conn;k}_{12} \\ &= \underbrace{2\sum_{ab;\mathbf{k}'}\mathcal{U}^{\mathbf{q}=\mathbf{0}}_{12ab}n^{\mathbf{k}'}_{ba} - \sum_{ab,\mathbf{q}}\mathcal{U}^{\mathbf{q}}_{1ba2}n^{\mathbf{k}-\mathbf{q}}_{ba}}_{\Sigma^{k}_{HF;12}} \underbrace{- \frac{1}{\beta}\sum_{\substack{abcdef\\ qk'}}\mathcal{U}^{q}_{a1bc}\chi^{qk'k'}_{0;cbed}F^{qk'k}_{de2f}G^{k-q}_{af}}_{\Sigma^{conn;k}_{12}},\end{aligned} \tag{3.55}$$

where the first two sums in Eq. (3.55) are coined Hartree- and Fock-terms, respectively, and the third term is the vertex part and only contains connected diagrams. A visual representation of the Schwinger-Dyson equation can be seen in Fig. 3.12. Here, $\mathcal{U}^{q}_{1234}$ is the full non-local Coulomb interaction, written in crossing-symmetric notation [50]

$$\mathcal{U}^{\mathbf{q}\mathbf{k}\mathbf{k}'}_{1234;\sigma\sigma'} = \underbrace{U_{1234} + V^{\mathbf{q}}_{1234}}_{\mathcal{U}^{\mathbf{q}}_{1234}} - \delta_{\sigma\sigma'}\Big(\underbrace{U_{2314} + V^{\mathbf{k}'-\mathbf{k}}_{2314}}_{\tilde{\mathcal{U}}^{\mathbf{k}'-\mathbf{k}}_{1234}}\Big), \tag{3.56}$$

containing the purely local interaction U and the purely non-local interaction V. Eq. (3.56) is diagrammatically displayed in Fig. 3.11. The ph and $\overline{\text{ph}}$ contributions of the connected part yield

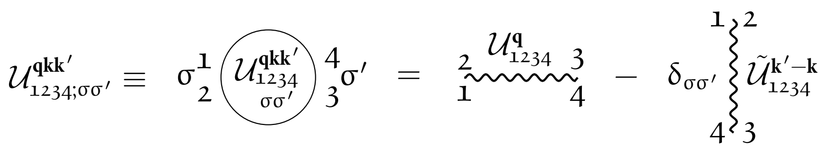


FIGURE 3.11 – Crossing-symmetric interaction, see Eq. (3.56).

separately (in ph notation)[11] [13]

$$\Sigma^{conn;k}_{ph;12} = -\frac{1}{\beta}\sum_{\substack{abcdef\\ qk'}}\mathcal{U}^{q}_{a1bc}\chi^{qk'k'}_{0;cbed}F^{qk'k}_{d;de2f}G^{k-q}_{af} \qquad \text{and} \tag{3.57a}$$

[10] In its most general form, the self-energy Σ also depends on the particle spin σ. In the SU(2)-symmetric case however we can omit the spin label, since $\Sigma_{\uparrow} = \Sigma_{\downarrow} = \Sigma$.
[11] With the help of Eq. (3.36)

$$\Sigma^{\text{conn;ph;k}}_{\overline{\text{ph}};12} = -\frac{1}{2\beta} \sum_{\substack{abcdef\\ qk'}} \tilde{\mathcal{U}}^{\mathbf{k'-k}}_{a1bc} \chi^{qk'k'}_{0;cbed} \left[F^{(k'-k)(k'-q)k'}_{d;2edf} + 3F^{(k'-k)(k'-q)k'}_{m;2edf} \right] G^{k-q}_{af}. \tag{3.57b}$$

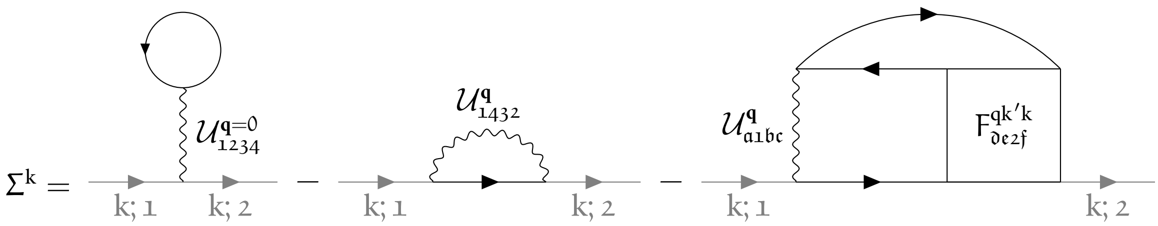


Figure 3.12 – Representation of the Schwinger-Dyson equation, see Eq. (3.55).

The self-energy Σ^k_{12} does not have to fulfill any crossing-symmetry relation, hence we do not use $\mathcal{U}^{\mathbf{qkk'}}_{1234;\sigma\sigma'}$ from Eq. (3.56) here but rather the two terms with $\mathcal{U}$ and $\tilde{\mathcal{U}}$ separately. The total self-energy is now the sum of the Hartree-Fock contribution and the ph/$\overline{\text{ph}}$ terms. We want to note that the Schwinger-Dyson equation can be rewritten in terms of three-leg vertices (also commonly called Hedin-vertices) instead of four-point Green's functions, see [51, 52] to eliminate the need of calculating the full vertex F via an inversion of the irreducible vertex Γ[12]. This equivalent formulation is commonly called the Hedin form. In this form, the SDE can be rewritten in terms of susceptibilities. We start by rewriting the full vertex as a combination of susceptibilities

$$F^{qkk'}_{1234;\sigma\sigma'} = \beta^2 \left[\left(\chi^{qk'}_{0;1234}\right)^{-1} - \sum_{k_1k_2;abcd} \left(\chi^{qk_1}_{0;12ba}\right)^{-1} \chi^{qk_1k_2}_{abcd;\sigma\sigma'} \left(\chi^{qk_2k'}_{0;dc34}\right)^{-1} \right]^{-1}. \tag{3.58}$$

Here and in the following, we will restrict ourselves to the ph channel reducibility and frequency notation if not stated otherwise. Next, we define the irreducible vertex as the first-order contribution — which is the bare Coulomb interaction in crossing-symmetric notation, see Eq. (3.56) — and the rest,

$$\Gamma^{qkk'}_{1234;\sigma\sigma'} = \mathcal{U}^{\mathbf{qkk'}}_{1234;\sigma\sigma'} + \delta\Gamma^{qkk'}_{1234;\sigma\sigma'}. \tag{3.59}$$

It is then straightforward to define an auxiliary susceptibility (denoted by the asterisk *)

$$\begin{aligned} \chi^{*;qkk'}_{1234;\sigma\sigma'} &= \left[\left(\chi^{qkk}_{0;1234}\right)^{-1} + \frac{1}{\beta^2}\delta\Gamma^{qkk'}_{1234;\sigma\sigma'} \right]^{-1} \\ &= \left[\left(\chi^{qkk}_{0;1234}\right)^{-1} + \frac{1}{\beta^2}\left(\Gamma^{qkk'}_{1234;\sigma\sigma'} - \mathcal{U}^{\mathbf{qkk'}}_{1234;\sigma\sigma'}\right) \right]^{-1}, \end{aligned} \tag{3.60}$$

with which we can rewrite the generalized susceptibility as

$$\chi^{qkk'}_{1234;\sigma\sigma'} = \chi^{*;qkk'}_{1234;\sigma\sigma'} - \sum_{\substack{k_1k_2;abcd\\ \sigma_1\sigma_2}} \chi^{*;qkk_1}_{12ba;\sigma\sigma_1} \mathcal{U}^{\mathbf{qk_1k_2}}_{abcd;\sigma_1\sigma_2} \chi^{qk_2k'}_{dc34;\sigma_2\sigma'}. \tag{3.61}$$

[12] In principle, one could retrieve the full vertex as $\boldsymbol{F}^q_r = \left[(\boldsymbol{\Gamma}^q_r)^{-1} + \chi^q_0\right]^{-1}$. However, as shown in Ref. [53], the (local) irreducible vertex Γ contains an infinite set of divergencies, hence inverting it is numerically unstable.

Previously, the BSE was formulated as a ladder with $\Gamma^{qkk'}_{1234}$ and $\chi^{qkk}_{0;1234}$ as building blocks. The rewritten BSE can now be understood as a ladder with $\chi^{*;qkk'}_{1234;\sigma\sigma'}$ and $\mathcal{U}^{qkk'}_{1234;\sigma\sigma'}$ as building blocks. Here, the auxiliary susceptibility $\chi^{*;qkk'}_{1234;\sigma\sigma'}$ can be seen as the irreducible part with respect to the bare Coulomb interaction, contrary to how $\Gamma^{qkk'}_{1234}$ is the set of irreducible diagrams with respect to cutting two Green function lines (or a bare susceptibility $\chi^{qkk}_{0;1234}$). Hence, this is commonly called interaction-reducibility [54]. The BSE displayed in terms of susceptibilities can be seen in Fig. 3.13. If one

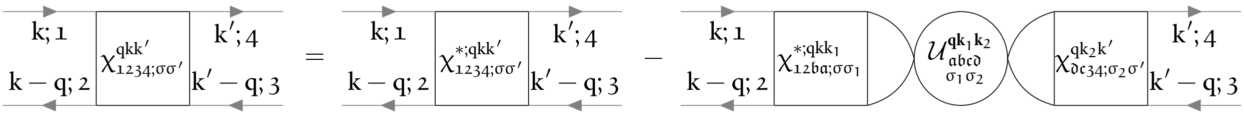


FIGURE 3.13 – Interaction-reducible form of the Schwinger-Dyson equation (3.61) with susceptibilities.

neglects the nonlocal $\overline{\text{ph}}$ contribution from the bare Coulomb interaction, one can rewrite the following expression in terms of sums of susceptibilities, where after some algebra,

$$\sum_{k';ab} F^{qkk'}_{12ab;\sigma\sigma'}\chi^{qk'k'}_{0;ba34} = \beta \sum_{abcd;\sigma_1} \gamma^{qk}_{12ab;\sigma\sigma'}\left(\mathbb{1}^{q}_{ba34} - \mathcal{U}^{q}_{badc;\sigma\sigma_1}\chi^{q}_{cd34;\sigma_1\sigma}\right), \tag{3.62}$$

where the Fermi-Bose three-leg vertex $\gamma^{qk}_{1234;\sigma\sigma'}$ is defined as

$$\gamma^{qk}_{1234;\sigma\sigma'} = \beta \sum_{k_1;ab}\left(\chi^{qkk}_{0;12ab}\right)^{-1}\chi^{*;qkk_1}_{ba34;\sigma\sigma'}. \tag{3.63}$$

This allows us to rewrite Eq. (3.55) as

$$\Sigma^{k}_{12} = \Sigma^{k}_{\mathrm{HF};12} - \sum_{\substack{q;abcd\\efgh}} \mathcal{U}^{q}_{a1bc}\left[\gamma^{qk}_{aced}\left(\mathbb{1}_{deh2} - \mathcal{U}^{q}_{defg}\chi^{q}_{gfh2}\right)\right] G^{k-q}_{bh}. \tag{3.64}$$

A visual representation of the rewritten SDE can be found in Fig. 3.14. This expression has some

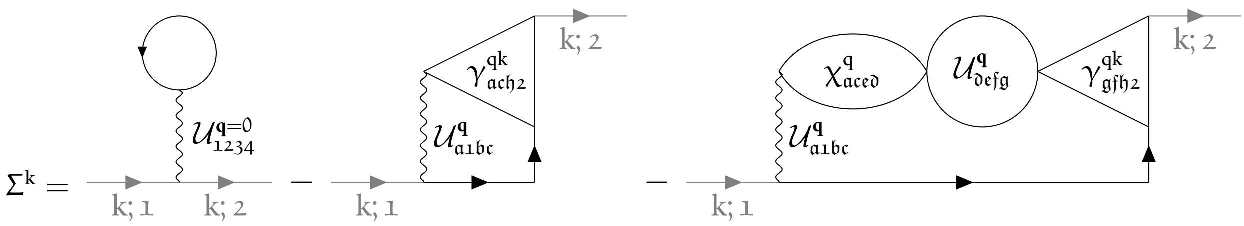


FIGURE 3.14 – Interaction-reducible form of the Schwinger-Dyson equation (3.64) with three-leg vertices.

numerical advantages, since the frequency dimension of the Fermi-Bose vertices needed for the calculation of the self-energy is one smaller compared to the approach with the full vertices, hence reducing the required memory storage.

For numerical calculations with the equations in the above sections one is always restricted to a certain frequency range where the input quantities are known, which is usually limited by time and the impurity solver's capabilities. However, the full and irreducible vertex functions outside this "core" frequency box are often not negligible[13]. For more accurate descriptions of materials, one can employ explicit asymptotics that should mimic the outer frequency structure of the vertex functions. These asymptotics can be done in many ways, from simply concatenating constant values to the region outside the calculated one[14] to more complicated formulations using so-called kernel functions. In this thesis we will introduce two treatments of the asymptotic behavior of vertex functions: one is the kernel-function approach — starting from the pioneering work of Ref. [55] that was later developed into a full local framework [43, 56–60] — which mimics the coarse frequency structure of the vertex function outside the core frequency region; and the other is an "RPA-esque" approach [61], where a constant value $\mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'}$ outside the core frequency region of the irreducible vertex is assumed and diagrams are generated by a ladder, similar to the BSE. We will outline the two methods in detail in the next section, explain their advantages and drawbacks and reason briefly when to choose either of them. Since the former method using kernel-functions has not been generalized to non-local quantities before at the writing of this thesis, we will derive equations that include non-local contributions to better incorporate these asymptotics in the computational frameworks that will be introduced in Ch. 5.

3.7 Vertex asymptotics

A full treatment of frequency and momentum dependence of multi-orbital vertex functions severely restricts the speed and memory consumption of numerical calculations. A step towards reducing the computational cost of numerical implementations of equations involving vertex functions, such as the Bethe-Salpeter equations, see Sec. 3.5, or the Schwinger-Dyson equation, see Sec. 3.6, thus requires an efficient treatment of the vertex for high Matsubara frequencies. As noted above, we will describe two ways of treating of vertex asymptotics in the following.

3.7.1 Kernel-function asymptotics

Shortly, we will see, that the full vertex in the high-frequency regime, which generally depends on three independent frequency parameters, can be approximated well as an object with only two frequency dimensions, thus drastically reducing the storage capacity required, while at the same time also reducing the computational cost after assembling these objects. Let us start with a sketch of the basic idea of how to take advantage of certain frequency structures when handling explicit vertex asymptotics. In Fig. 3.15, the frequency dependence of the local vertex functions $F^{\omega\nu\nu'}_{pp;\uparrow\downarrow}$, $\Lambda^{\omega\nu\nu'}_{pp;\uparrow\downarrow}$ and $\Phi^{\omega\nu\nu'}_{r;\uparrow\downarrow}$ calculated for a Single-Impurity Anderson Model (SIAM) are shown for vanishing transfer frequency ω [56]. From this, one can see three very prominent and distinct features of the full vertex F: (i) a constant background that differs from the (local and constant) Hubbard interaction U; (ii) two diagonal structures, the main ($\nu = \nu'$) and secondary ($\nu = -\nu'$) diagonal; and (iii) a "cross"-like structure along $\nu = \pm\frac{\pi}{\beta}$ and $\nu' = \pm\frac{\pi}{\beta}$. While only shown for the antiparallel spin combination $\uparrow\downarrow$ and vanishing transfer frequency $\omega = 0$, the features are analogous for the

[13] Especially not if one is interested in low-temperature behavior. Since the frequency spacing scales with $\sim \frac{1}{\beta} \sim T$, the lower the temperature, the smaller the effective frequency box at a fixed number of frequencies. Even though the full and irreducible vertex functions decay like $\frac{1}{\nu}$ as $\nu \to \infty$, the contributions outside the core frequency region are of importance.

[14] Like the bare Coulomb interaction $\mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'}$

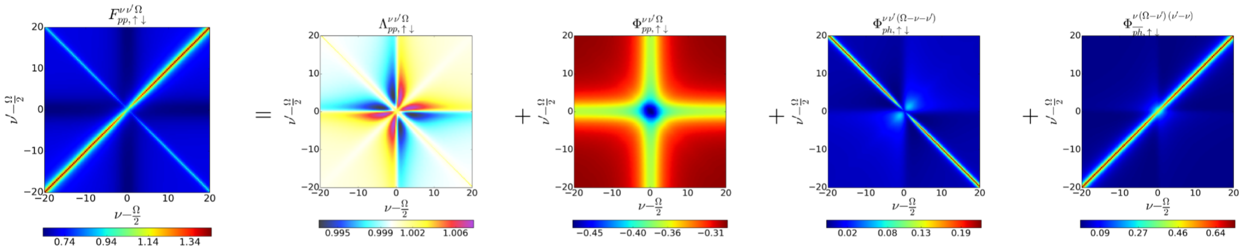


FIGURE 3.15 – Numerical SIAM results for all local vertices contained in the parquet equation Eq. (3.32) for vanishing transfer frequency $\Omega = 0$, $U = t$ and $\beta = 20$, taken from Ref. [56] where also further details can be found. In our convention, Ω should be replaced by ω.

parallel spin combination $\uparrow\uparrow$ and for finite $\omega \neq 0$. These three attributes do *not* decay for high Matsubara frequencies ν and ν' and therefore indicate complex asymptotic behavior. The individual asymptotic contributions of the fully irreducible vertex Λ and the reducible vertex in channel r to the full vertex F are written in detail in Ref. [56]. Summarized, Λ adds a constant background of size U, Φ_{ph} adds a secondary diagonal structure, $\Phi_{\overline{ph}}$ adds a diagonal structure and Φ_{pp} adds a constant background and a "cross"-like structure. Note that the description in Ref. [56] only refers to the SIAM model and considers solely the one-band case with local quantities and a constant local interaction U. An extension to a multi-orbital formalism and momentum-dependent vertices and interaction U follows naturally, see Ref. [60] for a purely local description, and will be discussed hereafter.
The simplest quantity to describe in terms of their asymptotics is the fully irreducible vertex $\Lambda^{qkk'}_{1234}$, for which

$$\Lambda^{\text{asympt};qkk'}_{1234;\sigma\sigma'} \approx \mathcal{U}^{qkk'}_{1234;\sigma\sigma'}. \tag{3.65}$$

Constructing d, m, s and t combinations of the full non-local interaction in Eq. (3.56) is useful for future calculations,

$$\begin{aligned}
\mathcal{U}^{qkk'}_{d;1234} &= 2\mathcal{U}^{q}_{1234} - \tilde{\mathcal{U}}^{k'-k}_{1234}, && (3.66a)\\
\mathcal{U}^{qkk'}_{m;1234} &= -\tilde{\mathcal{U}}^{k'-k}_{1234}, && (3.66b)\\
\mathcal{U}^{qkk'}_{s;1234} &= \mathcal{U}^{k+k'-q}_{1234} + \tilde{\mathcal{U}}^{k'-k}_{1234} \qquad \text{and} && (3.66c)\\
\mathcal{U}^{qkk'}_{t;1234} &= \mathcal{U}^{k+k'-q}_{1234} - \tilde{\mathcal{U}}^{k'-k}_{1234}. && (3.66d)
\end{aligned}$$

Next, we begin examining the diagrams contributing to Φ_r. A smart choice introduced in Ref. [56] is to classify these diagrams in each channel further into three categories:

- Class 1: These are diagrams which connect incoming and outgoing particles only by a single interaction vertex. These diagrams are bubble diagrams and hence only depend on a single bosonic frequency and momentum. The sum of all these diagrams will be encoded in the variable $\mathcal{K}^{(1);q}_{r;1234;\sigma\sigma'}$, see the first line in Fig. 3.16.

- Class 2: Diagrams corresponding to class 2 are identified by them having either the incoming *or* outgoing particles connected to the same interaction vertex. These diagrams depend on the bosonic transfer frequency and momentum and one fermionic frequency and momentum.

The sum of all these diagrams will be encoded in the variables $\mathcal{K}^{(2);qk}_{r;1234;\sigma\sigma'}$ and $\overline{\mathcal{K}}^{(2);qk}_{r;1234;\sigma\sigma'}$, depending on whether the outgoing particles ($\mathcal{K}$) or the incoming particles ($\overline{\mathcal{K}}$) are connected to the interaction vertex, see the second line in Fig. 3.16. $\mathcal{K}$ and $\overline{\mathcal{K}}$ are related to each other via time-reversal symmetry, which in the case of a system with additional SU(2)-symmetry corresponds to $\mathcal{K}^{(2);qk}_{r;1234;\sigma\sigma'} = \overline{\mathcal{K}}^{(2);qk}_{r;1234;\sigma\sigma'}$ [48].

- Class 3: These are diagrams where each external Green's function is connected to a different interaction vertex. These diagrams are parameterized by three independent frequency and momentum arguments and are encoded in the "rest" function $\mathcal{R}^{qkk'}_{r;1234;\sigma\sigma'}$.

In the following, we will refer to $\mathcal{K}^{(1)}$, $\mathcal{K}^{(2)}$ and $\overline{\mathcal{K}}^{(2)}$ as kernel functions. One now can — with the help of these classifications — write the reducible Φ_r-vertices as a decomposition of the different kernel functions and the "rest"-function

$$\Phi^{qkk'}_{r;1234;\sigma\sigma'} = \mathcal{K}^{(1);q}_{r;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk}_{r;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk'}_{r;1234;\sigma\sigma'} + \mathcal{R}^{qkk'}_{r;1234;\sigma\sigma'}. \tag{3.67}$$

This composition is *per se* exact. However, to take advantage of this formalism for numerical calculations, one discards the "rest"-function for the asymptotic, i.e. high-frequency part of the vertex. Their additional frequency dependence entering the inner propagators results in a quicker drop-off in all frequency directions. Thus, the approximation

$$\Phi^{\mathrm{asympt};qkk'}_{r;1234;\sigma\sigma'} \approx \mathcal{K}^{(1);q}_{r;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk}_{r;1234;\sigma\sigma'} + \mathcal{K}^{(2);qk'}_{r;1234;\sigma\sigma'} \tag{3.68}$$

serves as a valid expression for the high-frequency regime of Φ_r and becomes exact for $\nu, \nu' \to \infty$ [56]. As we can see above, the quantities on the right-hand side of Eq. (3.68) only depend on one (q) or two (q, k) Matsubara frequencies and momenta instead of three, reducing the numerical complexity and memory usage for Φ_r significantly. Using these explicit asymptotics results in a tremendous improvement of the numerical convergence over a simple truncation in Φ_r for high frequencies [56]. The explicit expressions for the kernel functions needed to calculate the reducible diagrams can be obtained as [26, 43, 62]

$$\mathcal{K}^{(1);q}_{ph;1234;\sigma\sigma'} = \sum_{\substack{abcd;k_1k_2\\ \sigma_1\sigma_2}} \mathcal{U}^{\mathbf{q}\mathbf{k}_1}_{12ab;\sigma\sigma_1} \chi^{qk_1k_2}_{badc;\sigma_1\sigma_2} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{cd34;\sigma_2\sigma'}, \tag{3.69}$$

$$\mathcal{K}^{(1);q}_{pp;1234;\sigma\sigma'} = \frac{1}{4} \sum_{\substack{abcd;k_1k_2\\ \sigma_1\sigma_2\sigma_3\sigma_4}} \mathcal{U}^{\mathbf{q}\mathbf{k}_1}_{\substack{1b3a\\ \sigma\sigma_2\sigma'\sigma_1}} \chi^{qk_1k_2}_{\substack{acbd\\ \sigma_1\sigma_3\sigma_2\sigma_4}} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{\substack{d2c4\\ \sigma_4\sigma\sigma_3\sigma'}}, \tag{3.70}$$

$$\begin{aligned}\mathcal{K}^{(2);qk}_{ph;1234;\sigma\sigma'} &= \sum_{\substack{ab;k_1\\ \sigma_1}} F^{qkk_1}_{12ab;\sigma\sigma_1} \chi^{qk_1k_1}_{0;baab} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{ba34;\sigma_1\sigma'} - \mathcal{K}^{(1);q}_{ph;1234;\sigma\sigma'}\\ &= -\sum_{\substack{abcd;k_1\\ \sigma_1}} \left(\chi^{qk}_{0;12dc}\right)^{-1} \chi^{qkk_1}_{cdab;\sigma\sigma_1} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{ba34;\sigma_1\sigma'} - \mathcal{K}^{(1);q}_{ph;1234;\sigma\sigma'}\end{aligned} \tag{3.71}$$

$$\begin{aligned}\mathcal{K}^{(2);qk}_{pp;1234;\sigma\sigma'} &= \frac{1}{2} \sum_{\substack{ab;k_1\\ \sigma_1\sigma_2\sigma_3\sigma_4}} F^{qkk_1}_{\substack{pp;1a3b\\ \sigma\sigma_1\sigma'\sigma_2}} \chi^{qk_1k_1}_{0;baab} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{\substack{b3a4\\ \sigma_2\sigma\sigma_1\sigma'}} - \mathcal{K}^{(1);q}_{pp;1234;\sigma\sigma'}\\ &= -\frac{1}{2} \sum_{\substack{abcd;k_1\\ \sigma_1\sigma_2\sigma_3\sigma_4}} \left(\chi^{pp;qk}_{0;1d2c}\right)^{-1} \chi^{qkk_1}_{\substack{pp;cadb\\ \sigma\sigma_1\sigma'\sigma_2}} \mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{\substack{b3a4\\ \sigma_2\sigma\sigma_1\sigma'}} - \mathcal{K}^{(1);q}_{pp;1234;\sigma\sigma'}\end{aligned} \tag{3.72}$$

where the subtraction of $\mathcal{K}^{(1)}$ in the kernel-2 function is to account for double-counting of diagrams and $\mathcal{U}^{\mathbf{qk}'}_{1234;\sigma\sigma'}$ is obtained from Eq. (3.56) by setting $\mathbf{k} = \mathbf{0}$. Both kernel functions are displayed diagrammatically for the ph-channel in Fig. 3.16. The density and magnetic contributions to the

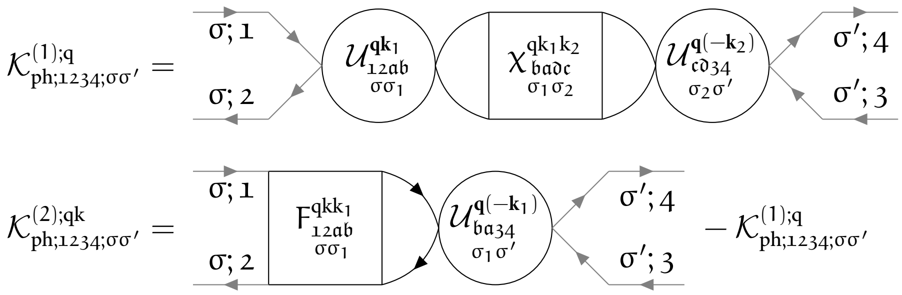


FIGURE 3.16 – Diagrammatic version of the kernel-1 and kernel-2 functions of Eq. (3.69) and Eq. (3.71).

kernel functions read[15]

$$\mathcal{K}^{(1);q}_{d;1234} = \sum_{\substack{abc\mathfrak{d}\\k_1k_2}} \mathcal{U}^{\mathbf{qk}_1}_{d;12ab}\chi^{qk_1k_2}_{d;ba\mathfrak{d}c}\mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{d;c\mathfrak{d}34}, \tag{3.73a}$$

$$\mathcal{K}^{(1);q}_{m;1234} = \sum_{\substack{abc\mathfrak{d}\\k_1k_2}} \mathcal{U}^{\mathbf{qk}_1}_{m;12ab}\chi^{qk_1k_2}_{m;ba\mathfrak{d}c}\mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{m;c\mathfrak{d}34}, \tag{3.73b}$$

$$\begin{aligned}\mathcal{K}^{(2);qk}_{d;1234} &= \sum_{ab;k_1} F^{qkk_1}_{d;12ab}\,\chi^{qk_1k_1}_{0;baab}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{d;ba34} - \mathcal{K}^{(1);q}_{d;1234}\\ &= -\sum_{abc\mathfrak{d};k_1}\left(\chi^{qk}_{0;12\mathfrak{d}c}\right)^{-1}\chi^{qkk_1}_{d;c\mathfrak{d}ab}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{d;ba34} - \mathcal{K}^{(1);q}_{d;1234} \qquad \text{and}\end{aligned} \tag{3.73c}$$

$$\begin{aligned}\mathcal{K}^{(2);qk}_{m;1234} &= \sum_{ab;k_1} F^{qkk_1}_{m;12ab}\,\chi^{qk_1k_1}_{0;baab}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{m;ba34} - \mathcal{K}^{(1);q}_{m;1234}\\ &= -\sum_{abc\mathfrak{d};k_1}\left(\chi^{qk}_{0;12\mathfrak{d}c}\right)^{-1}\chi^{qkk_1}_{m;c\mathfrak{d}ab}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{m;ba34} - \mathcal{K}^{(1);q}_{m;1234}.\end{aligned} \tag{3.73d}$$

The s/t combinations can be constructed analogously for the pp-channel (written in pp frequency notation):

$$\mathcal{K}^{(1);q}_{s;1234} = \frac{1}{4}\sum_{\substack{abc\mathfrak{d}\\k_1k_2}} \mathcal{U}^{\mathbf{qk}_1}_{s;1b3a}\chi^{qk_1k_2}_{s;acb\mathfrak{d}}\mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{s;\mathfrak{d}2c4} \overset{(3.43)}{=} \frac{1}{2}\sum_{\substack{abc\mathfrak{d}\\k_1k_2}} \mathcal{U}^{\mathbf{qk}_1}_{s;1b3a}\chi^{pp;qk_1k_2}_{acb\mathfrak{d};\uparrow\downarrow}\mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{s;\mathfrak{d}2c4}, \tag{3.74a}$$

$$\mathcal{K}^{(1);q}_{t;1234} = \frac{1}{4}\sum_{\substack{abc\mathfrak{d}\\k_1k_2}} \mathcal{U}^{\mathbf{qk}_1}_{t;1b3a}\chi^{qk_1k_2}_{t;acb\mathfrak{d}}\mathcal{U}^{\mathbf{q}(-\mathbf{k}_2)}_{t;\mathfrak{d}2c4} \overset{(3.43)}{=} 0, \tag{3.74b}$$

[15] Remember, that for density and magnetic contributions, frequencies should be in ph notation whereas for (s)inglet and (t)riplet contributions, frequencies should be in pp notation.

$$\begin{aligned}
\mathcal{K}^{(2);\mathrm{qk}}_{\mathrm{s};1234} &= \frac{1}{2}\sum_{\mathrm{ab};\mathrm{k}_1} \mathrm{F}^{\mathrm{qkk}_1}_{\mathrm{s};1\mathrm{a}3\mathrm{b}}\,\chi^{\mathrm{qk}_1\mathrm{k}_1}_{0;\mathrm{baab}}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{\mathrm{s};\mathrm{b}3\mathrm{a}4} - \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{s};1234} \\
&= -\frac{1}{2}\sum_{\mathrm{abcd};\mathrm{k}_1} \left(\chi^{\mathrm{pp};\mathrm{qk}}_{0;1\mathrm{d}2\mathrm{c}}\right)^{-1} \chi^{\mathrm{qkk}_1}_{\mathrm{s};\mathrm{cadb}}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{\mathrm{s};\mathrm{b}3\mathrm{a}4} - \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{s};1234} \\
&\overset{(3.43)}{=} -\sum_{\mathrm{abcd};\mathrm{k}_1} \left(\chi^{\mathrm{pp};\mathrm{qk}}_{0;1\mathrm{d}2\mathrm{c}}\right)^{-1} \chi^{\mathrm{pp};\mathrm{qkk}_1}_{\mathrm{cadb};\uparrow\downarrow}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{\mathrm{s};\mathrm{b}3\mathrm{a}4} - \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{s};1234} \qquad \text{and}
\end{aligned} \tag{3.74c}$$

$$\begin{aligned}
\mathcal{K}^{(2);\mathrm{qk}}_{\mathrm{t};1234} &= \frac{1}{2}\sum_{\mathrm{ab};\mathrm{k}_1} \mathrm{F}^{\mathrm{qkk}_1}_{\mathrm{t};1\mathrm{a}3\mathrm{b}}\,\chi^{\mathrm{qk}_1\mathrm{k}_1}_{0;\mathrm{baab}}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{\mathrm{t};\mathrm{b}3\mathrm{a}4} - \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{t};1234} \\
&= -\frac{1}{2}\sum_{\mathrm{abcd};\mathrm{k}_1} \left(\chi^{\mathrm{pp};\mathrm{qk}}_{0;1\mathrm{d}2\mathrm{c}}\right)^{-1} \chi^{\mathrm{qkk}_1}_{\mathrm{t};\mathrm{cadb}}\,\mathcal{U}^{\mathbf{q}(-\mathbf{k}_1)}_{\mathrm{t};\mathrm{b}3\mathrm{a}4} - \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{t};1234} \overset{(3.43)}{=} 0.
\end{aligned} \tag{3.74d}$$

Finally, the asymptotic part of the full vertex $\left(\mathrm{F}^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{r};1234;\sigma\sigma'}\right)$ can now be assembled using the asymptotic contributions of $\Lambda^{\mathrm{asympt};\mathrm{qkk}'}_{1234;\sigma\sigma'}$ and $\Phi^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{r};1234;\sigma\sigma'}$ through the parquet decomposition of Eq. (3.32). This results in

$$\mathrm{F}^{\mathrm{asympt};\mathrm{qkk}'}_{1234;\sigma\sigma'} - \mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'} = \Phi^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{ph};1234;\sigma\sigma'} + \Phi^{\mathrm{asympt};\mathrm{qkk}'}_{\overline{\mathrm{ph}};1234;\sigma\sigma'} + \Phi^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{pp};1234;\sigma\sigma'}, \tag{3.75}$$

or explicitly with kernel-1 and kernel-2 functions,

$$\begin{aligned}
\mathrm{F}^{\mathrm{asympt};\mathrm{qkk}'}_{1234;\sigma\sigma'} - \mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'} = \;& \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{ph};1234;\sigma\sigma'} + \mathcal{K}^{(2);\mathrm{qk}}_{\mathrm{ph};1234;\sigma\sigma'} + \mathcal{K}^{(2);\mathrm{qk}'}_{\mathrm{ph};1234;\sigma\sigma'} \\
& + \mathcal{K}^{(1);\mathrm{q}}_{\overline{\mathrm{ph}};1234;\sigma\sigma'} + \mathcal{K}^{(2);\mathrm{qk}}_{\overline{\mathrm{ph}};1234;\sigma\sigma'} + \mathcal{K}^{(2);\mathrm{qk}'}_{\overline{\mathrm{ph}};1234;\sigma\sigma'} \\
& + \mathcal{K}^{(1);\mathrm{q}}_{\mathrm{pp};1234;\sigma\sigma'} + \mathcal{K}^{(2);\mathrm{qk}}_{\mathrm{pp};1234;\sigma\sigma'} + \mathcal{K}^{(2);\mathrm{qk}'}_{\mathrm{pp};1234;\sigma\sigma'}.
\end{aligned} \tag{3.76}$$

The asymptotic behavior of the irreducible vertex in channel r follows from Eq. (3.33),

$$\Gamma^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{r};1234;\sigma\sigma'} = \mathrm{F}^{\mathrm{asympt};\mathrm{qkk}'}_{1234;\sigma\sigma'} - \Phi^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{r};1234;\sigma\sigma'} \tag{3.77}$$

and reads for the ph-channel explicitly

$$\begin{aligned}
\Gamma^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{ph};1234;\sigma\sigma'} &= \mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'} + \Phi^{\mathrm{asympt};\mathrm{qkk}'}_{\overline{\mathrm{ph}};1234;\sigma\sigma'} + \Phi^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{pp};1234;\sigma\sigma'} \\
&= \mathcal{U}^{\mathbf{qkk}'}_{1234;\sigma\sigma'} - \Phi^{\mathrm{asympt};(\mathrm{k}'-\mathrm{k})(\mathrm{k}'-\mathrm{q})\mathrm{k}'}_{\mathrm{ph};3214;\overline{\sigma'\sigma}} + \Phi^{\mathrm{asympt};(\mathrm{k}+\mathrm{k}'-\mathrm{q})\mathrm{kk}'}_{\mathrm{pp};1234;\sigma\sigma'},
\end{aligned} \tag{3.78}$$

where we used Eq. (3.35) to transform the $\overline{\mathrm{ph}}$-channel reducible vertex to a ph-channel reducible one. From this follows the irreducible vertex in the density and magnetic channel (in ph frequency notation) [57]

$$\begin{aligned}
\Gamma^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{d};1234} = \mathcal{U}^{\mathbf{qkk}'}_{\mathrm{d};1234} &- \frac{1}{2}\Phi^{\mathrm{asympt};(\mathrm{k}'-\mathrm{k})(\mathrm{k}'-\mathrm{q})\mathrm{k}'}_{\mathrm{d};3214} - \frac{3}{2}\Phi^{\mathrm{asympt};(\mathrm{k}'-\mathrm{k})(\mathrm{k}'-\mathrm{q})\mathrm{k}'}_{\mathrm{m};3214} \\
&+ \frac{1}{2}\Phi^{\mathrm{asympt};(\mathrm{k}+\mathrm{k}'-\mathrm{q})\mathrm{kk}'}_{\mathrm{s};1234} + \frac{3}{2}\Phi^{\mathrm{asympt};(\mathrm{k}+\mathrm{k}'-\mathrm{q})\mathrm{kk}'}_{\mathrm{t};1234},
\end{aligned} \tag{3.79a}$$

$$\begin{aligned}
\Gamma^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{m};1234} = \mathcal{U}^{\mathbf{qkk}'}_{\mathrm{m};1234} &- \frac{1}{2}\Phi^{\mathrm{asympt};(\mathrm{k}'-\mathrm{k})(\mathrm{k}'-\mathrm{q})\mathrm{k}'}_{\mathrm{d};3214} + \frac{1}{2}\Phi^{\mathrm{asympt};(\mathrm{k}'-\mathrm{k})(\mathrm{k}'-\mathrm{q})\mathrm{k}'}_{\mathrm{m};3214} \\
&- \frac{1}{2}\Phi^{\mathrm{asympt};(\mathrm{k}+\mathrm{k}'-\mathrm{q})\mathrm{kk}'}_{\mathrm{s};1234} + \frac{1}{2}\Phi^{\mathrm{asympt};(\mathrm{k}+\mathrm{k}'-\mathrm{q})\mathrm{kk}'}_{\mathrm{t};1234},
\end{aligned} \tag{3.79b}$$

where the $\Phi^{\mathrm{asympt};\mathrm{qkk}'}_{\mathrm{r};1234}$ are given by the kernel functions of Eq. (3.73) and Eq. (3.74). The irreducible vertices in channels $\overline{\mathrm{ph}}$ and pp in m/d and s/t combinations can be constructed analogously.

3.7.2 U-range asymptotics

Besides the kernel-function formalism introduced above, there also exist many other asymptotic schemes for vertex functions. The second one we would like to cover in this thesis is the "U-range" approach, where the irreducible vertex $\Gamma^{qkk'}_{1234;\sigma\sigma'}$ is padded with the bare non-local Coulomb interaction $\mathcal{U}^{\mathbf{qkk'}}_{1234;\sigma\sigma'}$, see Eq. (3.56), in the high-frequency regime. This in turn means that there is no frequency structure for the asymptotic region taken into account. We denote some equations that are necessary to employ these types of asymptotics and follow Ref. [61] with the addition of multi-orbital notation. First, we will split up the frequencies into a "core" and an "asymptotic" region and denote quantities that exist in those frequency regimes with an additional sub- or superscript $\in$ {core, asympt}. Generally speaking, we require $n_{\text{asympt}} \geq n_{\text{core}}$, where n is the number of fermionic Matsubara frequencies in the respective regime. Furthermore, if a quantity is written as $(A^{\text{asympt}})_{\text{core}}$ it means that the asymptotic value of A is cut to the core frequency region. Additionally, the size change and matrix inverse do *not* commute, i.e. $A^{-1}_{\text{core}} \equiv (A_{\text{core}})^{-1} \neq (A^{-1})_{\text{core}}$ for vertices that include asymptotics. Inversion and reduction to core region only commutes if the matrix A is block-diagonal.
The advantages of this method are not only the simpler way of treating asymptotics, as no complicated kernel-functions are needed, but also considers the ease of extending n_{asympt} without too much computational effort. As we will see shortly, we only need to solve the n_{asympt}-size of the local inverse BSE once and the rest of the equations can be performed within n_{core}. This allows us to increase n_{asympt} to very large sizes without sacrificing too much computational time and memory. Fortunately, as will be seen later in Sec. 4.2, this only needs to be done once at the beginning of the routine of the dynamical vertex approximation, since the local irreducible vertex is constant throughout the DΓA algorithm. Only the bare bubble susceptibility $\chi^{qk}_{0;1234}$ has to be known additionally, however, due to the simple frequency dependence of this quantity, it is easily available for n_{asympt} Matsubara frequencies.
We begin by extracting the local irreducible vertex $\Gamma^{\omega\nu\nu'}_{1234,\sigma\sigma'}$ from the generalized susceptibility and the bare bubble susceptibility obtained from the impurity solver in the core region,

$$\Gamma^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'} = \left(\chi^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} - \left(\chi^{\text{core};\omega\nu\nu'}_{0;1234}\right)^{-1}. \tag{3.80}$$

In order to incorporate the asymptotic bare (and local) Coulomb interaction $\mathcal{U}^{\text{asympt}}_{1234;\sigma\sigma'}$ into the local irreducible vertex, we need to solve the n_{asympt}-size BSE,

$$\delta\Gamma^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'} + \mathcal{U}^{\text{asympt}}_{1234;\sigma\sigma'} = \left(\chi^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} - \left(\chi^{\text{asympt};\omega\nu\nu'}_{0;1234}\right)^{-1}, \tag{3.81}$$

where $\delta\Gamma^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'}$ is the irreducible vertex minus the bare Coulomb interaction in the core region. Additionally, $\chi^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}$ must equal the generalized susceptibility obtained from the impurity solver in the n_{core} region. Since we do not know how the generalized susceptibility behaves outside the core frequency region at this stage, this approach does not look promising. However, since we have fixed the irreducible vertex to the bare Coulomb interaction for high frequencies, we can obtain the local (corrected) irreducible vertex with the help of an intermediary susceptibility, where

$$\tilde{\chi}^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'} = \left[\left(\chi^{\text{asympt};\omega\nu\nu'}_{0;1234}\right)^{-1} + \mathcal{U}^{\text{asympt}}_{1234;\sigma\sigma'}\right]^{-1}. \tag{3.82}$$

From the above two equations we can formulate the BSE as

$$\delta\Gamma^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'} = \left[\left(\chi^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1}\right]_{\text{core}} - \left[\left(\tilde{\chi}^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1}\right]_{\text{core}}, \tag{3.83}$$

where after some rewriting and algebra [61]

$$\Gamma^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'} + \mathcal{U}^{\text{core}}_{1234;\sigma\sigma'} = \left(\chi^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} - \left[\left(\tilde{\chi}^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}\right)_{\text{core}}\right]^{-1}. \tag{3.84}$$

We now have rewritten the irreducible vertex in terms of quantities that are already available from the impurity solver, such as $\chi^{\text{core};\omega\nu\nu'}_{1234;\sigma\sigma'}$, and the intermediate susceptibility, which is easily retrieved from the n_{asympt}-range bare bubble susceptibility and the bare Coulomb interaction. This can be seen as an asymptotic correction to the n_{core}-size BSE. Due to the frequency structure of $\tilde{\chi}^{\text{asympt};\omega\nu\nu'}_{1234;\sigma\sigma'}$, this does not require a lot of computational effort or memory and hence is a very resource efficient approach.

The extension to non-local asymptotics is straightforward and can be derived in a few steps, where we will only outline the most important results. For a more in-depth discussion, we refer the reader to Ref. [61]. At the beginning, we define

$$\Gamma^{X;\omega\nu\nu'}_{1234;\sigma\sigma'} = \left(\chi^{X;\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} - \left(\chi^{X;\omega\nu\nu'}_{0;1234}\right)^{-1} \qquad \text{and} \tag{3.85}$$

$$\left(\chi^{*;X;\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} = \left(\chi^{X;\omega\nu\nu'}_{1234;\sigma\sigma'}\right)^{-1} - \mathcal{U}^{X}_{1234;\sigma\sigma'}, \tag{3.86}$$

where $X \in \{\text{core}, \text{asympt}\}$ and $\chi^{*;X;\omega\nu\nu'}_{1234;\sigma\sigma'}$ is the same auxiliary susceptibility as already seen in Eq. (3.60). Both of these equations have to be satisfied for each frequency range n_X. From the lower equation we obtain [61]

$$\chi^{*;X;qkk'}_{1234;\sigma\sigma'} = \chi^{X;qkk'}_{1234;\sigma\sigma'} + \sum_{\substack{k_1 k_2;\mathfrak{abcdef} \\ \sigma_1\sigma_2\sigma_3\sigma_4}} \chi^{X;qkk_1}_{12\mathfrak{ba};\sigma\sigma_1} \mathcal{U}^{X;\mathbf{qk}_1\mathbf{k}_2}_{\mathfrak{abcd};\sigma_1\sigma_2} \chi^{X;qk_2k'}_{\mathfrak{dcef};\sigma_2\sigma_3} \left(\mathbb{1}_{\mathfrak{fe}34} - \mathcal{U}^{X;\mathbf{qk}_1\mathbf{k}_2}_{\mathfrak{fehg};\sigma_3\sigma_4} \chi^{X;qk_2k_1}_{\mathfrak{gh}34;\sigma_4\sigma'}\right)^{-1}. \tag{3.87}$$

From this one can arrive at an expression for the generalized physical susceptibility [61]

$$\chi^{\text{core};q}_{1234;\sigma\sigma'} = \sum_{kk'} \chi^{\text{core};qkk'}_{1234;\sigma\sigma'} = \sum_{kk'} \left[\left(\chi^{*;\text{core};qkk'}_{1234;\sigma\sigma'} + \chi^{\text{asympt};qkk'}_{0;1234;\sigma\sigma'} - \chi^{\text{core};qkk'}_{0;1234;\sigma\sigma'}\right)^{-1} + \mathcal{U}^{\text{core};\mathbf{qkk}'}_{1234;\sigma\sigma'}\right]^{-1}. \tag{3.88}$$

This is especially useful, since the generalized physical susceptibility enters the SDE (3.64) with the three-leg vertices of Eq. (3.63). Hence, this scheme proves to be particularly powerful [61, 63] for this explicit form of the equation of motion of Σ^{k}_{12}. Lastly, it can be shown [64], that for the three-leg vertex $\gamma^{qk}_{d/m;1234} \overset{\nu\to\infty}{\longrightarrow} \mathbb{1}_{1234}$ holds.

3.7.3 High-frequency Bethe-Salpeter equation with vertex asymptotics

Sometimes it is convenient to separate low- and high-frequency components of a quantity to simplify calculations. Some DMFT solvers [65] employ such a frequency separation to calculate the self-energy: a high-frequency asymptotic expansion derived from the spectral function's lowest moments is incorporated in the numerical solution of the Dyson equation at low frequencies. Inspired by this idea, we can proceed similarly for the irreducible vertex Γ, working with the framework of the Bethe-Salpeter equation instead of the Dyson equation. By explicitly using the SU(2)-symmetric combinations d, m — in which the BSE for the ph-channel becomes diagonal[16] — we are able

[16] The same can be done for the s, t combinations in the pp-channel, however only the ph-channel will be shown.

to reduce the computational effort by using a high-frequency expansion of the vertex functions. Following [55, 57, 59], the vertex in the low-frequency regime can be obtained by the regular BSE, where an additional high-frequency correction term is added. In the spirit of the end of Sec. 3.5, the BSE of Eq. (3.49) can be written as a matrix equation (in grouped indices nm) for spin combinations d, m in the following fashion

$$\chi^{q} = \chi_0^{q} - \frac{1}{\beta^2}\chi_0^{q}\boldsymbol{\Gamma}^{q}\chi^{q}. \tag{3.89}$$

Similar to Refs. [55, 57, 59], we will now split the matrices $\boldsymbol{F}$, $\boldsymbol{\Gamma}$ and χ in segments of low and high frequencies. We will denote this by an additional superscript $(\mathrm{L/H}, \mathrm{L/H})$, where (L, L) denotes the segment, where ν and ν' are in the low frequency region and, e.g. (L, H) denotes the segment, where ν is in the low- and ν' in the high-frequency region. We will therefore construct a set of low frequencies $\mathcal{I}_{\mathrm{L}}$ and a set of high frequencies $\mathcal{I}_{\mathrm{H}}$. This means, that for $\boldsymbol{\Gamma}^{\mathrm{LH}}$, $\nu \in \mathcal{I}_{\mathrm{L}}$ and $\nu' \in \mathcal{I}_{\mathrm{H}}$. In the LH, HL and HH blocks, where at least one frequency is considered large, we will employ explicit asymptotics for the vertex functions, as already discussed in Sec. 3.7. The goal is to calculate the vertex function with the BSE for the low-frequency region where we will expect a correction term arising from the low-frequency cutoff which employs the use of high-frequency blocks of the BSE components. The size of $\mathcal{I}_{\mathrm{L}}$ has to be at least the number of bosonic Matsubara frequencies ω [57] and might differ if non-local correlations are included [66]. Regarding $\mathcal{I}_{\mathrm{H}}$, there is no minimum size and the maximum size of the high-frequency region is only bounded by the availability of computational resources and memory.
Using this idea, we will solve Eq. (3.89) for $\boldsymbol{\Gamma}$ as

$$\begin{pmatrix} \boldsymbol{\Gamma}^{\mathrm{LL}} & \boldsymbol{\Gamma}^{\mathrm{LH}} \\ \boldsymbol{\Gamma}^{\mathrm{HL}} & \boldsymbol{\Gamma}^{\mathrm{HH}} \end{pmatrix} = \beta^2 \left[\begin{pmatrix} \chi^{\mathrm{LL}} & \chi^{\mathrm{LH}} \\ \chi^{\mathrm{HL}} & \chi^{\mathrm{HH}} \end{pmatrix}^{-1} - \begin{pmatrix} \chi_0^{\mathrm{LL}} & 0 \\ 0 & \chi_0^{\mathrm{HH}} \end{pmatrix}^{-1} \right] \tag{3.90}$$

where we omitted frequency labels for brevity. Multiplying on the left by χ and solving this matrix-like equation for $\boldsymbol{\Gamma}^{\mathrm{LL}}$, we find

$$\boldsymbol{\Gamma}^{\mathrm{LL}} = \beta^2 \left[\left(\chi^{\mathrm{LL}}\right)^{-1} - \left(\chi_0^{\mathrm{LL}}\right)^{-1} \right] - \left(\chi^{\mathrm{LL}}\right)^{-1} \chi^{\mathrm{LH}} \boldsymbol{\Gamma}^{\mathrm{HL}}, \tag{3.91}$$

with

$$\chi^{\mathrm{LH}} = -\chi^{\mathrm{LL}} \boldsymbol{\Gamma}^{\mathrm{LH}} \left[\boldsymbol{\Gamma}^{\mathrm{HH}} + \beta^2 \left(\chi_0^{\mathrm{HH}}\right)^{-1} \right]^{-1}. \tag{3.92}$$

Combining these two equations yields for $\boldsymbol{\Gamma}^{\mathrm{LL}}$

$$\boldsymbol{\Gamma}^{\mathrm{LL}} = \beta^2 \left[\left(\chi^{\mathrm{LL}}\right)^{-1} - \left(\chi_0^{\mathrm{LL}}\right)^{-1} \right] + \boldsymbol{\Gamma}^{\mathrm{LH}} \left[\boldsymbol{\Gamma}^{\mathrm{HH}} + \beta^2 \left(\chi_0^{\mathrm{HH}}\right)^{-1} \right]^{-1} \boldsymbol{\Gamma}^{\mathrm{HL}}. \tag{3.93}$$

The last term in the sum of Eq. (3.93) can be thought of as a correction to the inversion of the BSE in the low-frequency subspace, which otherwise reads

$$\boldsymbol{\Gamma}^{\mathrm{LL}} = \beta^2 \left[\left(\chi^{\mathrm{LL}}\right)^{-1} - \left(\chi_0^{\mathrm{LL}}\right)^{-1} \right]. \tag{3.94}$$

The correction term can be obtained at a significantly lower cost compared to the full vertex — since for the high-frequency regimes of $\boldsymbol{\Gamma}$ we use either (i) kernel-1 and kernel-2 functions, which depend on one or two momentum and frequency dimensions less than the full vertex (or the generalized susceptibility), see Sec. 3.7.1 or (ii) the even more resource efficient "U-range" approach of Sec. 3.7.2 — and is therefore easily available for much larger asymptotic frequency ranges. Other sets of equations for $\boldsymbol{\Gamma}^{\mathrm{LL}}$[17] can be constructed easily by manipulating the original BSE (3.89). These methods would require the calculation of the full vertex for a large frequency range[18], which is more storage-heavy than the use of kernel-1 and kernel-2 functions or a padding of the bare Coulomb interaction. Thus, the approach outlined in this thesis is more efficient and is going to be a more practical one to use for multi-orbital calculations.

Since we have now assembled all necessary tools to describe the properties of correlated electrons on a lattice, let us introduce the DΓA's purely local predecessor: the dynamical mean-field theory (DMFT).

[17] One can already use Eq. (3.91) instead of Eq. (3.93), cf. "Method 2" in Ref. [57].

[18] Retrieving the vertex functions in a high-frequency regime is *per se* not difficult; they can be obtained for arbitrary frequency ranges from the impurity solver of choice. However, the computing time increases drastically the larger the frequency box and thus this method is not preferred over the one mentioned above.

Chapter 4 Modern methods for correlated electron systems and superconductivity

4.1 Dynamical mean-field theory

Dynamical mean-field theory is based on the idea that in the limit of infinite spatial dimension ($d \to \infty$), the self-energy Σ^{k}_{12} becomes *purely local* [67], where the Kronecker delta reduces the non-local self-energy to a purely local one[1]. In high dimensions, the hopping amplitude must scale appropriately to remain finite. As a result, non-local correlations between lattice sites i and j are suppressed[2]. Starting from this discovery, it was shown [5] that there is a mapping between an auxiliary single-impurity Anderson model (SIAM) and the Hubbard model in infinite dimensions, which allowed for a derivation of a self-consistency cycle determining the purely local self-energy. This self-consistency can be calculated using approximate[3] or numerically exact[4] methods [68, 69]. A modern implementation of the DMFT procedure — which is used in this thesis to calculate input data for the dynamical vertex approximation, more on that later — is w2dynamics [70], a continuous-time hybridization expansion (CT-HYB) QMC solver for the SIAM.

Let us briefly outline the idea and self-consistency equations of DMFT in the following, see [68] for a more in-depth discussion. In high spatial dimensions, the number of neighboring sites Z_{ij} grows as $Z_{ij} \sim d$. To ensure a well-defined competition between kinetic energy and Coulomb interaction, the hopping amplitudes must therefore scale as

$$t_{im,jn} \to t^{*}_{im,jn} = \frac{t_{im,jn}}{\sqrt{Z_{ij}}}, \tag{4.1}$$

where m, n denote origin and destination orbitals, respectively. Renormalizing the hopping elements automatically renormalizes the corresponding Green's function, which is written as

$$G^{k}_{12} = \left[(i\nu + \mu)\delta_{12} - \varepsilon^{\mathbf{k}}_{12} - \Sigma^{\nu}_{\mathrm{DMFT};12}\right]^{-1}. \tag{4.2}$$

The Fourier transforms of the hopping amplitudes $\varepsilon^{\mathbf{k}}_{12}$ enter the Green's function, hence it scales similarly.

[1] Due to the spatial locality of the self-energy, a Fourier transform to momentum space results in a flat dispersion of $\Sigma^{\mathbf{k}}_{\mathrm{DMFT}}$.

[2] Indeed, one can show that the diagrams contributing to the non-local part of the self-energy are suppressed by $\frac{1}{\sqrt{Z_{ij}}}$, where Z_{ij} is the number of neighbors at a given distance $\|\mathbf{R}_i - \mathbf{R}_j\|$ from lattice site i.

[3] e.g. self-consistent perturbation theory, iterated perturbation theory or non-crossing approximation

[4] e.g. quantum Monte Carlo (QMC), numerical renormalization group or exact diagonalization

Diagrammatically, DMFT includes the local contribution of all topologically inequivalent Feynman diagrams which can also be obtained via the SIAM, however only if the interaction of the SIAM Hamiltonian has the same form as the original Hubbard Hamiltonian, i.e.

$$\hat{\mathcal{H}}_{\mathrm{AIM}} = \sum_{ab;\mathbf{k}} \varepsilon^{\mathbf{k}}_{ab} \hat{c}^{\dagger}_{\mathbf{k};a} \hat{c}_{\mathbf{k};b} + \sum_{ab} \varepsilon^{d}_{ab} \hat{d}^{\dagger}_{a} \hat{d}_{b} + \frac{1}{2} \sum_{abcd} U_{abcd} \hat{d}^{\dagger}_{a} \hat{d}^{\dagger}_{c} \hat{d}_{d} \hat{d}_{b} + \sum_{ab;\mathbf{k}} \left[V^{\mathbf{k}}_{ab} \hat{d}^{\dagger}_{a} \hat{c}_{\mathbf{k};b} + \mathrm{h.c.} \right]. \tag{4.3}$$

Here, $\hat{d}^{\dagger}_{a}$ ($\hat{d}_{a}$) creates (annihilates) an electron on the impurity, whereas the operators $\hat{c}^{\dagger}_{\mathbf{k};a}$ ($\hat{c}_{\mathbf{k};a}$) create (annihilate) bath — i.e. non-interacting conduction — electrons, which hybridize with the localized impurity electrons via $V^{\mathbf{k}}_{ab}$. For a successful mapping of the Hubbard model onto the AIM, there are three independent equations in DMFT that describe a self-consistency cycle [68]. The fundamental requirement is that the impurity Green's function must equal the local Green's function of the corresponding lattice model. Since the local Green's function is not known a priori, the problem is solved self-consistently. We first either guess the starting Green's function or calculate it with the self-energy from the previous iteration, see Eq. (4.4a), calculate the effective non-interacting Green's function of the impurity with Eq. (4.4b), and use this to solve the impurity problem in Eq. (4.4c) (with any of the methods previously mentioned). This yields a new, interacting Green's function. We then invert Eq. (4.4b), to obtain Eq. (4.4d), which is our new "guess" for the next iteration. Once we have, that $\mathcal{G}^{\nu}_{0;12} = \sum_{\mathbf{k}} G^{\mathbf{k}}_{\mathrm{DMFT};12}$, we successfully acquired a self-consistent solution. Mathematically the self-consistency cycle consists of four equations

$$G^{\nu}_{12} = \frac{1}{V_{\mathrm{BZ}}} \int_{\mathrm{BZ}} \mathbf{dk}\, \underbrace{\left[(i\nu + \mu)\delta_{12} - \varepsilon^{\mathbf{k}}_{12} - \Sigma^{\nu}_{\mathrm{DMFT};12} \right]^{-1}}_{G^{\mathbf{k}}_{\mathrm{DMFT};12}}, \tag{4.4a}$$

$$\mathcal{G}^{\nu}_{0;12} = \left[\left(G^{\nu}_{12} \right)^{-1} + \Sigma^{\nu}_{\mathrm{DMFT};12} \right]^{-1}, \tag{4.4b}$$

$$G^{\nu}_{12} = \left\langle \mathcal{T} \hat{d}_{1} \hat{d}^{\dagger}_{2} \right\rangle \quad \text{and} \tag{4.4c}$$

$$\Sigma^{\nu}_{\mathrm{DMFT};12} = \left(\mathcal{G}^{\nu}_{0;12} \right)^{-1} - \left(G^{\nu}_{12} \right)^{-1}. \tag{4.4d}$$

Here, $\mathcal{G}^{\nu}_{0;12}$ denotes the non-interacting Green's function of the impurity. The numerically challenging part is the computation of the AIM's Green's function in Eq. (4.4c) for which a number of different solvers are available, see the beginning of Sec. 4.1. Eq. (4.4a) is the lattice Dyson equation, Eqs. (4.4b) and (4.4d) are the impurity Dyson equation.

4.2 Dynamical vertex approximation

DMFT — as a local framework — is very capable of describing strong quasiparticle renormalization and the Mott-Hubbard metal-to-insulator transition in heavy fermion systems and transition metal oxides accurately. Its capabilities however extend way beyond that, successful descriptions of (anti-)ferromagnetic effects including metamagnetism, influence on ferromagnetic properties through the lattice and Hund's coupling, scattering at spins or impurities and many more have been reported [68, 71]. As broad as the range of DMFT's capabilities is, it comes with a major limitation: by construction, it neglects non-local correlations. Non-local correlations are responsible for numerous physical phenomena, such as the formation of valence bonds, pseudogaps, and d-wave superconductivity, as well as (para-)magnons and quantum critical behavior near phase transitions. These effects are particularly important in quasi-two-dimensional materials, where the infinite-dimensional limit of DMFT is far from realistic and a proper treatment of momentum or spatial structure is necessary to capture, for example, d-wave pairing.

There are two main routes to extend DMFT beyond the local scope to also include non-local correlations [72] (i) cluster extensions and (ii) diagrammatic extensions. Cluster extensions make use of a small cluster of impurities that are embedded in a bath of non-interacting conduction electrons [73]. Non-local interaction effects within this cluster are then taken into account, making it possible to capture short-range fluctuations. Diagrammatic extensions on the other hand extend DMFT by explicitly adding non-local diagrammatic contributions. The dynamical vertex approximation falls into the latter one and will be in the focus of this thesis, more specifically the self-consistent "ladder" variant [74]. For other flavors of DΓA, we refer the reader to Refs. [64, 75–81]. For a first introduction to the "standard" description of the dynamical vertex approximation, we refer the reader to Ref. [6].

Here, we will extend the original description by including the purely non-local Coulomb interaction to the irreducible vertex and follow the derivation outlined in Ref. [13] closely. For completeness, we will also write down the set of equations analogously to [63] for the case of non-local interactions. For clarity, we will split those two separate derivation branches into their own section. The reason we include both derivations in this thesis is simple: Previously, we have written down two distinct ways to treat explicit vertex asymptotics. The first one, where explicit kernel-functions are used, see Sec. 3.7.1, works well if one explicitly employs the full vertex F in the equations, as [13] does. However, the second derivation, where we do not need the full vertex in the equations explicitly, benefits when treating vertex asymptotics with the "U-range" approach, see Sec. 3.7.2. The numerical implementation of the latter is more resource efficient and one does not need local quantities on a very large frequency grid, i.e. it is easier to increase the size of the asymptotic range compared to the kernel-function approach. During the work on this thesis we found that the second approach is easier to realize numerically, and the code developed in the course of this thesis employs the "U-range" asymptotics as formulated by [82], which is also tested against the implementation of [15].

The DΓA uses a converged DMFT solution, thus including all local diagrams as its starting point. It then calculates a non-local approximation of the vertex, which in turn is inserted into the Schwinger-Dyson equation (3.55) to obtain a new self-energy. We will present the "ladder"-approach, which reduces the numerical complexity by assuming the (ph, $\overline{\text{ph}}$ and fully) irreducible vertices to be purely local and equal to those of DMFT. Furthermore, it also assumes pure locality of pp-reducible diagrams[5]. For the following, we assume that we already have an existing converged DMFT solution at hand, containing all local diagrams. The quantities we require for the execution of the dynamical vertex approximation are the DMFT self-energy and the local two-particle Green's function in the particle-hole channel[6].
At the heart of DΓA is the irreducible vertex in the particle-hole channel[7], which we extend by

[5] This is not a priori justified, however, results of the full (parquet) DΓA equations yield a negligible contribution of the particle-particle channel to the full vertex F in the presence of strong spin fluctuations close to the antiferromagnetic phase transition in two dimensions [78]. It is therefore reasonable to neglect the non-locality of the particle-particle channel diagrams. For superconducting properties on the other hand, these non-local diagrams are often essential.

[6] If one would use the asymptotics of the vertex functions as described in Sec. 3.7.1, one also requires the local two-particle Green's function in the pp-channel.

[7] From now on, if not explicitly specified otherwise, we will always assume the ph-frequency notation and channel

adding the purely non-local Coulomb interaction,

$$\Gamma^{qkk'}_{1234;\sigma\sigma'} = \Gamma^{\omega\nu\nu'}_{1234;\sigma\sigma'} + \mathcal{V}^{\mathbf{qkk}'}_{1234;\sigma\sigma'}, \tag{4.5}$$

where $\mathcal{V}^{\mathbf{qkk}'}_{1234;\sigma\sigma'}$ is given by the crossing-symmetric expression (similar to Eq. (3.56))

$$\mathcal{V}^{\mathbf{qkk}'}_{1234;\sigma\sigma'} = V^{\mathbf{q}}_{1234} - \delta_{\sigma\sigma'} V^{\mathbf{k}'-\mathbf{k}}_{1432}. \tag{4.6}$$

The irreducible vertex can now be seen as containing all local irreducible diagrams plus the non-locality of the lowest order, providing a natural extension to this framework. This means that we have now included the first-order term — which is also the major contributor to the irreducible vertex — fully momentum-dependent. We will, similar to Ref. [13], exclude from now on the explicit transversal particle-hole contribution to the non-local irreducible vertex in Eq. (4.5), similar to the GW approach [83], which will reduce the complexity of the DΓA equations considerably, since two out of three momentum indices vanish; the only leftover one from now on is the transfer momentum **q**. This yields for the irreducible vertex in ph-channel for the d/m spin combinations

$$\Gamma^{q\nu\nu'}_{d;1234} = \Gamma^{\omega\nu\nu'}_{d;1234} + 2V^{\mathbf{q}}_{1234} \qquad \text{and} \tag{4.7a}$$
$$\Gamma^{q\nu\nu'}_{m;1234} = \Gamma^{\omega\nu\nu'}_{m;1234}. \tag{4.7b}$$

From this, the Bethe-Salpeter equation in the ph-channel of Eq. (3.48) reduces to

$$F^{q\nu\nu'}_{d/m;1234} = \Gamma^{q\nu\nu'}_{d/m;1234} - \frac{1}{\beta}\sum_{abcd;\nu_1} \Gamma^{q\nu\nu_1}_{d/m;12ab}\chi^{q\nu_1\nu_1}_{0;badc} F^{q\nu_1\nu'}_{d/m;cd34}, \tag{4.8}$$

where

$$\chi^{q\nu\nu'}_{0;1234} = \sum_{\mathbf{kk}'} \chi^{qkk'}_{0;1234} = -\beta \sum_{\mathbf{kk}'} G^{k}_{14} G^{k'-q}_{32}. \tag{4.9}$$

From now on we will sometimes work with grouped indices, as described at the end of Sec. 3.5, to simplify the following derivation of the DΓA equations. Note that the grouped indices are now $\{n_1, n_2, \nu\}$ and $\{n_4, n_3, \nu'\}$, respectively, and do not include any momenta anymore. We will denote quantities written in grouped indices as bold quantities, similar to before. We can now see that the numerical implementation of the DΓA equations is easier, since the vertex matrices decrease in size from $(N_{\mathbf{k}} \cdot N_\nu \cdot N^2_{\text{bands}}) \times (N_{\mathbf{k}} \cdot N_\nu \cdot N^2_{\text{bands}})$ to $(N_\nu \cdot N^2_{\text{bands}}) \times (N_\nu \cdot N^2_{\text{bands}})$, which reduces the storage requirement by $N^2_{\mathbf{k}}$ and the number of operations by $N^2_{\mathbf{k}}$ for matrix-vector multiplication and $N^3_{\mathbf{k}}$ for matrix-matrix multiplication and matrix inversion compared to the full parquet approach.

In principle, it would be straightforward to retrieve the full vertex via $\boldsymbol{F}^q_r = \left[(\boldsymbol{\Gamma}^q_r)^{-1} + \boldsymbol{\chi}^q_0\right]^{-1}$ from Eq. (3.89). This poses numerical problems, however, as already stated at the end of Sec. 3.6. Thus, we rewrite the inversion of the irreducible vertex $\boldsymbol{\Gamma}$ using the local variant of the Bethe-Salpeter equation of Eq. (3.48) and Eq. (4.7) and after some algebra, the full vertex in the density and magnetic spin combinations reads (see [13] for a derivation)

$$\boldsymbol{F}^q_d = \left[\boldsymbol{F}^\omega_d + 2\boldsymbol{V}^{\mathbf{q}}\left(1 - \boldsymbol{\chi}^\omega_0 \boldsymbol{F}^\omega_d\right)\right]\left[\mathbb{1} + \boldsymbol{\chi}^{\text{nl};q}_0 \boldsymbol{F}^\omega_d - 2\boldsymbol{\chi}^q_0 \boldsymbol{V}^{\mathbf{q}}\left(1 - \boldsymbol{\chi}^\omega_0 \boldsymbol{F}^\omega_d\right)\right]^{-1} \quad \text{and} \tag{4.10a}$$

reducibility of Γ and Φ to be particle-hole.

$$\boldsymbol{F}^{q}_{m} = \boldsymbol{F}^{\omega}_{m}\left[\mathbb{1} + \chi_0^{nl;q}\boldsymbol{F}^{\omega}_{m}\right]^{-1}. \tag{4.10b}$$

Here, the purely non-local susceptibility $\chi_0^{nl;q}$ is given by

$$\chi_0^{nl;q} = \chi_0^{q} - \chi_0^{\omega}. \tag{4.11}$$

Rewriting the full vertex this way avoids the aforementioned numerical difficulties that lie in the inversion of the irreducible vertex $\boldsymbol{\Gamma}$, however we still have to deal with the inclusion of the transversal particle-hole channel, since the above equations were generated through the BSE (3.48) in the ph-channel only. The BSE only carries over the swapping symmetry of the two-particle Green's function to the (ir-)reducible vertices $\boldsymbol{\Gamma}$ and $\boldsymbol{\Phi}$. The crossing symmetry of the full vertex can be restored by explicitly adding diagrams that belong to the transversal particle-hole channel. We therefore modify the parquet-decomposition in Eq. (3.32) in the following way

$$F^{qkk'}_{d;1234} = F^{\omega\nu\nu'}_{d;1234} + 2V^{\mathbf{q}}_{1234} + \left(\Phi^{q\nu\nu'}_{d;1234} - \Phi^{\omega\nu\nu'}_{d;1234}\right) + \left(\Phi^{qkk'}_{d;\overline{ph};1234} - \Phi^{\omega\nu\nu'}_{d;\overline{ph};1234}\right) \quad \text{and} \tag{4.12}$$

$$F^{qkk'}_{m;1234} = F^{\omega\nu\nu'}_{r;1234} + \left(\Phi^{q\nu\nu'}_{m;1234} - \Phi^{\omega\nu\nu'}_{m;1234}\right) + \left(\Phi^{qkk'}_{m;\overline{ph};1234} - \Phi^{\omega\nu\nu'}_{m;\overline{ph};1234}\right). \tag{4.13}$$

Here, all diagrams in the pp-channel and all fully irreducible diagrams (except the bare interaction $V^{\mathbf{q}}_{r;1234}$) are local. We furthermore find via the local and non-local BSE,

$$\left(\Phi^{q\nu\nu'}_{d;1234} - \Phi^{\omega\nu\nu'}_{d;1234}\right) = F^{nl;q\nu\nu'}_{d;1234} - 2V^{\mathbf{q}}_{1234} \quad \text{and} \tag{4.14a}$$

$$\left(\Phi^{q\nu\nu'}_{m;1234} - \Phi^{\omega\nu\nu'}_{m;1234}\right) = F^{nl;q\nu\nu'}_{m;1234}, \tag{4.14b}$$

where $F^{nl;q\nu\nu'}_{r;1234}$ is defined analogously to the purely non-local susceptibility

$$F^{nl;q\nu\nu'}_{r;1234} = F^{q\nu\nu'}_{r;1234} - F^{\omega\nu\nu'}_{r;1234}. \tag{4.15}$$

We can express the difference of the transversal particle-hole diagrams of Eq. (4.12) with the help of Eq. (3.36), yielding

$$\left(\Phi^{qkk'}_{d;\overline{ph};1234} - \Phi^{\omega\nu\nu'}_{d;\overline{ph};1234}\right) = -\frac{1}{2}F^{nl;(k'-k)(\nu'-\omega)\nu'}_{d;3214} - \frac{3}{2}F^{nl;(k'-k)(\nu'-\omega)\nu'}_{m;3214} \quad \text{and} \tag{4.16a}$$

$$\left(\Phi^{qkk'}_{m;\overline{ph};1234} - \Phi^{\omega\nu\nu'}_{m;\overline{ph};1234}\right) = -\frac{1}{2}F^{nl;(k'-k)(\nu'-\omega)\nu'}_{m;3214} + \frac{1}{2}F^{nl;(k'-k)(\nu'-\omega)\nu'}_{m;3214}. \tag{4.16b}$$

Combining the results of Eq. (4.12) yields

$$F^{qkk'}_{d;1234} = F^{\omega\nu\nu'}_{d;1234} + F^{nl;q\nu\nu'}_{d;1234} - \frac{1}{2}F^{nl;(k'-k)(\nu'-\omega)\nu'}_{d;3214} - \frac{3}{2}F^{nl;(k'-k)(\nu'-\omega)\nu'}_{m;3214} \quad \text{and} \tag{4.17a}$$

$$F^{qkk'}_{m;1234} = F^{\omega\nu\nu'}_{m;1234} + F^{nl;q\nu\nu'}_{m;1234} - \frac{1}{2}F^{nl;(k'-k)(\nu'-\omega)\nu'}_{d;3214} + \frac{1}{2}F^{nl;(k'-k)(\nu'-\omega)\nu'}_{m;3214}. \tag{4.17b}$$

Within the approximations introduced above, the connected part of the self-energy (see Eq. (3.55) and below) is zero for the momentum-dependent contribution from the non-local bare interaction $V^{\mathbf{k}'-\mathbf{k}}_{1234}$ and now reads

$$\Sigma^{k}_{12} = \Sigma^{\mathbf{k}}_{HF;12} - \frac{1}{\beta}\sum_{\substack{abcdef\\ q\nu'}}\left[\mathcal{U}^{\mathbf{q}}_{a1bc}\chi^{q\nu'\nu'}_{0;cbed}F^{q\nu'\nu}_{d;de2f} - \frac{1}{2}\tilde{\mathcal{U}}_{a1bc}\chi^{q\nu'\nu'}_{0;cbed}\left[F^{nl;q\nu'\nu}_{d;2edf} + 3F^{nl;q\nu'\nu}_{m;2edf}\right]\right]G^{k-q}_{af}. \tag{4.18}$$

It is now time to split up the derivation into two paths, one devoted to the steps outlined in [13], coined “Approach 1”, and one devoted to the steps outlined in [63], coined “Approach 2”.

Approach 1 — From the purely local part of Eq. (4.18), which is contained in the first term in the brackets, one can extract the DMFT self-energy, since the ph-channel contribution with the local U_{1234} can be rewritten as

$$\Sigma^{k}_{\mathrm{ph};U;12} = \Sigma^{\mathbf{k}}_{\mathrm{HF};12} - \frac{1}{\beta}\sum_{q\nu'} U_{a1bc}\left(\chi^{\mathrm{nl};q\nu'\nu'}_{0;cbed} + \chi^{\omega\nu'\nu'}_{0;cbed}\right) F^{\omega\nu'\nu}_{d;de2f} G^{k-q}_{af} \tag{4.19}$$

$$= \Sigma^{\nu}_{\mathrm{DMFT};12} + \Sigma^{\mathrm{nl};\mathbf{k}}_{\mathrm{HF};12} - \frac{1}{\beta}\sum_{q\nu'} U_{a1bc}\chi^{\mathrm{nl};q\nu'\nu'}_{0;cbed} F^{\omega\nu'\nu}_{d;de2f} G^{k-q}_{af}, \tag{4.20}$$

where $\Sigma^{\mathrm{nl};\mathbf{k}}_{\mathrm{HF};12}$ is the purely non-local Hartree-Fock term with the contributions from $V^{\mathbf{q}}_{1234}$ but not U_{1234}, see Eq. (3.55). We can take advantage of the momentum and frequency structure in the SDE above to simplify the numerical calculation of the self-energy. Therefore, we — in analogy to Ref. [13] — define three Hedin vertices in $r \in \{d, m\}$ spin combinations,

$$\zeta^{\omega\nu}_{r;1234} = \sum_{ab;\nu'} \chi^{\omega\nu'\nu'}_{0;12ab} F^{\omega\nu'\nu}_{r;ba34}, \tag{4.21}$$

$$\zeta^{q\nu}_{r;1234} = \sum_{ab;\nu'} \chi^{\mathrm{nl};q\nu'\nu'}_{0;12ab} F^{\omega\nu'\nu}_{r;ba34} \qquad \text{and} \tag{4.22}$$

$$\xi^{q\nu}_{r;1234} = \sum_{ab;\nu'} \left[\chi^{q\nu'\nu'}_{0;12ab} F^{q\nu'\nu}_{r;ba34} - \chi^{\omega\nu'\nu'}_{0;12ab} F^{\omega\nu'\nu}_{r;ba34}\right]. \tag{4.23}$$

To not confuse these three-leg vertices with our definition of Eq. (3.63), we labeled them differently here. The completely local Hedin vertex can straightforwardly be retrieved from DMFT. The vertex $\xi^{q\nu}_{1234}$ in spin combinations d/m can be calculated efficiently with a matrix inversion

$$\boldsymbol{\xi}^{q}_{d} = \left(\vec{\mathbb{1}} + \zeta^{\omega}_{d}\right)\left(\left[\mathbb{1} - \boldsymbol{\chi}^{\mathrm{nl};q}_{0}\boldsymbol{F}^{\omega}_{d} - 2\boldsymbol{\chi}^{q}_{0}\boldsymbol{V}^{\mathbf{q}}\left(\vec{\mathbb{1}} + \zeta^{\omega}_{d}\right)\right]^{-1} - \mathbb{1}\right) \qquad \text{and} \tag{4.24a}$$

$$\boldsymbol{\xi}^{q}_{m} = \left(\vec{\mathbb{1}} + \zeta^{\omega}_{m}\right)\left(\left[\mathbb{1} - \boldsymbol{\chi}^{\mathrm{nl};q}_{0}\boldsymbol{F}^{\omega}_{m}\right]^{-1} - \mathbb{1}\right), \tag{4.24b}$$

where $\vec{\mathbb{1}} = \vec{\mathbb{1}}_{1234} = \delta_{14}\delta_{23}$, contrary to $\mathbb{1}$, which is given by $\mathbb{1}^{\nu\nu'}_{1234} = \delta_{14}\delta_{23}\delta_{\nu\nu'}$. Combining these expressions and simplifying it [13] yields the DΓA self-energy

$$\begin{aligned}\boldsymbol{\Sigma}^{k}_{\mathrm{D\Gamma A}} = \boldsymbol{\Sigma}^{\nu}_{\mathrm{DMFT}} + \boldsymbol{\Sigma}^{\mathrm{nl};\mathbf{k}}_{\mathrm{HF}} - \frac{1}{\beta}\sum_{q}\left(\boldsymbol{U} + \boldsymbol{V}^{\mathbf{q}} - \frac{\tilde{\boldsymbol{U}}}{2}\right)\boldsymbol{\xi}^{q}_{d}\boldsymbol{G}^{k-q} \\ + \frac{3}{2\beta}\sum_{q}\tilde{\boldsymbol{U}}\boldsymbol{\xi}^{q}_{m}\boldsymbol{G}^{k-q} - \frac{1}{\beta}\sum_{q}\left(\boldsymbol{V}^{\mathbf{q}}\zeta^{\omega}_{d} - \boldsymbol{U}\zeta^{q}_{d}\right)\boldsymbol{G}^{k-q}.\end{aligned} \tag{4.25}$$

Approach 2 — For an alternative formulation using the three-leg vertices already derived in Sec. 3.6, see Eq. (3.63), we have to rewrite Eq. (4.18) with the full ladder vertices $F^{q\nu\nu'}_{d/m;1234}$,

$$F^{\mathrm{nl};q\nu'\nu}_{d/m;1234} = F^{q\nu'\nu}_{d/m;1234} - F^{\omega\nu'\nu}_{d/m;1234}, \tag{4.26}$$

such that

$$\Sigma^{k}_{12} = \Sigma^{\mathbf{k}}_{\mathrm{HF};12} - \frac{1}{\beta}\sum_{\substack{abcdef\\ q\nu'}}\left[\mathcal{U}^{\mathbf{q}}_{a1bc}\chi^{q\nu'\nu'}_{0;cbed}F^{q\nu'\nu}_{d;de2f} - \frac{1}{2}\tilde{\mathcal{U}}_{a1bc}\chi^{q\nu'\nu'}_{0;cbed}\left[F^{q\nu'\nu}_{d;2edf} + 3F^{q\nu'\nu}_{m;2edf}\right]\right]G^{k-q}_{af} - \Sigma^{k}_{dc;12}, \tag{4.27}$$

where $\Sigma^{k}_{dc;12}$, which is merely a double-counting correction, is given by

$$\Sigma^{k}_{dc;12} = \frac{1}{2\beta}\sum_{\substack{abcdef\\ q\nu'}}\tilde{\mathcal{U}}_{a1bc}\chi^{q\nu'\nu'}_{0;cbed}\left[F^{\omega\nu'\nu}_{d;2edf} + 3F^{\omega\nu'\nu}_{m;2edf}\right]G^{k-q}_{af}. \tag{4.28}$$

With the definition of the three-leg vertex from Eq. (3.63), we can write the upper equation as

$$\begin{aligned}
\Sigma^{k}_{D\Gamma A;12} &= \Sigma^{\mathbf{k}}_{\mathrm{HF};12} - \frac{1}{2}\sum_{\substack{q;abcd\\ efgh}}\Bigg[\mathcal{U}^{\mathbf{q}}_{d;a1bc}\gamma^{q\nu}_{d;cbed}\left(\mathbb{1}_{de2h} - \mathcal{U}^{\mathbf{q}}_{d;defg}\chi^{q}_{d;gf2h}\right) \\
&\qquad + 3\,\mathcal{U}^{\mathbf{q}}_{m;a1bc}\gamma^{q\nu}_{m;cbed}\left(\mathbb{1}_{de2h} - \mathcal{U}^{\mathbf{q}}_{m;defg}\chi^{q}_{m;gf2h}\right) - 2\,\mathcal{U}^{\mathbf{q}}_{m;a12h}\Bigg]G^{k-q}_{ah} - \Sigma^{k}_{dc;12} \\
&= \Sigma^{\mathbf{k}}_{\mathrm{HF};12} - \frac{1}{2}\sum_{\substack{q;abcd\\ efgh}}\Bigg[\left(2U_{a1bc} + 2V^{\mathbf{q}}_{a1bc} - U_{acb1}\right)\gamma^{q\nu}_{d;cbed}\times \\
&\qquad \times\left(\mathbb{1}_{de2h} - \left(2U_{defg} + 2V^{\mathbf{q}}_{defg} - U_{dgfe}\right)\chi^{q}_{d;gf2h}\right) \\
&\qquad - 3U_{acb1}\gamma^{q\nu}_{m;cbed}\left(\mathbb{1}_{de2h} + U_{dgfe}\chi^{q}_{m;gf2h}\right) + 2U_{ah21}\Bigg]G^{k-q}_{ah} - \Sigma^{k}_{dc;12}.
\end{aligned} \tag{4.29}$$

4.3 Self-consistent dynamical vertex approximation

The self-energy obtained from a simple "one-shot" calculation with the ladder-DΓA does not (always) show the correct asymptotic behavior. This deficiency is even more pronounced for higher values of the physical susceptibilities [74]. In addition, the susceptibilities that can be derived from the full vertex diverge at the DMFT Néel temperature. This is concerning, since it therefore violates the Mermin-Wagner theorem [84] for the case of two-dimensional systems. This can be resolved by applying a so-called λ-correction [85], where enforcing a specific sum rule of the spin (and charge) susceptibilities yields a renormalization parameter λ, which is in turn used to renormalize physical susceptibilities in the DΓA routine.

The formalism for the λ-correction is currently only available in the one-band regime, an extension to multi-orbital systems has not been derived yet. The reasons being (i) λ would become a matrix object with as many entries as spin-orbit combinations exist, yielding a multidimensional optimization problem; and (ii) the solution of this multidimensional problem can be non-unique and there are no rules of which of these solutions to choose. Hence, the approach using a regularization scheme with the help of a λ-correction is not suitable for multi-orbital systems, which are in the focus of this thesis. Therefore, to "repair" the violation of the Mermin-Wagner theorem, we will employ a different scheme, a so-called self-consistency iteration [74], more on that later.

Since the λ-correction is currently not possible for multi-band systems, simply using a non-renormalized one-shot ladder-DΓA calculation might be a first approach. The issue is that it does not provide a viable description any longer in parameter regimes where non-local correlations are large [74], e.g. where the DMFT susceptibilities diverge due to a phase transition. The source of this is two-fold: (i) due to the assumed locality of the particle-particle reducible diagrams in the ladder-DΓA equations, which would dampen the particle-hole fluctuations [8], the ladder diagrams of the particle-hole channel lack their non-local contributions; and (ii) the self-energy entering the Bethe-Salpeter equations is only the local DMFT self-energy, and any non-local contributions due to the self-energy are simply missing in the BSE equations. The latter problem can be solved by applying a self-consistency scheme within the DΓA equations as shown in Fig. 4.1 [74]. The steps to acquire a converged DΓA routine with an appropriate self-energy and vertex functions are as follows:

1. Initially, the method requires a band structure $\varepsilon^{\mathbf{k}}_{12}$ and a (non-local) interaction $\mathcal{U}^{\mathbf{q}}_{1234}$, which can come from model studies or, for example, from DFT or cRPA. Using this momentum-dependent band structure and interaction, a DMFT calculation is performed with the chosen (imaginary-time) impurity solver. Once the DMFT calculation has converged, the necessary local one- and two-particle quantities — such as the local self-energy, the local irreducible vertex, and the full vertex — are extracted[8]. Each of these local quantities remains fixed throughout the subsequent self-consistency cycle. This is particularly important for the irreducible vertex Γ and the full vertex F, which are required for the Bethe-Salpeter equation and the double-counting correction to Σ, respectively. All of these quantities then serve as input for the following DΓA calculation.

2. The first step in the sc-DΓA routine is the calculation of the one-particle Green's function via the Dyson equation (3.17). This is done with the local self-energy from DMFT in the first iteration and then with the non-local self-energy calculated from the non-local SDE of the previous iteration.

3. The next step is to update the chemical potential from the newly acquired Green's function to make sure the particle number stays constant.

4. We then proceed with the calculation of the necessary (non-local) three-leg vertices $\gamma^{q\nu}_{d/m;1234}$, see Eq. (3.63), and the generalized physical susceptibilities $\chi^{q}_{d/m;1234}$ from Eq. (3.88). While we keep $\Gamma^{\omega\nu\nu'}_{d/m;1234}$ constant throughout the whole process, $\gamma^{q\nu}_{d/m;1234}$ and $\chi^{q}_{d/m;1234}$ are changing each iteration due to the updated Green's function (and bare bubble) and its feedback into these non-local quantities.

5. The self-energy of the current iteration is finally calculated via Eq. (4.25) using the previously acquired vertex functions and susceptibilities. To complete the cycle, the self-energy then enters the Green's function again via the Dyson equation, see step 2.

To ensure a steady convergence, the self-energy that enters the Green's function is either mixed linearly with a parameter α, such that $\Sigma_n = \alpha\Sigma_n + (1-\alpha)\Sigma_{n-1}$, where n denotes the iteration number and α is typically chosen around $0.2 - 0.4$ or with other mixing schemes, such as Pulay mixing [86]. This accelerated and predictive mixing scheme has been used previously in numerical

[8] In principle, one does not need to calculate the local self-energy through the local Schwinger-Dyson equation with the vertices from DMFT, as it should be equal to the DMFT self-energy by construction. However, the local SDE is an easy way to benchmark the implementation of our code against the self-energy results obtained from DMFT.

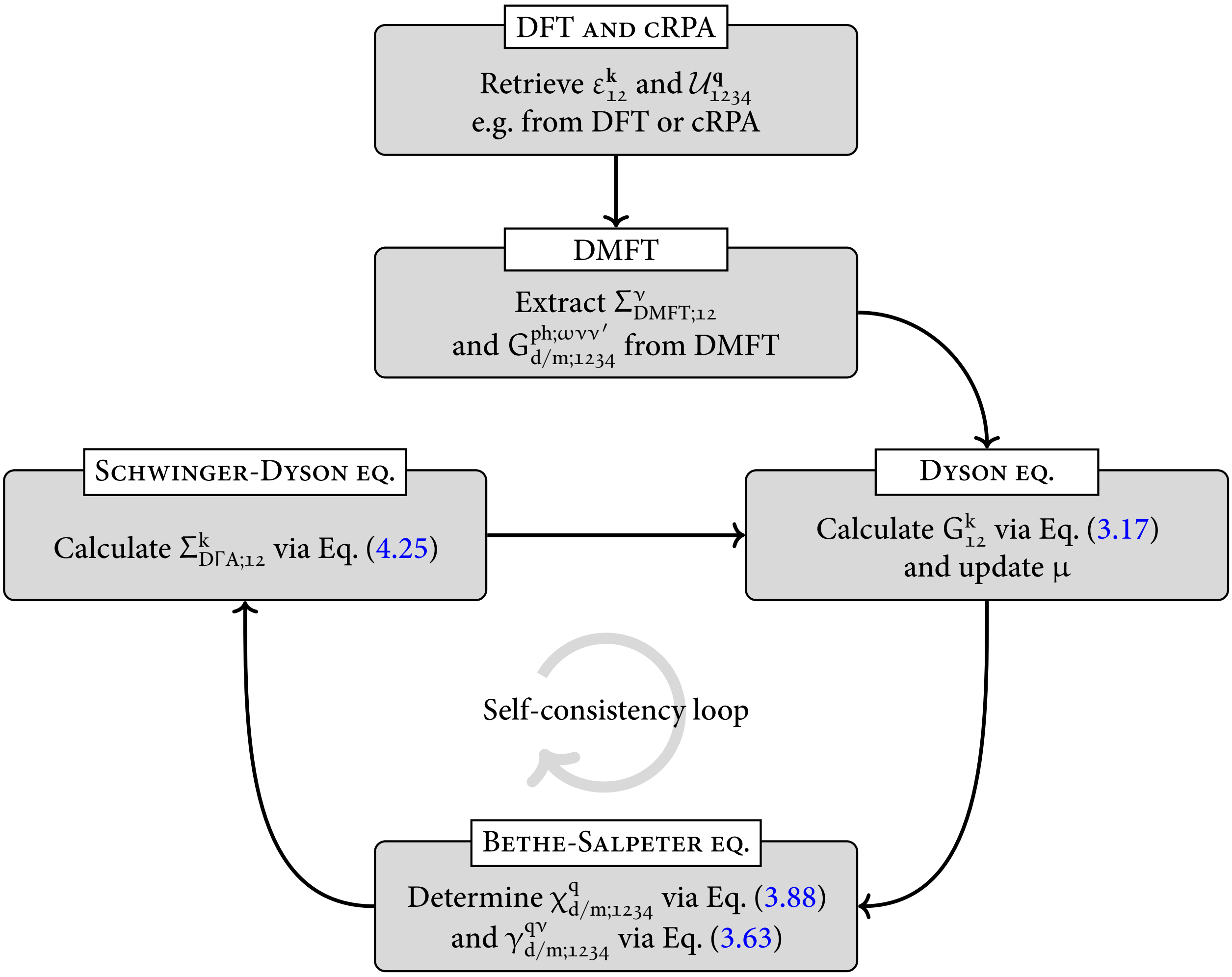


Figure 4.1 – Flowchart of the DΓA self-consistency cycle. The first box is concerned with the bandstructure and interaction calculations which can be obtained from simple model studies or for actual materials from DFT and cRPA. The next box denotes the DMFT calculation which is performed using the previously acquired bandstructure and interaction. From the converged DMFT calculation we extract the DMFT self-energy and the local two-particle Green's function. The three boxes below describe the self-consistency cycle of the DΓA approach. The bandstructure, interaction and output quantities from DMFT, i.e. all purely local (ir-)reducible vertex functions are kept constant throughout the whole DΓA self-consistency cycle.

calculations [87] with significant improvement in convergence speed and behavior over the linear mixing strategy, despite the additional computational and storage cost.

To determine whether the self-energy is converged, a simple criterion comparing the absolute values $|\Sigma^{k}_{n} - \Sigma^{k}_{n-1}| < \varepsilon$ is used. Here, n denotes the current iteration and ε is of $\sim \mathcal{O}\left(10^{-4}\right)$.

From the one-particle Green's function, as well as the vertices and susceptibilities that are directly accessible at this stage, one can extract additional physical quantities, including the optical conductivity and superconducting properties. For instance, the critical temperature can be estimated and the symmetry of the superconducting gap can be determined by solving the Eliashberg equation, which is treated as a post-processing step described in the following section.

4.4 Eliashberg equation

After a successful converged ladder-DΓA routine, we have the most important non-local one- and two-particle quantities available. Starting from this, we can further study superconductivity with the so-called Eliashberg equations [88, 89]. In this context, we introduce, additionally to the one-particle Green's function of Eq. (3.1), two *anomalous* Green's functions [90]

$$F^k_{12} = -\langle \mathcal{T}\left[\hat{c}_{k;1}\hat{c}_{-k;2}\right]\rangle \qquad \text{and} \tag{4.30a}$$

$$\overline{F}^k_{12} = -\left\langle \mathcal{T}\left[\hat{c}^\dagger_{k;1}\hat{c}^\dagger_{-k;2}\right]\right\rangle . \tag{4.30b}$$

Here, F^k_{12} and $\overline{F}^k_{12}$ are not to be confused with the full vertex $F^{qkk'}_{1234}$. The anomalous Green's functions are rather unusual contractions of either two electrons or two holes, contrary to the regular Green's function, which is a contraction of an electron and a hole. These two contractions will vanish in the normal state because of particle conservation, however they are of finite value in the superconducting phase.
At this point it is helpful to switch to the Nambu formalism [47], where the basis is spanned by the spinors

$$\hat{\psi}^k_1 = \left(\hat{c}_{k;1;\uparrow}, \hat{c}_{k;1;\downarrow}, \hat{c}^\dagger_{-k;1;\uparrow}, \hat{c}^\dagger_{-k;1;\downarrow}\right)^T . \tag{4.31}$$

In this basis, the anomalous Green's functions are (block-)off-diagonal entries of the (generalized) Gro'kov single-particle propagator [91, 92]

$$\begin{aligned}\hat{\boldsymbol{G}}^k_{12} &= -\left\langle \mathcal{T}\left[\hat{\psi}^k_1\hat{\psi}^{k;\dagger}_2\right]\right\rangle \\ &= \begin{pmatrix} G^k_{12} & -F^k_{12} \\ -\overline{F}^k_{12} & \overline{G}^k_{12}\end{pmatrix},\end{aligned} \tag{4.32}$$

where G^k_{12} and F^k_{12} are $2n \times 2n$ matrices in spin-orbital space, and n denotes the number of bands. The diagonal entries of the Gro'kov single-particle propagator are the regular single-particle (G^k_{12}) and single-hole ($\overline{G}^k_{12} = -G^{-k}_{12}$) propagators. After a Fourier transform to Matsubara frequencies, the generalized Dyson equation can be written as follows:

$$\hat{\boldsymbol{G}}^k_{12} = \hat{\boldsymbol{G}}^k_{0;12} + \sum_{ab} \hat{\boldsymbol{G}}^k_{0;1a}\hat{\boldsymbol{\Sigma}}^k_{ab}\hat{\boldsymbol{G}}^k_{b2}, \tag{4.33}$$

where

$$\hat{\boldsymbol{G}}^k_{0;12} = \begin{pmatrix} G^k_{0;12} & 0 \\ 0 & G^k_{0;12}\end{pmatrix} \tag{4.34}$$

is the non-interacting generalized propagator and

$$\hat{\boldsymbol{\Sigma}}^k_{12} = \begin{pmatrix} \Sigma^k_{12} & \Delta^k_{12} \\ \overline{\Delta}^k_{12} & \overline{\Sigma}^k_{12}\end{pmatrix} \tag{4.35}$$

denotes the generalized self-energy, where the diagonal entries Σ^{k}_{12} and $\overline{\Sigma}^{k}_{12}$ are the single-particle and single-hole self-energies, respectively and the off-diagonal *pairing* function Δ^{k}_{12} is coined the superconducting *gap function*. Δ^{k}_{12} is directly connected to the anomalous propagators and therefore can capture the relevant Cooper-pairing nature of the electrons in the superconducting phase. While the gap function is non-zero only in the superconducting phase, it is nevertheless possible to obtain a solution directly from the normal state: The phase transition from the normal to the superconducting state is characterized by a divergence in the respective particle-particle susceptibility, which can be studied through an eigenvalue problem also known as the Eliashberg equation [93]. We will start with the BSE written in terms of singlet and triplet susceptibilities in the particle-particle channel, see Eq. (3.49c),

$$\chi^{qkk'}_{s/t;1234} = \chi^{qkk'}_{s/t;0;1234} - \frac{1}{2\beta^2} \sum_{k_1 k_2;abcd} \chi^{qkk_1}_{s/t;0;1c3a} \Gamma^{qk_1k_2}_{s/t;abcd} \left(\chi^{qk_2k'}_{s/t;d2b4} + \chi^{qk_2k'}_{s/t;0;d2b4} \right), \tag{4.36}$$

where the bare bubble susceptibility in the singlet and triplet channel is given by [94]

$$\chi^{qkk'}_{s/t;0;1234} = \mp \chi^{qkk'}_{0;1234}. \tag{4.37}$$

When describing the superconducting state, one usually employs quantities written in the Nambu basis, i.e. we are now using the generalized Gro'kov single-particle propagator in the equation for the susceptibility, not the regular Green's function [47, 95]. We will denote these Green's functions by an additional superscript, $G^{sc;k}_{12}$, and call them *superconducting* Green's functions. The corresponding Eliashberg equation can then be written as

$$\lambda_{s/t} \Delta^{k}_{s/t;12} = \pm \frac{1}{2} \sum_{k_1 k_2;abcd} \Gamma^{qkk_1}_{s/t;1b2a} \chi^{sc;qk_1k_2}_{s/t;0;acbd} \Delta^{k_2}_{s/t;dc}, \tag{4.38}$$

yielding an eigenvalue problem with $\lambda_{s/t}$ as the corresponding eigenvalue. The irreducible vertex $\Gamma^{qkk'}_{s/t;1234}$ appearing in Eq. (4.38) describes the scattering of two electrons with spin-orbit index 1, 2 and momenta $(k + q, -k)$ to electrons with spin-orbit index 3, 4 and momenta $(k' + q, -k')$. Here, the bare bubble susceptibility $\chi^{sc;qkk'}_{0;1234}$ is now built from the superconducting Green's function $G^{sc;k}_{12}$ and therefore contains the anomalous self-energy in the following form

$$G^{sc;k}_{12} = \frac{i\nu\delta_{12} + \varepsilon^{\mathbf{k}}_{12} + \Sigma^{k}_{12}}{(i\nu\delta_{12} + \varepsilon^{\mathbf{k}}_{12} + \Sigma^{k}_{12})(i\nu\delta_{12} + \varepsilon^{\mathbf{k}}_{12} - \Sigma^{k}_{12}) - |\Delta^{k}_{12}|^2}, \tag{4.39}$$

which results from the generalized Dyson equation of Eq. (4.33)[9]. A divergence of $\chi^{qkk'}_{s/t;1234}$ occurs when any eigenvalue of $-\Gamma^{qkk_1}_{s/t;1b2a}\chi^{sc;qk_1k_2}_{s/t;0;acbd}$ reaches unity[10]. In other words, as the temperature is lowered, the superconducting instability occurs at T_c, where $\lambda_{s/t} \longrightarrow 1$[11]. Eq. (4.38) is the full Eliashberg equation, which represents a non-linear implicit equation for the superconducting gap function Δ^{k}_{12}. To solve this equation numerically, one often linearizes the Eliashberg equation and

[9] $\Delta^{k}_{s/t;12}$, as already mentioned above, is the superconducting gap function that transitions smoothly into the anomalous self-energy below the critical temperature T_c.

[10] This eigenvalue does not necessarily have to be the largest one. Some eigenvalues may exceed unity but remain irrelevant if they are not the ones that first trigger the divergence in the channel under consideration.

[11] A common numerical approach to determine T_c is to compute $\lambda_{s/t}$ at several low temperatures and extrapolate to the point where $\lambda_{s/t} \to 1$.

neglects the gap function in the denominator of the superconducting Green's function. This is then commonly coined the *linearized* Eliashberg equation and reads

$$\lambda_{s/t}\Delta^{k}_{s/t;12} = \pm\frac{1}{2}\sum_{k_1 k_2;abcd} \Gamma^{qkk_1}_{s/t;1b2a}\chi^{qk_1k_2}_{s/t;0;acbd}\Delta^{k_2}_{s/t;dc}, \tag{4.40}$$

where the superconducting Green's function transitions to the regular Green's function for a vanishing Δ^{k}_{12}. Eq. (4.40) is diagrammatically depicted for the case $q = 0$ in Fig. 4.2. Choosing this single point of vanishing transfer momentum simplifies the problem, as Cooper pairs form at zero total energy (i.e. in the centre-of-mass system), making the equation effectively uniform and time-independent (c.f. Migdal-Eliashberg theory). Finite q can indicate competing (pair-)density wave states [96].

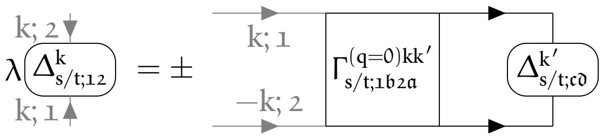


FIGURE 4.2 – Linearized Eliashberg equation for vanishing transfer frequency $q = 0$, see Eq. (4.40).

Due to Pauli's principle requiring Δ to be antisymmetric under the exchange of two particles, the combined symmetry operation of (i) spin flip ($\hat{S}; \sigma \to \overline{\sigma}$), (ii) parity ($\hat{P}; \mathbf{k} \to -\mathbf{k}$), (iii) orbital exchange ($\hat{O}; n_1 \leftrightarrow n_2$) and (iv) time reversal ($\hat{T}; \nu \to -\nu$) has to yield [97]

$$\hat{S}\hat{P}\hat{O}\hat{T}\Delta^{k}_{12} = -\Delta^{-k}_{21}. \tag{4.41}$$

The pairing type of the Cooper pairs can be uniquely determined by individually applying the symmetry operations (i)-(iv). The (anti-)symmetric behavior of Δ^{k}_{12} under these operations of course limits the solutions of the linearized Eliashberg equations to a small subset, which is important to be aware of when solving them numerically. Additionally, the corresponding gap function for singlet pairing fulfills $\Delta^{k}_{12} = \Delta^{-k}_{21}$, whereas the corresponding gap function for triplet pairing fulfills $\Delta^{k}_{12} = -\Delta^{-k}_{21}$ [98].

One can solve the linearized Eliashberg equation through a power iteration on $\boldsymbol{M} = -\boldsymbol{\Gamma}\boldsymbol{\chi}$, e.g. a Lanczos algorithm. There, one specifies a (normalized) starting gap function $\boldsymbol{\Delta}_0$ and then iterates through the following equations self-consistently:

$$\boldsymbol{\Delta}_n = \boldsymbol{M}\boldsymbol{\Delta}_{n-1}, \tag{4.42a}$$

$$\boldsymbol{\Delta}_n = \frac{\boldsymbol{\Delta}_n}{||\boldsymbol{\Delta}_n||} \quad \text{and} \tag{4.42b}$$

$$\lambda_n = \boldsymbol{\Delta}_n^{\dagger}\boldsymbol{M}\boldsymbol{\Delta}_n, \tag{4.42c}$$

where n denotes the iteration number. Solving this set of equations is not difficult, the computationally demanding part of solving the linearized Eliashberg equation is the calculation of the irreducible vertex in the particle-particle channel. Following the derivation outlined in the supplemental material of [82] and generalizing the equations to account for multiple orbitals yields for the full vertex in ladder-DΓA for $r \in \{d/m\}$ spin combinations

$$\begin{aligned} F^{q\nu\nu'}_{r;1234} = \beta^2 \Big[\left(\chi^{q\nu\nu'}_{0;1234}\right)^{-1} - \sum_{\nu_1\nu_2;abcd} \left(\chi^{q\nu\nu_1}_{0;12ba}\right)^{-1} \chi^{*;q\nu_1\nu_2}_{r;abcd} \left(\chi^{q\nu_2\nu'}_{0;dc34}\right)^{-1} \Big] \\ + \sum_{\substack{abcd\\efgh}} \gamma^{q\nu}_{r;12ab} \mathcal{U}^{q}_{r;badc} \left(\mathbb{1}_{cdgh} - \mathcal{U}^{q}_{r;cdef}\chi^{q}_{r;fegh}\right) \gamma^{q\nu'}_{r;hg34}, \end{aligned} \tag{4.43}$$

with the auxiliary susceptibility $\chi^{*;q\nu\nu'}_{r;1234}$ of Eq. (3.60) and the three-leg vertex $\gamma^{q\nu}_{r;1234}$ of Eq. (3.63). In the following section, we will present a short derivation of the particle-particle pairing vertex for the case of $q = 0$ within the framework of ladder-DΓA.

4.4.1 Derivation of the pairing vertex Γ_{pp} within ladder-DΓA

The vertex in Eq. (4.43) does not fulfill any crossing-symmetry relation, as it is generated solely from the particle-hole BSE[12]. Consequently, it only possesses particle-swapping symmetry [13]. In contrast, the full vertex in ladder-DΓA, given in Eq. (4.17), satisfies crossing symmetry. For the $\uparrow\downarrow$-component, we have [13, 48]

$$F^{D\Gamma A;qkk'}_{1234;\uparrow\downarrow} = \frac{1}{2}\left(F^{q\nu\nu'}_{d;1234} - F^{q\nu\nu'}_{m;1234}\right) - F^{(k-k')\nu(\nu-\omega)}_{m;1432} - \frac{1}{2}\left(F^{\omega\nu\nu'}_{d;1234} - F^{\omega\nu\nu'}_{m;1234}\right). \tag{4.44}$$

The $F^{(k-k')\nu(\nu-\omega)}_{m;1432}$-term arises from the ph $\longrightarrow \overline{ph}$ transformation that restores crossing symmetry, introducing an orbital permutation and frequency shifts. This expression is written only for the $\uparrow\downarrow$-component. Since we are interested in the singlet ($\uparrow\downarrow - \overline{\uparrow\downarrow}$) and triplet ($\uparrow\downarrow + \overline{\uparrow\downarrow}$) spin combinations, the $\overline{\uparrow\downarrow}$-component is also needed. Fortunately, one can obtain the $\overline{\uparrow\downarrow}$-contribution via a crossing symmetry relation of the $\uparrow\downarrow$-component in the following way:

$$F^{qkk'}_{1234;\overline{\uparrow\downarrow}} = -F^{(k-k')k(k-q)}_{1432;\uparrow\downarrow}. \tag{4.45}$$

One therefore obtains for the $\overline{\uparrow\downarrow}$-component via Eq. (4.45)

$$\begin{aligned} F^{D\Gamma A;qkk'}_{1234;\overline{\uparrow\downarrow}} = -\frac{1}{2}\left(F^{(k-k')\nu(\nu-\omega)}_{d;1432} - F^{(k-k')\nu(\nu-\omega)}_{m;1432}\right) + F^{q\nu\nu'}_{m;1234} \\ + \frac{1}{2}\left(F^{(\nu-\nu')\nu(\nu-\omega)}_{d;1432} - F^{(\nu-\nu')\nu(\nu-\omega)}_{m;1432}\right). \end{aligned} \tag{4.46}$$

This yields for the full vertices in ph-notation for the singlet

$$\begin{aligned} F^{D\Gamma A;qkk'}_{s;1234} = \frac{1}{2}\left(F^{q\nu\nu'}_{d;1234} - 3F^{q\nu\nu'}_{m;1234}\right) - \frac{1}{2}\left(F^{\omega\nu\nu'}_{d;1234} - F^{\omega\nu\nu'}_{m;1234}\right) \\ + \frac{1}{2}\left(F^{(k-k')\nu(\nu-\omega)}_{d;1432} - 3F^{(k-k')\nu(\nu-\omega)}_{m;1432}\right) - \frac{1}{2}\left(F^{(\nu-\nu')\nu(\nu-\omega)}_{d;1432} - F^{(\nu-\nu')\nu(\nu-\omega)}_{m;1432}\right) \end{aligned} \tag{4.47}$$

and the triplet spin combination

$$\begin{aligned} F^{D\Gamma A;qkk'}_{t;1234} = \frac{1}{2}\left(F^{q\nu\nu'}_{d;1234} + F^{q\nu\nu'}_{m;1234}\right) - \frac{1}{2}\left(F^{\omega\nu\nu'}_{d;1234} - F^{\omega\nu\nu'}_{m;1234}\right) \\ - \frac{1}{2}\left(F^{(k-k')\nu(\nu-\omega)}_{d;1432} + F^{(k-k')\nu(\nu-\omega)}_{m;1432}\right) + \frac{1}{2}\left(F^{(\nu-\nu')\nu(\nu-\omega)}_{d;1432} - F^{(\nu-\nu')\nu(\nu-\omega)}_{m;1432}\right). \end{aligned} \tag{4.48}$$

[12] It is solely the ladder-vertex and not the full vertex in the ladder-DΓA formalism.

We now want to evaluate this expression for $q_{pp} = 0$ with variable momenta k_{pp} and k'_{pp} where the subscript pp refers to the particle-particle frequency convention (the ph momenta in the particle-hole convention are related to the pp momenta as: $q_{ph} = k_{pp} + k'_{pp}$, $k_{ph} = k_{pp}$ and $k'_{ph} = k'_{pp}$ [13]). Inserting these frequencies and momenta yields for the singlet channel

$$\begin{aligned} F^{D\Gamma A;(q_{pp}=0)kk'}_{s;1234} &= \frac{1}{2}\left(F^{(k+k')\nu\nu'}_{d;1234} - 3F^{(k+k')\nu\nu'}_{m;1234}\right) - \frac{1}{2}\left(F^{(\nu+\nu')\nu\nu'}_{d;1234} - F^{(\nu+\nu')\nu\nu'}_{m;1234}\right) \\ &\quad + \frac{1}{2}\left(F^{(k-k')\nu(-\nu')}_{d;1432} - 3F^{(k-k')\nu(-\nu')}_{m;1432}\right) - \frac{1}{2}\left(F^{(\nu-\nu')\nu(-\nu')}_{d;1432} - F^{(\nu-\nu')\nu(-\nu')}_{m;1432}\right) \\ &= \underbrace{\frac{1}{2}\left(F^{(k+k')\nu\nu'}_{d;1234} - 3F^{(k+k')\nu\nu'}_{m;1234}\right) - F^{(\nu+\nu')\nu\nu'}_{1234;\uparrow\downarrow}}_{\tilde{F}^{D\Gamma A;(q=0)kk'}_{s;1234}} + \underbrace{\left(k' \to -k' \;\&\; 1234 \to 1432\right)}_{\tilde{F}^{D\Gamma A;(q=0)k(-k')}_{s;1432}}. \end{aligned} \tag{4.49}$$

For the triplet channel we analogously find

$$\begin{aligned} F^{D\Gamma A;(q_{pp}=0)kk'}_{t;1234} &= \frac{1}{2}\left(F^{(k+k')\nu\nu'}_{d;1234} + F^{(k+k')\nu\nu'}_{m;1234}\right) - \frac{1}{2}\left(F^{(\nu+\nu')\nu\nu'}_{d;1234} - F^{(\nu+\nu')\nu\nu'}_{m;1234}\right) \\ &\quad - \frac{1}{2}\left(F^{(k-k')\nu(-\nu')}_{d;1432} + F^{(k-k')\nu(-\nu')}_{m;1432}\right) + \frac{1}{2}\left(F^{(\nu-\nu')\nu(-\nu')}_{d;1432} - F^{(\nu-\nu')\nu(-\nu')}_{m;1432}\right) \\ &= \underbrace{\frac{1}{2}\left(F^{(k+k')\nu\nu'}_{d;1234} + F^{(k+k')\nu\nu'}_{m;1234}\right) - F^{(\nu+\nu')\nu\nu'}_{1234;\uparrow\downarrow}}_{\tilde{F}^{D\Gamma A;(q=0)kk'}_{t;1234}} - \underbrace{\left(k' \to -k' \;\&\; 1234 \to 1432\right)}_{\tilde{F}^{D\Gamma A;(q=0)k(-k')}_{t;1432}}. \end{aligned} \tag{4.50}$$

We can construct the pairing vertex in either singlet or triplet channel by subtracting the purely *local* particle-particle reducible diagrams. This yields

$$\Gamma^{(q=0)kk'}_{s/t;1234} = F^{D\Gamma A;(q=0)kk'}_{s/t;1234} - \phi^{(\omega=0)\nu\nu'}_{s/t;1234}. \tag{4.51}$$

The local reducible diagrams can be subtracted straightforwardly from $\tilde{F}^{D\Gamma A;(q_{pp}=0)k(\pm k')}_{s/t;1234}$ in the following fashion

$$\begin{aligned} \Gamma^{(q=0)kk'}_{s/t;1234} &= \tilde{F}^{D\Gamma A;(q=0)kk'}_{s/t;1234} \pm \tilde{F}^{D\Gamma A;(q=0)k(-k')}_{s/t;1432} - \phi^{(\omega=0)\nu\nu'}_{s/t;1234} \\ &= \tilde{F}^{D\Gamma A;(q=0)kk'}_{s/t;1234} \pm \tilde{F}^{D\Gamma A;(q=0)k(-k')}_{s/t;1432} - \left(\phi^{(\omega=0)\nu\nu'}_{pp;1234;\uparrow\downarrow} \mp \phi^{(\omega=0)\nu\nu'}_{pp;1234;\overline{\uparrow\downarrow}}\right) \\ &= \tilde{F}^{D\Gamma A;(q=0)kk'}_{s/t;1234} \pm \tilde{F}^{D\Gamma A;(q=0)k(-k')}_{s/t;1432} - \left(\phi^{(\omega=0)\nu\nu'}_{pp;1234;\uparrow\downarrow} \pm \phi^{(\omega=0)\nu(-\nu')}_{pp;1432;\uparrow\downarrow}\right) \\ &= \underbrace{\tilde{F}^{D\Gamma A;(q=0)kk'}_{s/t;1234} - \phi^{(\omega=0)\nu\nu'}_{pp;1234;\uparrow\downarrow}}_{\tilde{\Gamma}^{(q=0)kk'}_{s/t;1234}} \pm \underbrace{\left(\tilde{F}^{D\Gamma A;(q=0)k(-k')}_{s/t;1432} - \phi^{(\omega=0)\nu(-\nu')}_{pp;1432;\uparrow\downarrow}\right)}_{\tilde{\Gamma}^{(q=0)k(-k')}_{s/t;1432}} \\ &= \tilde{\Gamma}^{(q=0)kk'}_{s/t;1234} \pm \left(k' \to -k' \;\&\; 1234 \to 1432\right), \end{aligned} \tag{4.52}$$

where the + (−) sign refers to the singlet (triplet) channel.

[13] The frequency notation for the particle-particle channel used in this thesis has been written down in Eq. (3.28c).

4.4.2 Symmetry of the gap function

We can show that the gap function inherits the correct symmetry upon using the symmetrized vertices of Eq. (4.52). Recall that the multi-orbital Eliashberg equation for the case $q = 0$ is given by

$$\lambda_{s/t}\Delta^{k}_{s/t;12} = \pm\frac{1}{2}\sum_{k';abcd}\Gamma^{(q=0)kk'}_{s/t;1b2a}\chi^{(q=0)k'}_{s/t;0;acbd}\Delta^{k'}_{s/t;dc} \tag{4.53}$$

for $q = 0$ and the singlet and triplet gap functions should fulfill $\Delta^{k}_{s;12} = \Delta^{-k}_{s;21}$ and $\Delta^{k}_{t;12} = -\Delta^{-k}_{t;21}$, respectively. We can write Γ in the symmetrized form with the irreducible vertices $\tilde{\Gamma}$ from Eq. (4.52) above, i.e.

$$\Gamma^{(q=0)kk'}_{s/t;1234} = \tilde{\Gamma}^{(q=0)kk'}_{s/t;1234} \pm \tilde{\Gamma}^{(q=0)k(-k')}_{s/t;1432}, \tag{4.54}$$

where $\tilde{\Gamma}^{(q=0)kk'}_{s/t;1234}$ contains both the non-local and local contribution to the irreducible vertex. To confirm that this combination enforces the correct symmetry of the gap function, we need to show that

$$\Delta^{k}_{s;12} - \Delta^{-k}_{s;21} = 0 \qquad \text{and} \tag{4.55a}$$

$$\Delta^{k}_{t;12} + \Delta^{-k}_{t;21} = 0. \tag{4.55b}$$

These expressions for $q = 0$ read for the singlet and triplet channels (using the gap functions from Eq. (4.53))

$$\Delta^{k}_{s;12} - \Delta^{-k}_{s;21} \propto \sum_{k';abcd}\left[\Gamma^{(q=0)kk'}_{s;1b2a} - \Gamma^{(q=0)(-k)k'}_{s;2b1a}\right]\chi^{(q=0)k'}_{s;0;acbd}\Delta^{k'}_{s;dc} \qquad \text{and} \tag{4.56a}$$

$$\Delta^{k}_{t;12} + \Delta^{-k}_{t;21} \propto \sum_{k';abcd}\left[\Gamma^{(q=0)kk'}_{t;1b2a} + \Gamma^{(q=0)(-k)k'}_{t;2b1a}\right]\chi^{(q=0)k'}_{t;0;acbd}\Delta^{k'}_{t;dc}. \tag{4.56b}$$

The full non-local ladder vertices of Eq. (4.49) and Eq. (4.50) fulfill the swapping symmetry, which is passed along to the pairing vertex $\Gamma^{(q=0)kk'}_{s/t;1234}$ and reads in pp-notation for $q = 0$

$$\Gamma^{(q=0)kk'}_{s/t;1234} = \Gamma^{(q=0)(-k)(-k')}_{s/t;3412} \overset{(4.54)}{=} \pm\Gamma^{(q=0)(-k)k'}_{s/t;3214}. \tag{4.57}$$

Applying Eq. (4.57) to Eq. (4.56a) then yields for the singlet channel

$$\Delta^{k}_{s;12} - \Delta^{-k}_{s;21} \propto \sum_{k';abcd}\left[\Gamma^{(q=0)kk'}_{s;1b2a} - \Gamma^{(q=0)kk'}_{s;1b2a}\right]\chi^{(q=0)k'}_{s;0;acbd}\Delta^{k'}_{s;dc} = 0 \tag{4.58}$$

and analogously for the triplet channel, see Eq. (4.56b),

$$\Delta^{k}_{t;12} + \Delta^{-k}_{t;21} \propto \sum_{k';abcd}\left[\Gamma^{(q=0)kk'}_{t;1b2a} - \Gamma^{(q=0)kk'}_{t;1b2a}\right]\chi^{(q=0)k'}_{t;0;acbd}\Delta^{k'}_{t;dc} = 0, \tag{4.59}$$

confirming that the symmetrization in Eq. (4.54) leads to the correct symmetry of the gap function.

4.4.3 Efficient solving of the Eliashberg equation via a Fourier transform

To avoid having to deal with two momentum indices for the pairing vertex $\Gamma_{1234}^{(q=0)kk'}$ in the calculation, one can perform a Fourier transform to real space. However, this is only of advantage in the ladder approximation of the vertex, since then $\Gamma_{s/t;1234}^{(q=0)kk'} = \Gamma_{s/t;1234}^{\mathbf{k}-\mathbf{k}',(\omega=0)\nu\nu'} \pm \Gamma_{s/t;1432}^{\mathbf{k}+\mathbf{k}',(\omega=0)\nu(-\nu')}$. The sum over k' of Eq. (4.40) is essentially a convolution in momentum space, which reduces to a multiplication in Fourier space. We thus take the Fourier transform of Eq. (4.40) in the case of $q = 0$ with respect to $\mathbf{k}$

$$\lambda_{s/t} \sum_{\mathbf{k}} e^{i\mathbf{k}\mathbf{x}} \Delta_{s/t;12}^{k} = \pm\frac{1}{2} \sum_{kk';abcd} e^{i\mathbf{k}\mathbf{x}} \left[\Gamma_{s/t;1b2a}^{\mathbf{k}-\mathbf{k}'(\omega=0)\nu\nu'} \pm \Gamma_{s/t;1a2b}^{\mathbf{k}+\mathbf{k}'(\omega=0)\nu(-\nu')}\right] \chi_{s/t;0;acbd}^{k'} \Delta_{s/t;dc}^{k'}, \tag{4.60}$$

where $\mathbf{x} = (x_1, \ldots, x_D)$ is a real-space vector for systems in D dimensions. If we introduce

$$\tilde{\Delta}_{s/t;ba}^{k'} = \sum_{cd} \chi_{s/t;0;acbd}^{k'} \Delta_{s/t;dc}^{k'} \tag{4.61}$$

and use the shifts $\mathbf{k} = \mathbf{q} + \mathbf{k}'$ for the left term and $\mathbf{k} = \mathbf{q} - \mathbf{k}'$ for the right term, we can rewrite Eq. (4.60) as

$$\lambda_{s/t} \sum_{\mathbf{k}} e^{i\mathbf{k}\mathbf{x}} \Delta_{s/t;12}^{k} = \pm\frac{1}{2} \sum_{\nu' ab} \sum_{\mathbf{q}k'} e^{i\mathbf{q}\mathbf{x}} \left[\Gamma_{s/t;1b2a}^{\mathbf{q}(\omega=0)\nu\nu'} e^{i\mathbf{k}'\mathbf{x}} \tilde{\Delta}_{s/t;ba}^{k'} \pm \Gamma_{s/t;1a2b}^{\mathbf{q}(\omega=0)\nu(-\nu')} e^{-i\mathbf{k}'\mathbf{x}} \tilde{\Delta}_{s/t;ba}^{-\mathbf{k}'\nu'}\right]. \tag{4.62}$$

This reads in Fourier-transformed quantities as

$$\lambda_{s/t} \Delta^{\mathbf{x}\nu} = \pm\frac{1}{2} \sum_{\nu' ab} \left[\Gamma_{s/t;1b2a}^{\mathbf{x}(\omega=0)\nu\nu'} \tilde{\Delta}_{s/t;ba}^{\mathbf{x}\nu'} \pm \Gamma_{s/t;1a2b}^{-\mathbf{x}(\omega=0)\nu(-\nu')} \tilde{\Delta}_{s/t;ba}^{-\mathbf{x}\nu'}\right]. \tag{4.63}$$

In essence, we have reduced the convolution of Eq. (4.40) in momentum space to a product of the quantities in real space, effectively reducing the object from two momentum dimensions $N_{\mathbf{k}} \times N_{\mathbf{k}}$ to a single real-space dimension $N_{\mathbf{x}} (= N_{\mathbf{k}})$, reducing the object size by a factor $N_{\mathbf{k}}$. Numerically, solving the equation now can be done by performing a discrete Fourier transform on the quantities before executing the power iteration to calculate the gap function on the real-space axis. To afterward obtain the momentum-dependent gap function, one simply applies the inverse transformation to the result.

Chapter 5
The code

The main part of this thesis is the implementation of a multi-orbital self-consistent ladder-DΓA code with an emphasis on usability and minimal setup effort. The code package is called moLDGA[1] and the source code is publicly available on GitHub[2]. Please note that there already exist two relatively modern implementations of a ladder-DΓA code, one concerned with a one-shot multi-orbital calculation without asymptotics [14], and one with a λ-corrected approach for single bands using the "U-range" asymptotics described in Sec. 3.7.2 [15]. With the new code, we aim to combine both their functionalities, employing explicit multi-orbital asymptotics, a self-consistency loop with a prediction-based self-energy mixing scheme, and an additional calculation of the superconducting eigenvalues through the multi-orbital Eliashberg equation. This allows for an improvement over the multi-orbital AbinitioDΓA code from Ref. [14] for low-temperature data in order to study physically relevant multi-orbital materials such as the recently discovered bilayer nickelates $La_3Ni_2O_7$ in the near future. Both of these previous codes have had an influence on the code described in this chapter and they also heavily influenced the design choices as discussed in Sec. 5.1 in more detail.
The code is designed to be both accessible and performant, targeting fast execution through efficient parallelization and extensive memory optimizations. It is written in Python and relies mainly on NumPy for numerical computations, making use of vectorized operations and optimized linear algebra routines. In addition, the code carefully exploits symmetries inherent to the physical problem to further reduce computational cost and memory usage. In this chapter, we present the implementation of the computational framework developed to achieve a fully converged self-energy through the self-consistency algorithm of the ladder-DΓA equations and a subsequent calculation of the superconducting eigenvalues and gap functions for both singlet and triplet electron pairing.

Building upon the underlying theoretical foundations which were established already in the previous chapters, this numerical implementation aims to translate the formal expressions into a working algorithm. Special attention has been given to structuring the code in a modular and efficient manner, ensuring that it can be used not only for the ladder-DΓA equations, but also for other algorithms that require multi-point vertex functions within the Matsubara frequency framework in condensed matter physics.

The package will be provided under the MIT license, making it freely available for use, modification, and distribution. This permissive license promotes collaboration and allows academics and developers to make changes on the code without placing restrictions. By publishing the implementation open-source, we hope to encourage accessibility, reproducibility, and community development.

[1] Short for multi-orbital ladder dynamical vertex approximation.
[2] https://github.com/Julpe/moLDGA

In this chapter, we start with an overview of the overall structure and design choices of the code, followed by a detailed discussion of its core components and algorithms. We then describe the typical workflow for using the code, including input handling, computational steps and output generation. Next, we show benchmark results against the already existing DGApy code in the single-orbital case and perform consistency checks for the situation of fully disconnected bands and for an orbital-space-rotated basis. Finally, we discuss potential improvements for future work.

5.1 Structure and design choices

As already mentioned above, there already exist several programs that allow the calculation of vertex functions and the self-energy through ladder-DΓA, with two of the modern versions being (i) AbinitioDΓA [13, 14, 26], which is a one-shot multi-orbital calculation of the DΓA self-energy and (ii) DGApy [15], which is the one-band version of it with a λ-correction, where additional quantities, such as the superconducting eigenvalues and real-frequency Green's functions, are calculated. The former is written in Fortran, whereas the latter has been developed in Python. The purpose of the code developed in the course of this thesis is to provide the functionality of treating multi-orbital systems like AbinitioDΓA, while being easily accessible and quick to set up, similar to DGApy. Therefore, we decided to write the program in Python. A main advantage of this programming language is its low entry-barrier and the easy set-up and code-execution. In theory, all that is needed is a Python environment with the necessary packages[3], DMFT input files from a preceding DMFT calculation of one-particle configuration and the vertex, and the configuration file.
So far, DGApy has successfully been used in several instances, for example in Refs. [9, 10], to only mention a few. Since it is already used among researchers, the transition from DGApy to this code should be as simple as possible, hence the configuration file and some internal modules are designed similarly.

5.1.1 Parallelization via MPI

To lower the computation time, we employ the use of the mpi4py library in Python, which enables distributed computing across multiple processes in a straightforward and Pythonic manner. This allows the program to run inside a process pool, where objects can easily be sliced and passed to or gathered from other processes. While an ideal speed-up factor of n for n processes is theoretically possible, in practice, communication overhead and memory transfer costs typically limit the effective gain. Nevertheless, for large-scale computations involving high-dimensional vertex functions or large frequency boxes, this parallelization leads to significant reductions in runtime.
The central class handling parallelization is the MpiDistributor, which is initialized at the point where the explicitly non-local parts of the ladder-DΓA algorithm begin. This class abstracts the communication logic and provides a consistent interface for scattering and gathering objects across processes. The current implementation slices the data along the transferred momentum $\mathbf{q}$, which is possible due to the independent parametric dependence of all equations up to the Schwinger-Dyson equation (3.64). Each MPI process is responsible for a specific set of $\mathbf{q}$-points of all quantities

[3] For a detailed list of requirements including package versions, see the file requirements.txt in the source code directory.

throughout the DΓA cycle, allowing for highly parallel and independent evaluation of vertex contractions.
Internally, the slicing is guided by knowledge of the irreducible Brillouin zone, and the distributor automatically determines the mapping from global **q**-grid indices to process-local ones. This separation between parallelization logic and physics routines enhances maintainability and flexibility of the code. Further MPI optimizations, such as asynchronous communications or load balancing across uneven grid sizes, could be explored in future versions.

5.1.2 Memory reduction through symmetry

The memory demands of a full ladder-DΓA calculation can be substantial, especially for multi-orbital models and large frequency ranges. To mitigate this, the code incorporates several symmetry-based optimizations that significantly reduce the memory footprint and enable efficient scaling to larger systems.
First, the calculation of non-local quantities is performed exclusively within the irreducible Brillouin zone. Thanks to the crystal symmetries of common lattice structures (e.g. square or quasi-1D lattices), only a small subset of the full momentum grid needs to be explicitly computed. The code determines this set via symmetry operations specified in the brillouin_zone module, which ensures compatibility with arbitrary lattice symmetries defined in the configuration. For instance, on a $16 \times 16 \times 1$ grid, only 45 out of 256 **q**-points are needed for a square lattice, which directly reduces both computational and memory load by more than a factor of five and even higher for larger grid sizes.
To further save memory, we keep most of the objects in half of their bosonic Matsubara frequency range: Due to the symmetry $F_{1234}^{\omega\nu\nu'} = (F_{4321}^{(-\omega)(-\nu')(-\nu)})^* = (F_{1234}^{(-\omega)\nu\nu'})^*$[4], we can restrict the objects to their positive bosonic Matsubara frequency region only. This saves half of the memory and thus a very large amount for the biggest objects, such as the auxiliary susceptibility or the full vertex.
These symmetry-based reductions are implemented transparently for the user: the internal representation is compressed, but the high-level API provides access to the full data. This allows for both memory efficiency and usability, enabling the user to work with large systems that would otherwise be intractable on a single machine.

5.1.3 Object-oriented API

We tried to keep a very object-oriented approach when developing the toolbox for the program, as opposed to both DGApy and AbinitioDΓA. Hence instead of directly working with NumPy arrays, we work with (Local)FourPoint, (Local)Interaction, SelfEnergy, GreensFunction and GapFunction classes. The reason being that in Python, it is very easy to overload operators, i.e. with so-called "dunder" methods, hence we implemented the most common operators "+/-/@/~"[5], alongside multiplication and division with numbers, for these objects to help make the implementation of equations easier. This might not be an obvious improvement at first, however the underlying structure allows a very simple high-level handling of these objects, where most of the logic is performed in the background and only has to be coded once. We will discuss the explicit implementation of these dunder methods later.

[4] The transformation from $-\nu', -\nu$ to ν, ν' and the orbital swap are possible due to the symmetrized nature of the two-particle Green's function extracted from DMFT and the inherited vertex functions.
[5] Corresponding to addition, subtraction, multiplication and inversion.

Compared to DGApy, we stripped down the module responsible for the Matsubara frequency handling, since most of the features now can be called directly from the objects themselves. Instead of calling a method from the matsubara_frequencies module to cut an objects' frequency range, it is now a method on the object itself, e.g. object.cut_niv(10). Most of these methods are only implemented once in a base class from which all subtypes inherit. These base classes already cover most of the objects' functionality, where the subclass only implements features unique to the specific object type.

All modules should be very self-explanatory, their names already provide a good description of what the module does and what its responsibilities are. For any details, we refer the reader to the documentation of the code inside the repository.

One benefit of the object-oriented approach is the now easy implementation of various equations. For example, writing an arbitrary equation with arbitrary (non-local) vertex functions A, B, C and D like

$$D^{q\nu\nu'}_{1234} = \frac{1}{2\beta}\sum_{\nu_1;ab} A^{q\nu\nu_1}_{12ab}\left(B^{\omega\nu_1}_{ba34} + C^{\omega\nu_1\nu'}_{ba34}\right) \tag{5.1}$$

can be done with a call of D = 1 / (2 * beta) * A.matmul(B.add(C)) or the dunder methods for matrix multiplication "@" and addition "+", i.e. D = 1 / (2 * beta) * A @ (B + C), where the index contraction, frequency handling and summation will be done in the background. This eases readability in the code and reduces redundancy. Furthermore, it allows for very straightforward optimization of contractions and multiplications through this architecture. Additionally, inverting a vertex function is also very easy now, all that needs to be called is the obj.invert() function (or with a "~" prefix, i.e. ~obj) on any object and the code performs the inversion in the background using NumPy's vectorization and inversion by converting the object to compound indices. Moreover, other operations, such as addition/subtraction in the form of A.add(B)/A.sub(B) or just simply $A \pm B$ are possible and automatically handle different amounts of frequency dimensions, etc. There are several other convenience methods available, where we would like to refer the interested reader to the code and its documentation for more details.

5.1.4 Setup and execution

The code is very easy to set up as it is available as an installable Python package, called moldga. After installing the necessary packages denoted in the requirements.txt file to the Python environment, one simply has to run "pip install -e ." from the repository directory in the terminal to install the package in editable mode. This allows the user to now access any module and class inside the moldga package. The main entry point to the program is the file dga_main.py file, which can be started with either "python dga_main.py" for single-core execution (mostly used for testing purposes) or "mpiexec -np <n_proc> python dga_main.py" for multi-core processing. There are two additional command line parameters available to append to the execution:

- "-p": With "-p <path>" one can specify the path to the configuration file, which contains all run-specific parameters. This is useful if one wants to store multiple configuration files in different directories. If this parameter is not set, the path defaults to the location of the repository directory.

- "-c": With "-c <config name>" one can specify the name of the configuration file one wants to load. This defaults to dga_config.yaml.

As an example, the following shell command runs the code using 8 MPI processes and loads the configuration file my_config.yaml from the path /configs/:

"mpiexec -np 8 python dga_main.py -p /configs/ -c my_config.yaml".

Additionally, we feature extensive logging: every important step in the calculation will be logged. If it is started from a terminal, the logging will be done to the standard output. If the code is executed on a slurm-based cluster, one will find the logs in the job output file. The reason we employ a lot of logging is the ease of finding errors that might occur during a calculation.

5.1.5 Configuration

The configuration file is split into small blocks of configuration sections, which will be explained in detail in the following. For those entries where only a single data type is expected, the code will try to parse the input to this specific type. If the parsing was unsuccessful, a warning is logged and a default value for this variable is used. Note that the order of the different configuration sections in the file does not matter for a successful read-in of the settings, and we discuss the order of the blocks as they are in the default dga_config.yaml file.

The first block of the configuration is concerned with the number of Matsubara frequencies one wants to use for the calculation:

```yaml
box_sizes:
  niw_core: 50   # int, default: -1
  niv_core: 30   # int, default: -1
  niv_shell: 20  # int, default:  0
```

The "core" region defines the frequency box used for explicitly solving the Bethe-Salpeter and Schwinger-Dyson equations, while the "shell" region sets the size of the asymptotic tails needed for vertex reconstruction through the method described in Sec. 3.7.2. Here, niw_core defines the number of positive bosonic Matsubara frequencies, whereas niv_core determines the number of positive fermionic Matsubara frequencies[6]. It is possible to set the core frequencies to "-1", which means that the number of positive Matsubara frequencies from the DMFT calculation will be taken. If the number of fermionic or bosonic Matsubara frequencies is smaller than the frequency box from DMFT, the DMFT vertices will be cut to the specified frequency box size.

The next section is concerned with the Hamiltonian of the system, which currently is a little bit more complex, since we wanted to adapt the existing configuration file from DGApy and added new features. It defines the lattice symmetries, the momentum grid sizes, the non-interacting and interacting part of the Hamiltonian:

[6]The total number of frequencies of the objects are $2 * \text{niw\_core} + 1$ and $2 * \text{niv\_core}$, respectively.

```
lattice:
  symmetries: "two_dimensional_square"
                              # string, default: "two_dimensional_square"
  type: "from_wannier90"      # string, default: "from_wannier90"
  hr_input: "/path/to/file" # string | list[float], default: None
  interaction_type: "one_band_from_dmft"
                              # string, default: "one_band_from_dmft"
  interaction_input: "/path/to/file"
                              # string, default: ""
  nk: [16, 16, 1]             # list[int], default: [16, 16, 1]
  nq: [16, 16, 1]             # list[int], default: [16, 16, 1]
```

Here, we have a bunch of settings available that can be combined in different ways. First, we can define the lattice symmetry, where one has to enter a set of predefined symmetry sets. It is possible to extend the symmetries the code supports by adding the corresponding symmetry operations and symmetry sets in the code. The available symmetry sets are two_dimensional_square, quasi_one_dimensional_square, simultaneous_x_y_inversion and quasi_two_dimensional_square_symmetries. These symmetries are implemented in the brillouin_zone module and we refer to the documentation of the code for more details. Next is the type of input to the non-interacting part of the Hamiltonian we require. It can either be from_wannier90, from_wannierHK or t_tp_tpp (only for single-band input). The first input type is used if the subsequent field hr_input references to a file, where the hopping elements are written in real space, whereas from_wannierHK is used if the Hamiltonian in momentum space is available as a file[7]. The last option is only used for single-band input, where the band distribution is that of a t, t' and t'' hopping model only. If this type is specified, then a list of three float values in hr_input is expected.

Since the moLDGA code is capable of performing multi-orbital calculations with some non-local interactions, we require the definition of an interaction type and an input. The available interaction types are one_band_from_dmft, kanamori_from_dmft and custom. If one_band_from_dmft is specified, we expect a single-band input, where the interaction strength is read in from the DMFT input files. Similarly, the kanamori_from_dmft option takes the interaction values from DMFT and constructs the correct interaction matrix with the Kanamori-specific entries. In both cases, the field interaction_input will not be read. Lastly, if one specifies custom as the interaction type, the code expects a path to a file in interaction_input, which is structured similarly to a real-space Hamiltonian file but with four orbital entries instead of just two. It is possible to include non-local interactions there. Furthermore, it is possible to set the sizes of both the $\mathbf{k}$- and $\mathbf{q}$-grid in the fields nk and nq, respectively. It is possible to not set nq, where nk will be used for the $\mathbf{q}$-grid. Note, that the execution of the Eliashberg equation is only possible if both the $\mathbf{k}$- and the $\mathbf{q}$-grid are of the same size.

The next configuration section is concerned with the self-consistency calculation and defines the convergence behavior and mixing strategy of the self-consistency loop:

[7] Note that for the former, it is possible to use arbitrary $\mathbf{k}$-grid sizes, since the Fourier transform will be performed in the code, whereas for the latter one the correct grid size corresponding to the supplied $\mathcal{H}(\mathbf{k})$ has to be entered.

```
self_consistency:
  max_iter: 20                        # int, default: 20
  save_iter: True                     # bool, default: True
  epsilon: 1e-6                       # float, default: 1e-4
  mixing: 0.3                         # float, default: 0.2
  mixing_strategy: "pulay"            # string, default: "linear"
  mixing_history_length: 2            # int, default: 3
  previous_sc_path: "path/to/files"  # string, default: "./"
```

The field max_iter is to set an upper boundary on the number of iterations during the self-consistency cycle. If the self-energy does not converge within max_iter iterations, the iterative scheme will stop. When setting save_iter to True, the code will save the non-local DΓA self-energy for each iteration. The next parameter, epsilon, is set to determine the absolute value difference allowed for the last two calculated sigmas to be considered converged. The convergence will be tested on the full $\mathbf{k}$-grid and the first $\nu_{n,\text{max}} = \left\lceil \frac{\text{niv\_core}}{5} \right\rceil$ frequencies[8], i.e. the self-energy is considered converged after l iterations when for all values $\mathbf{k}$ and $\nu_n \in \{1, \ldots, \nu_{n,\text{max}}\}$ the expression $|\Sigma_l(\mathbf{k}, \nu_n) - \Sigma_{l-1}(\mathbf{k}, \nu_n)| \leq$ epsilon holds.

Next, we can specify the mixing parameter that will be used in the mixing scheme and is a floating point number between $[0, 1]$. We can furthermore specify the mixing scheme in the next parameter, called mixing_strategy. Here, we can choose from either linear or pulay mixing [86], where if pulay is specified, it takes mixing_history_length number of previous self-energy results to construct a prediction. If the number of iterations is less than the history length, the code will use linear mixing for these iterations and switch to Pulay mixing afterward. The last parameter, previous_sc_path allows us to specify the path to a previous self-consistency iteration folder, which might not have been converged and needs to be converged further or with different parameters. The program then takes the last iterations of the already performed calculation and uses these as a starting point for the next calculation. If the number of iterations from the previous calculation is equal or larger than mixing_history_length and pulay is specified, then Pulay mixing will be applied, otherwise we mix linearly.

Given the improved speed and scalability of the moLDGA code compared to DGApy, we have also implemented two λ-correction schemes which only work for single-band cases. This can be configured with the following short section:

```
lambda_correction:
  perform_lambda_correction: True # bool, default: False
  type: "spch"                    # string, default: "spch"
```

Here, setting perform_lambda_correction to True will perform a λ-correction if the input is a single-band model. Otherwise, an error is raised and the code exits. Note that if one calculates the one-shot lambda-corrected self-energy and Eliashberg equation, the variables max_iter and mixing in the self_consistency section will be set to "1" during execution. The other parameter is the type of λ-correction which will be performed. Setting this to spch will renormalize both the density and magnetic susceptibilities, whereas setting it to sp will only renormalize the magnetic susceptibility.

The next section is a very important one, since it is concerned with the location of the

[8]This is just an estimate and currently hard-coded in the program, since for low frequencies the self-energy does not display significant statistical noise carried over from DMFT.

DMFT files and mainly defines their location and file names:

```
dmft_input:
  type: "w2dyn"                  # string, default: "w2dyn"
  input_path: "/path/to/files"   # string, default: "./"
  fname_1p: "filename"           # string, default: "1p-data.hdf5"
  fname_2p: "filename"           # string, default: "g4iw_sym.hdf5"
  do_sym_v_vp: True              # bool, default: True
```

Currently, the type argument only supports w2dyn as input type, which refers to the w2dynamics code, however this could be extended in the future to also support input files from other sources with other file structures. Next, input_path needs to be set to the folder that contains the output data from w2dynamics. We require two files, one containing a converged DMFT self-energy and impurity Green's function and one containing two-particle quantities (i.e. the two-particle Green's function in both density and magnetic channels); their filenames can be set in fname_1p and fname_2p, respectively. Note that w2dynamics outputs a large Vertex.hdf5-file, where we have to extract the relevant worm components beforehand. This can be done with the script symmetrize.py, which is also part of the repository[9]. This will automatically extract all relevant worm components of the two-particle Green's function and writes it to a separate file containing the density and magnetic contributions required for fname_2p, while the file for fname_1p can be directly taken from the one-particle DMFT calculation. A straightforward extraction of the two-particle Green's function from the Vertex.hdf5 file is not possible. Lastly, we have the option to symmetrize the two-particle Green's function in ν and ν' before processing them further. This might be necessary sometimes to ensure numerical symmetry of the vertex functions.

The second to last configuration section we will cover here is the output section which, as the name already suggests, covers the program's output settings, such as the output path and plotting instructions:

```
output:
  output_path: "/path"                # string, default: "./"
  do_plotting: True                   # bool, default: "True"
  plotting_subfolder_name: "Plots"    # string, default: "Plots"
  save_quantities: True               # bool, default: True
```

We are able to choose the path where the output of the program is saved to. If this field is chosen empty, the output will be put into a sub-folder[10] inside the DMFT input folder. If we specify do_plotting, then we plot a selective set of quantities and save them to the sub-folder specified in plotting_subfolder_name. We then have the possibility to additionally save almost all quantities that are calculated during the main program as .npy files, except for the pairing vertex and the full ladder-DΓA vertex. The output is separately handled in the Eliashberg section, since they are only calculated if one wants to perform the calculation of the superconducting eigenvalue and gap function.

Finally, the last configuration section is concerned with the Eliashberg equation and is configuring the eigenvalue and gap function-finding through a Lanczos method:

[9] In the corresponding repository of Ref. [14] a file with a similar name is included, however the functionality between both versions is slightly different.

[10] Currently, the folder name is LDGA_Nk<nk>_Nq<nq>_wc<niw_core>_vc<niv_core>_vs<niv_shell>.

```
eliashberg:
  perform_eliashberg: True      # bool, default: False
  save_pairing_vertex: False    # bool, default: False
  save_fq: False                # bool, default: False
  n_eig: 4                      # int, default: 4
  epsilon: 1e-9                 # float, default: 1e-6
  symmetry: "random"            # string, default: "random"
  include_local_part: False     # bool, default: False
  subfolder_name: "Eliashberg"  # string, default: "Eliashberg"
```

Here, the first setting is to let the program know if one wants to perform the Eliashberg equation to obtain the superconducting singlet and triplet eigenvalues and gap functions, respectively. If this setting is set to "False", then the program will exit after the self-energy has been calculated. The next two settings allow saving the (non-symmetrized) pairing vertex $\tilde{\Gamma}^{\mathbf{q}=\mathbf{k}-\mathbf{k}';\nu\nu'}$ and the full ladder vertex $F^{q\nu\nu'}$ to a file. These are set to "False" per default, as these objects can be quite large, especially the full vertex, which can reach up to hundreds of gigabytes. Both will be saved for the irreducible Brillouin zone only to reduce the storage space needed. The following three settings, n_eig, epsilon and symmetry are concerned with the power iteration that is performed to solve the Eliashberg equation. n_eig allows the user to specify the number of largest eigenvalues and corresponding eigenvectors to be calculated using a Lanczos method. Here, epsilon determines the target precision of the power iteration which is performed and symmetry determines the starting vector. There are a couple of options available for symmetry that may help speeding up the power iteration convergence if one knows the pairing symmetry of the gap function a priori: random, p-wave-x, p-wave-y and d-wave. In almost all cases it should suffice to pick a random starting vector for the power iteration.

In principle one also needs the *purely local* full and reducible vertices for the pairing vertex, however these can be omitted if one expects p- or d-wave symmetry of the gap function, where the local vertex functions do not contribute to the eigenvalue and gap function due to symmetry. Setting include_local_part to "False" does exactly that. Not calculating the local part allows for a slightly larger frequency box and a better estimation of the eigenvalue and gap function for p- or d-wave symmetry cases. Please note that at the time of writing this thesis, the implementation of the local part has not been fully tested.

The last setting is the sub-folder name where the Eliashberg output will be saved to.

5.1.6 Generating the input files

After the vertex calculation has been performed with w2dynamics [70], we have to extract a couple of key quantities, such as the local self-energy, Green's function, interaction and two-particle Green's functions from the output files of w2dynamics. There is a script located in the repository, called symmetrize.py, which allows for the straightforward extraction of the two-particle Green's function from the vertex file of w2dynamics. A simple execution (with "python symmetrize.py") of this script yields a new file, where the density and magnetic components of $G^{\omega\nu\nu'}_{r;1234}$ are written to. This is handy, since the vertex file from w2dynamics is usually very large, and it can be deleted after the symmetrization process. Then one is left with the 1p-data.hdf5 and g4iw_sym.hdf5 files, which contain the one-particle and two-particle (symmetrized) quantities. These are, apart from a file containing the real-space (e.g. wannier_hr.dat) or Fourier-space (e.g. wannier.hk) Hamiltonian and the configuration file, the only input files needed for the execution of the DΓA algorithm. Notice that no file name is fixed, and they can be specified in the corresponding dga_config.yaml file.

Since we only care about the density and magnetic spin combinations for the two-particle vertex functions and can disregard any spin combination where there is an odd number of $\uparrow$ and $\downarrow$ or any of the form $\uparrow\downarrow\uparrow\downarrow$, we only have to sample a reduced set of worm components. Out of the possible $16 \times n_o^4$ (total number of spin-orbit combinations), there only exist $6 \times n_o^4$ non-zero worm components. This reduces the computational time of the QMC run by more than a factor of 2.5. The reduced worm component list can be retrieved by a call to the function get_worm_components in the file symmetrize.py. This means that for a single band we only have to sample 6 out of 16 (6/16), for two bands 96/256 and for three bands 486/1296 components. Since in principle for SU(2) symmetry the number of spin components can be reduced to two[11], the number of worm components can be further reduced to $2 \times n_o^4$ at the cost of stochastic noise which is lowered by combining more spin combinations.

5.2 Numerical effort

In the following, we will briefly analyze the numerical scaling of the implemented DΓA algorithm, which is in principle the same as in [14]: For low temperatures, the calculation of the local vertex functions through the CT-HYB QMC solver is very often the most time-consuming part of a DΓA calculation. The numerical cost of evaluating the vertex scales as $\mathcal{O}(\beta^5 n_o^4)$ with a considerable prefactor due to the Monte Carlo sampling of w2dynamics [70]. Here, n_o is the number of orbitals (bands). This scaling arises because the local vertex contains $\mathcal{O}(n_\omega n_\nu^2 n_o^4)$ components[12], while updating the CT-HYB hybridization matrix itself requires $\mathcal{O}(\beta^2)$ operations: the average expansion order thus grows linearly with β.

To mitigate the cost, the asymptotic high-frequency form of the irreducible vertex, see 3.7.2, can be calculated with only two frequencies and therefore scales as $\mathcal{O}(\beta^4 n_o^4)$, allowing for a smaller frequency-box size of the vertices in the QMC calculation. This makes calculations near superconducting transition temperatures feasible.

In the new code, the calculations are parallelized over the system's $\mathbf{q}$-grid and are diagonal in the bosonic Matsubara frequency ω, yielding a numerical effort that scales with $\mathcal{O}(n_\mathbf{q} n_\omega)$. For each $\mathbf{q}$ and ω entry, the dominant cost stems from operations using vectors and matrices whose dimensions are $N = n_\nu n_o^2$. Therefore, a matrix-matrix multiplication or straightforward inversion of a single matrix scales as $\mathcal{O}(N^3)$[13], whereas a matrix-vector multiplication scales as $\mathcal{O}(N^2)$. Consequently, the overall computational cost for these operations behaves as $\mathcal{O}(n_\mathbf{q} n_\omega n_\nu^3 n_o^6)$ and $\mathcal{O}(n_\mathbf{q} n_\omega n_\nu^2 n_o^4)$, respectively.

The evaluation of the self-energy through the Schwinger-Dyson equation has a different scaling, $\mathcal{O}(n_\mathbf{k} n_\mathbf{q} n_\omega n_\nu n_o^6)$ and only becomes dominant for very dense momentum grids.

Furthermore, the whole algorithm scales linearly with the number of self-consistency iterations, adding a factor of n_{iter} to the full one-shot DΓA cycle.

[11] $\{\uparrow\uparrow\uparrow\uparrow, \uparrow\uparrow\downarrow\downarrow\}$ instead of $\{\uparrow\uparrow\uparrow\uparrow, \downarrow\downarrow\downarrow\downarrow, \uparrow\uparrow\downarrow\downarrow, \downarrow\downarrow\uparrow\uparrow, \uparrow\downarrow\downarrow\uparrow, \downarrow\uparrow\uparrow\downarrow\}$

[12] In general, we assume that the number of Matsubara frequencies scales with β, i.e. $n_\omega \sim n_\nu \sim \beta$.

[13] In our implementation, we use the inversion provided by the Python package NumPy, which employs the LAPACK functions cgetrf/cgetri or zgetrf/zgetri, depending on the precision required. These algorithms scale as $\mathcal{O}(N^3)$, whereas other algorithms with a lower complexity exist, such as Strassen-like or $N^{2.5}$-type inversions. However, these are typically only available in specialized libraries and are rarely used for dense numerical inversion in scientific computing because of stability and memory tradeoffs.

5.3 Benchmarks and internal consistency checks

In order to check if the program's output is correct, we chose a few simple test cases to benchmark the algorithm and check our code for internal consistency:

(i) First, we verify that the new code reproduces the same results as DGApy. While this comparison is limited to single-band models, it remains a crucial benchmark and ensures that our implementation yields results in agreement with what is found in literature. Since the new implementation also supports λ-correction for single-band data, we compare both codes across two λ-correction schemes. Further details are provided in Sec. 5.3.1.

(ii) Next, we assess the correctness of the multi-orbital implementation. To this end, we constructed an artificial two-orbital dataset in which both bands (orbitals) are identical, fully disconnected and inter-orbital interactions are absent. Using the same parameters as in test (i), we compare the results for each orbital with those from the single-band case. Aside from statistical noise, we expect the results to be identical. This consistency check is discussed in more detail in Sec. 5.3.2.

(iii) Finally, we test the invariance of the results under basis transformations in orbital space; here we take orbital-space rotations. Starting from the previous test case, we rotate the DMFT input data — including the local one- and two-particle Green's functions, the DMFT self-energy, the Hamiltonian and the interaction matrix — in orbital space. After performing the calculation, we apply the inverse rotation to the output and compare the result with the unrotated case. As detailed in Sec. 5.3.3, the two results have to coincide within numerical precision.

For the single-band input, we used $\beta = 12.5/t$, a filling of $n = 0.85$ and tight-binding parameters representative of the Ni-$d_{x^2-y^2}$ orbital in $Nd_{1-x}Sr_xNiO_2$ and related compounds: $t = 1, t' = -0.25t$ and $t'' = 0.12t$, with an interaction strength of $U = 8t$. Analogous parameters were used for the artificial two-band input, where now the total filling was set to $n = 1.7$, corresponding to 0.85 electrons per band. The Hamiltonian was constructed to be orbital-diagonal, using identical hopping amplitudes and interaction strengths for both bands. All input data were generated using the CT-HYB quantum Monte Carlo solver w2dynamics, run on the Vienna Scientific Cluster.
Note that for the upcoming plots in this section all energies will be given in units of the hopping amplitude t and the lattice spacing is set to $a = 1$.

5.3.1 Benchmark against DGApy for the single-orbital model

We performed a full one-shot DΓA calculation using both DGApy and our new implementation, employing identical input files and parameters. Across all computed quantities, our implementation reproduced the results of the DΓA algorithm with floating-point accuracy. This agreement held for both the basic one-shot calculation without λ-corrections and for cases with λ-corrections applied — whether in the spin channel alone or in both spin and charge channels.
We noticed a tiny subtlety in the implementation of DGApy. It uses a different double-counting correction, which reads for the single-band case

$$\Sigma^{k}_{dc} = \frac{1}{\beta} \sum_{q\nu'} U \chi_0^{q\nu'} F_m^{\omega\nu'\nu} G^{k-q}, \tag{5.2}$$

compared to the expression containing a combination of the full vertex in the density and magnetic channel, see our definition in Eq. (4.28). This does not add a significant numerical difference here, but it should nevertheless be noted as it becomes more relevant for lower temperatures. The usage of Eq. (5.2) in DGApy is possible for the special case of single-band data, where one can show that

$$\sum_{q\nu'} \chi_0^{q\nu'} \left[F_d^{\omega\nu'\nu} + F_m^{\omega\nu'\nu}\right] G^{k-q} \propto \sum_{q\nu'} \chi_0^{q\nu'} F_{\uparrow\uparrow}^{\omega\nu'\nu} G^{k-q} = 0, \tag{5.3}$$

which also leads to the correct asymptotic behavior of the self-energy calculated with DΓA. We carried out the single-band calculation with Eq. (5.2) and Eq. (4.28) and compared them to the results of DGApy separately, both of which can be seen in Fig. 5.1 and Fig. 5.2, respectively. These calculations were done with a **k**-grid of $64 \times 64 \times 1$, $n_\omega = 60$, $n_{\nu;\text{core}} = 40$ and $n_{\nu;\text{asympt}} = 20$ without any λ-correction. The absolute differences in the self-energies between the two implementations did

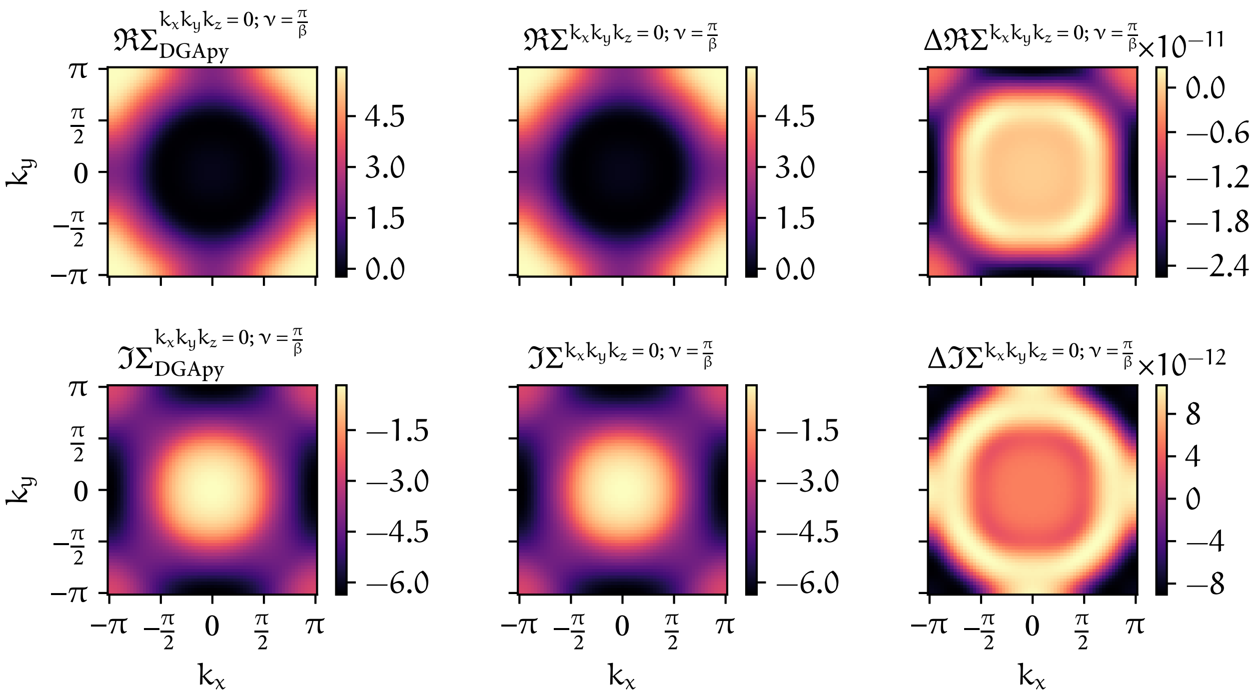


FIGURE 5.1 – Difference between the one-band self-energy of DGApy (first column) and moLDGA with the double-counting kernel of Eq. (5.2) (second column) for $k_z = 0$ and $\nu = 0$. The difference between both, i.e. $\Delta\Sigma = \Sigma_{\text{DGApy}} - \Sigma$, can be seen in the last column. The rows indicate the real- and imaginary part of all self-energies, respectively.

not exceed $\mathcal{O}(10^{-11})$, and the five largest eigenvalues of the pairing vertex in the Eliashberg equation differed by at most $\mathcal{O}(10^{-6})$ when implementing the double-counting kernel of Eq. (5.2). This level of agreement was consistently achieved across different **k**- and **q**-grids and varying numbers of Matsubara frequencies n_ω and n_ν.
For an implementation with the double-counting kernel of Eq. (4.28) we found an agreement of $\mathcal{O}(10^{-3})$ in the self-energies, where the largest superconducting eigenvalues differed by $\mathcal{O}(10^{-4})$. A comparison between the *local* irreducible vertices for both calculations and both density and magnetic spin combinations can be seen in Fig. 5.3 and Fig. 5.4, respectively.

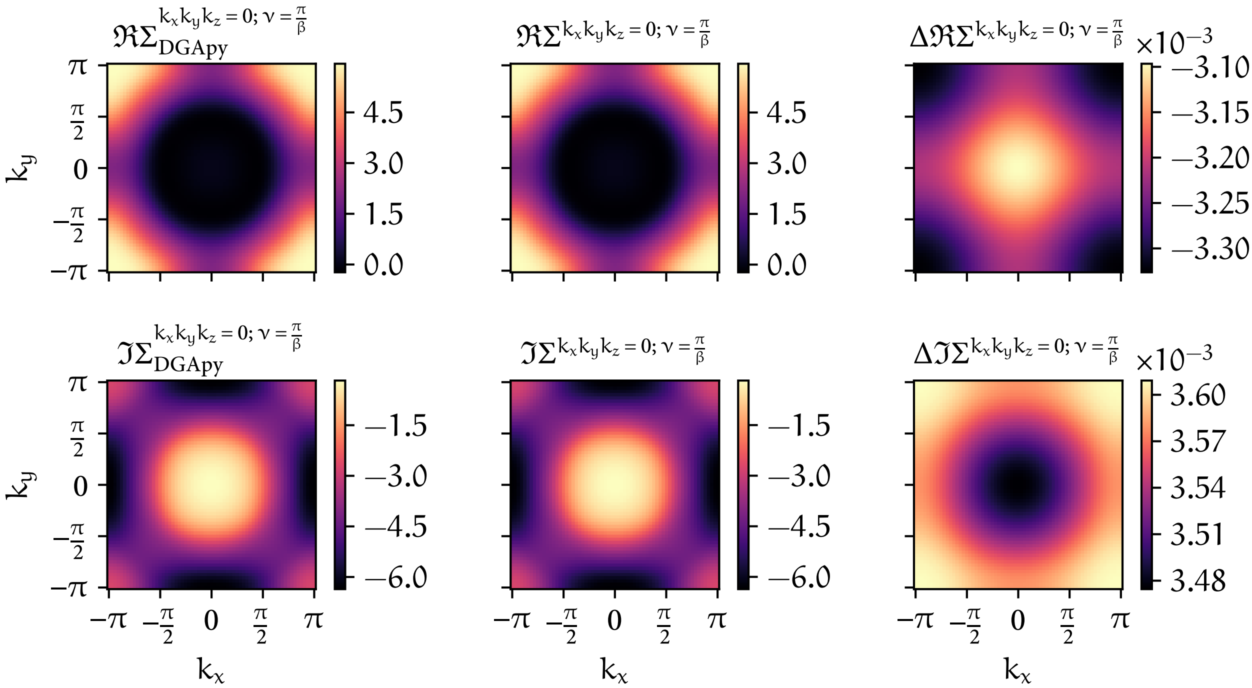


Figure 5.2 – Analogous to Fig. 5.1, but with the double-counting kernel of Eq. (4.28) applied for moLDGA (second column).

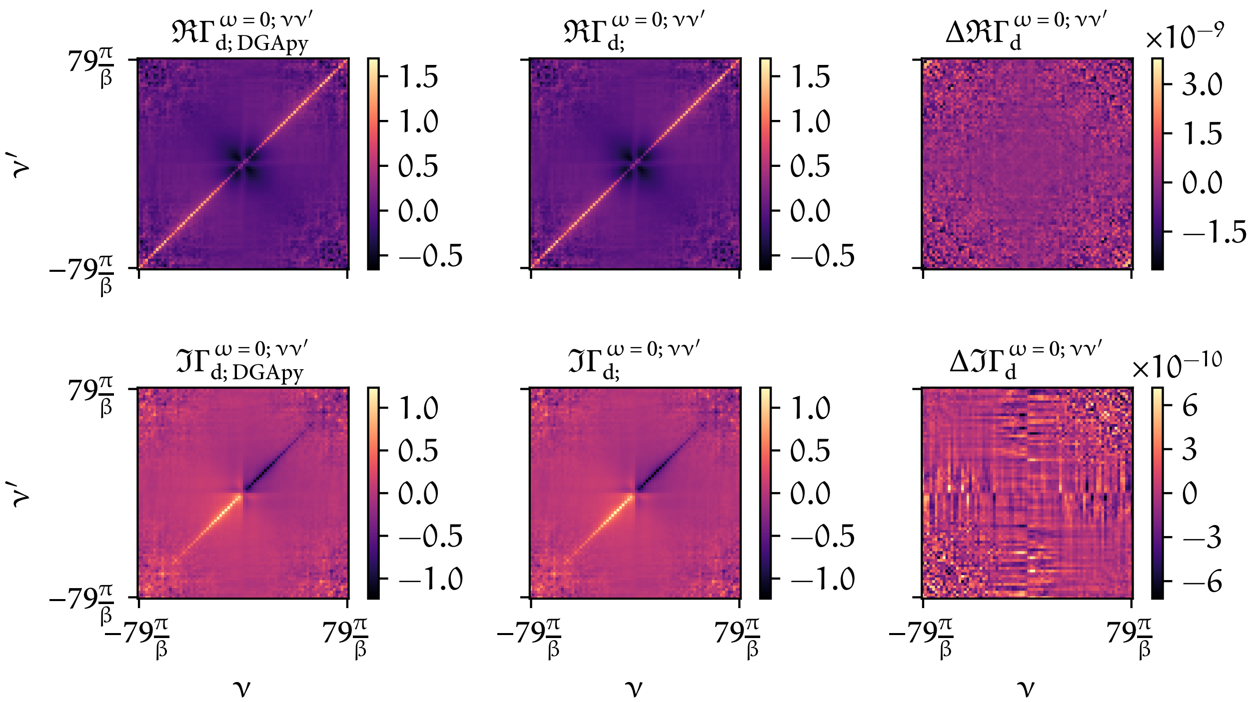


Figure 5.3 – Real and imaginary part of the *local* irreducible vertex in the density channel calculated with DGApy (first column) and moLDGA (second column). Their differences can be seen in the third column.

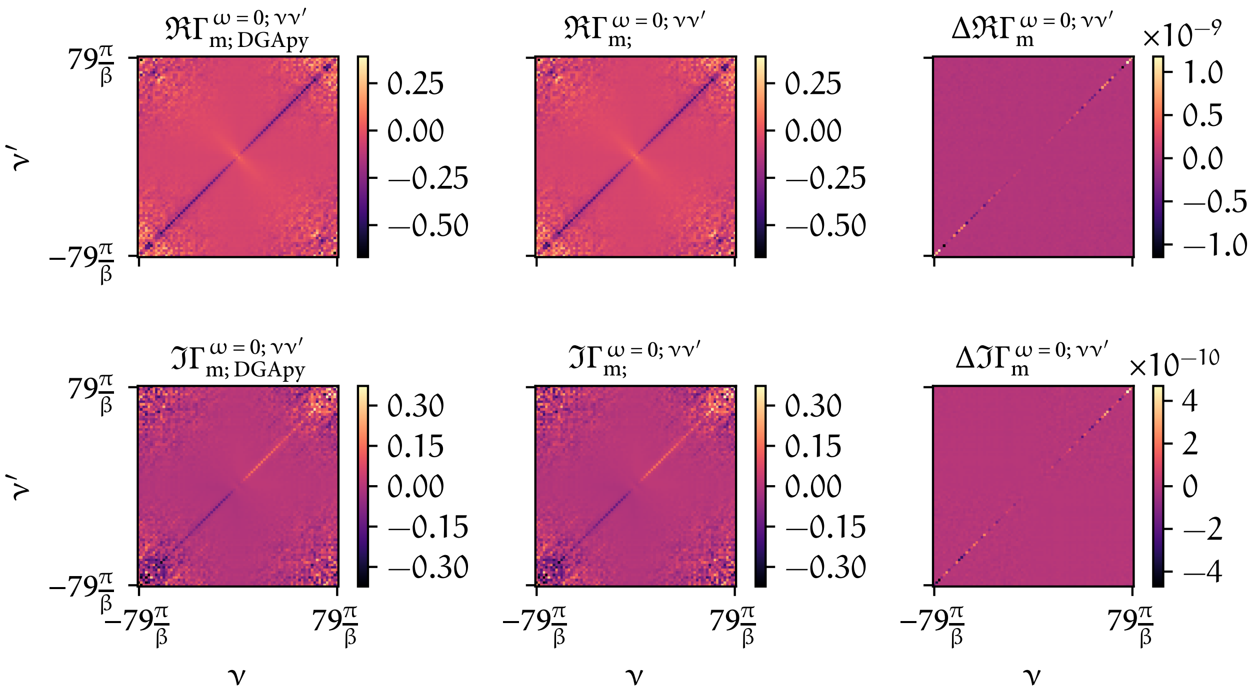


FIGURE 5.4 – Analogous to Fig. 5.3 but for the magnetic spin combination.

5.3.2 Disconnected two-band model

Next, we show the successful consistency checks for the multi-orbital implementation of the code using the artificial two-band input discussed earlier. The purpose of these tests was to verify that our code treats such a system in the expected way, namely as two independent orbitals with identical physical properties, and that no bugs have been introduced by the handling of multi-orbital input data. Based on this setup, we carried out three consistency checks:

The first test consisted of projecting out a single orbital of the disconnected two-band dataset generated by w2dynamics, performing a single-band DΓA calculation and comparing it to the corresponding orbital of the calculation of the full two-band model. Since both calculations start from the same underlying data, one expects perfect agreement between the two. Indeed, this is exactly what we observed: both the self-energy and the two-particle vertex functions matched up to numerical precision. This confirmed that our implementation correctly respects the orbital structure of the disconnected two-band system and does not introduce any discrepancies (due to off-diagonal components of the vertex functions in the two-band calculations) when picking a specific orbital and comparing it to the single-band results. Furthermore, the code was also able to keep the purely diagonal structure of the two-orbital self-energy and the off-diagonal entries are exactly zero. Since this test case reproduced the same numerical results (up to floating point precision) it will not be shown here explicitly.

As a second test, we carried out the inverse procedure. Starting from the single-band dataset, we artificially embedded it into a disconnected two-band structure[14] and then compared the DΓA

[14]This means, for example, that $H^{\mathbf{k}} = \begin{pmatrix} H^{\mathbf{k}}_{11} & 0 \\ 0 & H^{\mathbf{k}}_{22} = H^{\mathbf{k}}_{11} \end{pmatrix}$ or $G^{\nu}_{0} = \begin{pmatrix} G^{\nu}_{0;11} & 0 \\ 0 & G^{\nu}_{0;22} = G^{\nu}_{0;11} \end{pmatrix}$ in orbital space.

results of one of the diagonal orbitals from this construction to the original single-band data calculated with the program. Again, this comparison did not involve two independent w2dynamics calculations, but rather served as a stringent internal consistency check on the code's ability to handle single- and multi-band inputs in a uniform manner using identical inputs. This time our input data had less noise since we only had to perform single-orbital calculations in w2dynamics. These tests further give insights that helped to disentangle the results for the third test, where we use different w2dynamics calculation runs to check whether differences arise from the DMFT input or the implementation. As would be expected, the agreement was again exact up to numerical precision for both the self-energy and the vertex functions, confirming that the code faithfully reproduces the equivalence between the single-band description and its trivial two-band embedding. Since this test case reproduced the exact numerical results as expected, we will similarly as above not show any results.

The third test, in contrast, did involve two independent input datasets. Here, we instead performed two independent DMFT calculations using w2dynamics to generate data for both the single-band and the disconnected two-band systems separately and then compared the results of two separate DΓA calculations. The results of both datasets agreed very well overall, demonstrating that the expected equivalence between the two models is also preserved when the input is generated independently. In this comparison, however, small deviations were visible. This behavior is fully explained by the stochastic nature of the continuous-time quantum Monte Carlo method employed in w2dynamics. This interpretation is also supported by the first two consistency checks, as well as the fact that these two independent calculations display varying differences depending on the number of QMC measurements used to generate the input. Statistical noise is an unavoidable feature of CT-QMC data and its presence here is therefore not surprising. Importantly, these statistical fluctuations did not obscure the overall agreement between the single- and two-band results. The comparison between the single- and two-band self-energies and local irreducible vertices for a lower QMC measurement statistics run can be seen in Fig. 5.5, Fig. 5.6 and Fig. 5.7, respectively. To briefly illustrate the numerical differences introduced by the w2dynamics calculations, we also compared those quantities for a four-times increase of the QMC measurement number in Fig. 5.8, Fig. 5.9 and Fig. 5.10, respectively. This calculation was done for the artificial two-band model with a **k**-grid of $24 \times 24 \times 1$, $n_\omega = 60$, $n_{\nu;\mathrm{core}} = 40$ and $n_{\nu;\mathrm{asympt}} = 0$. For the artificial two-band model, we show the (11)-component of the self-energy and the (1111)-component of the irreducible vertex for the comparison. The other orbital-diagonal entry is — except for small stochastic noise — indistinguishable from the first one.

Finally, we also compare the frequency dependence of the one- and two-band self-energy. For this, we plotted the single-band self-energy compared to the first diagonal (11) component of the artificial two-band calculation for both lower (Fig. 5.11(a)) and higher (Fig. 5.11(b)) number of QMC measurements, and $\mathbf{k} = \mathbf{0}$.

Taken together, these three tests confirm that our implementation treats disconnected multi-band systems consistently. The first two tests established internal consistency within a single dataset by verifying the equivalence of single- and two-band representations derived from the same calculation. The third test then demonstrated that this equivalence is preserved even when comparing truly independent datasets generated by w2dynamics. The results provide strong evidence that our code robustly respects the expected physics of disconnected orbital systems and correctly reproduces the equivalence of two independent copies of the single-band model.

This is done similarly for the DMFT self-energy and local two-particle Green's function.

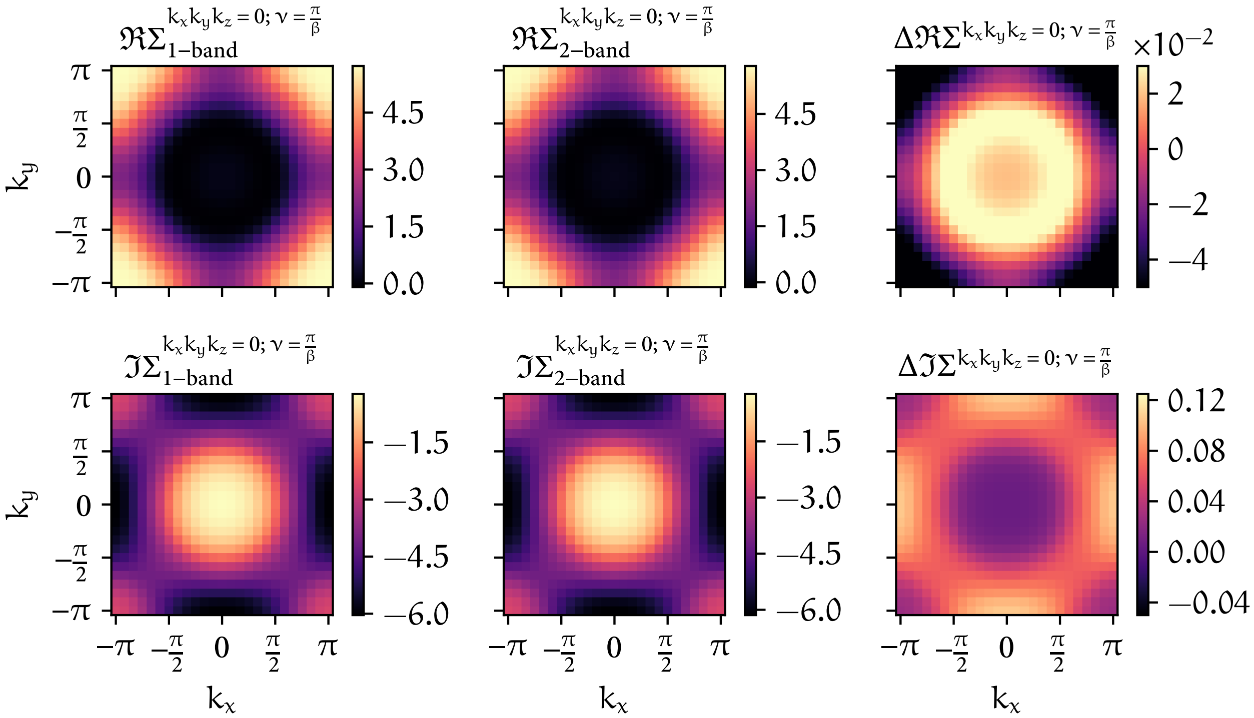


FIGURE 5.5 – One-band self-energy (first column) and the first diagonal component of the self-energy from the artificial two-band system (second column) calculated using the inputs from separate w2dynamics runs. The difference between both can be seen in the last column and the rows indicate the real- and imaginary part of all self-energies, respectively.

5.3.3 Rotated disconnected two-band model

For this test, we use the disconnected two-band model introduced earlier and applied rotations in orbital space. The goal is to verify that the moLDGA code also works correctly if orbital-off-diagonal components are included, while at the same time the simple rotation guarantees that we have results to compare to. After rotating back the orbitals must again be properly separated and not mixed in unintended ways. To this end, we implemented simple rotation routines in the code, which allow us to rotate the one- and two-particle vertex functions in orbital space. For the one-particle Green's function and self-energy this looks like

$$G^{k;\vartheta}_{12;\sigma} = \sum_{ab} \left(S^{\vartheta}_{a1}\right)^{\dagger} G^{k}_{ab;\sigma}\, S^{\vartheta}_{b2}, \tag{5.4}$$

and similarly for $\Sigma^{k;\vartheta}_{12;\sigma}$, whereas for any four-point vertex the rotation is performed like

$$F^{qkk';\vartheta}_{1234;\sigma\sigma'} = \sum_{abcd} \left(S^{\vartheta}_{a1}\right)^{\dagger} \left(S^{\vartheta}_{c3}\right)^{\dagger} F^{qkk'}_{abcd;\sigma\sigma'}\, S^{\vartheta}_{b2} S^{\vartheta}_{d4}. \tag{5.5}$$

Here, the indices $\{i\}$ are representing orbital indices only. Under the hood, this transforms the creation and annihilation operators of the objects with S being a two-dimensional rotation matrix, which simply looks like

$$S^{\vartheta} = \begin{pmatrix} \cos(\vartheta) & \sin(\vartheta) \\ -\sin(\vartheta) & \cos(\vartheta) \end{pmatrix} \qquad \text{and} \qquad \left(S^{\vartheta}_{12}\right)^{\dagger} = S^{\vartheta}_{21} = S^{-\vartheta}_{12}. \tag{5.6}$$

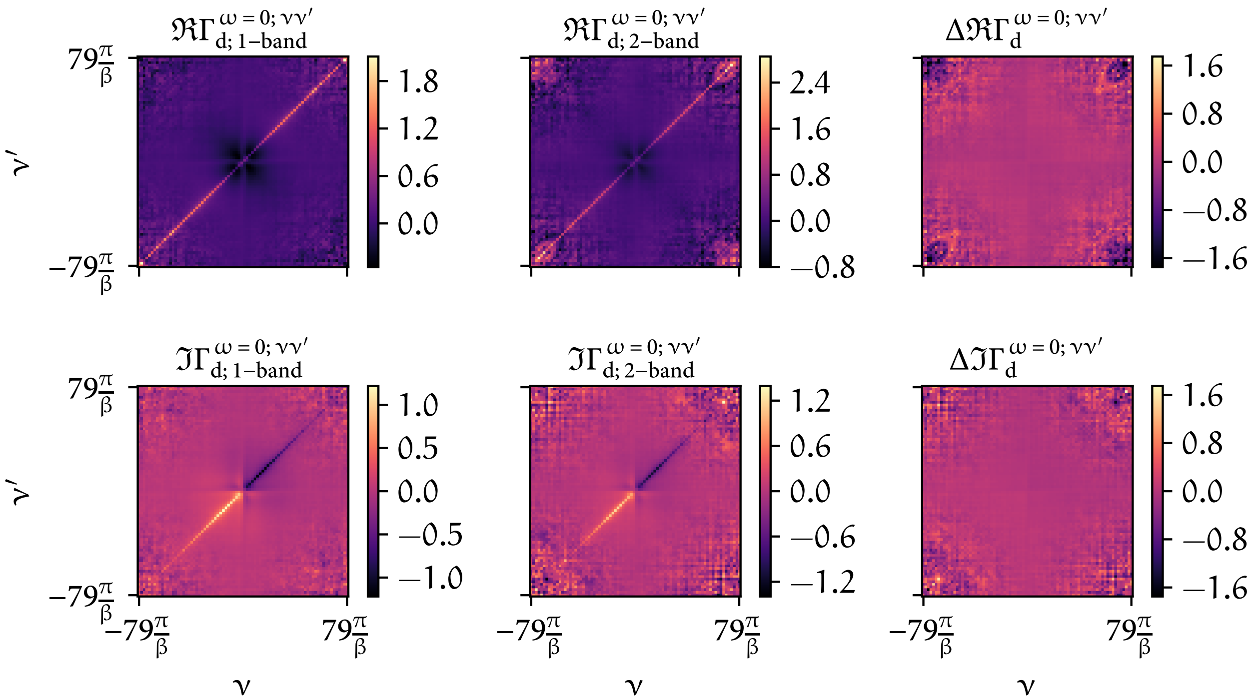


FIGURE 5.6 – Real and imaginary part of the *local* irreducible vertex in the density channel of the one-band dataset (first column) and the artificial two-band dataset (second column) calculated using the inputs from separate w2dynamics runs. Their differences are shown in the third column.

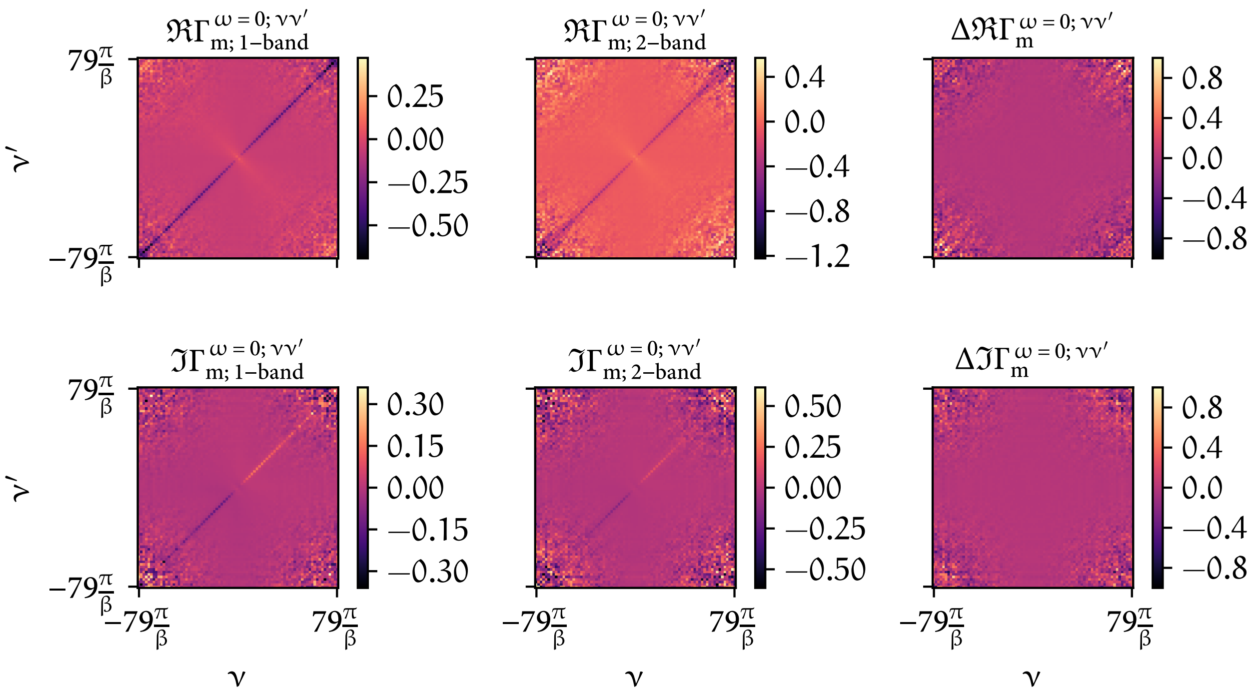


FIGURE 5.7 – Analogous to Fig. 5.6 but for the magnetic spin combination.

To perform this test, we applied a rotation by an angle ϑ to all input quantities from DMFT and the Hamiltonian. The DΓA algorithm was then used to compute the non-local self-energy, which we

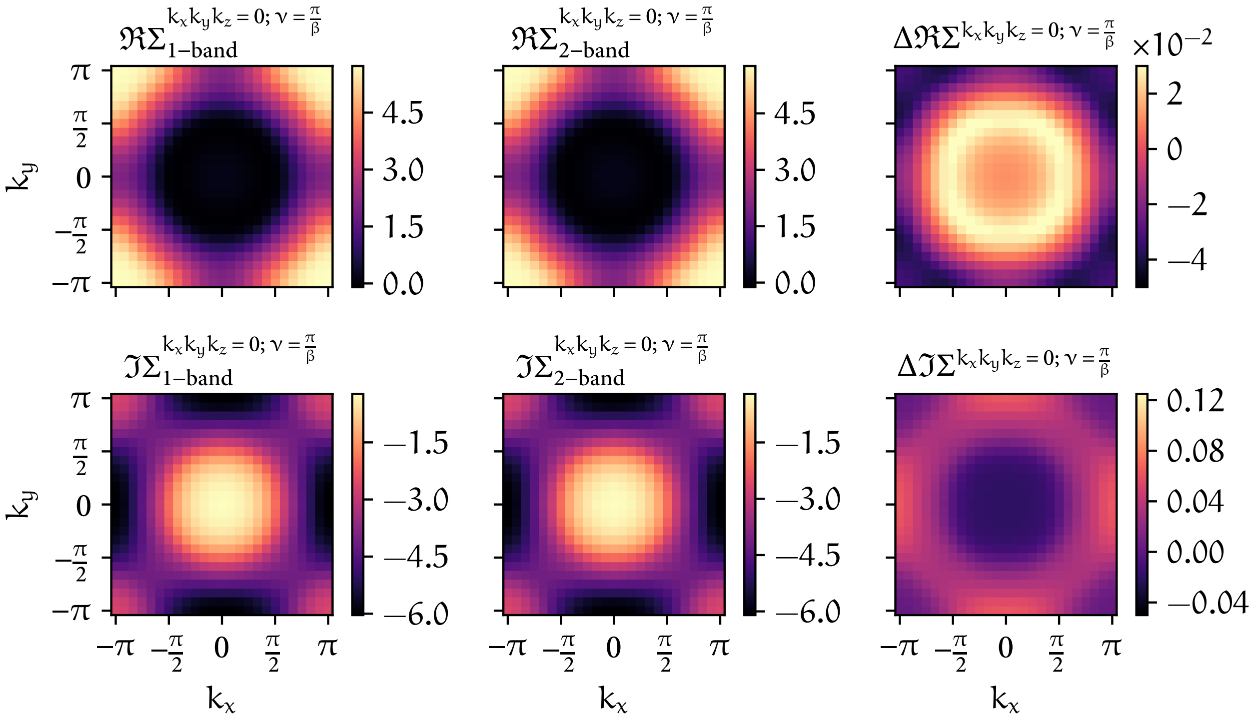


Figure 5.8 – Similar to Fig. 5.5 but this time the number of QMC measurements for the one- and two-particle calculations in w2dynamics were increased by a factor of four.

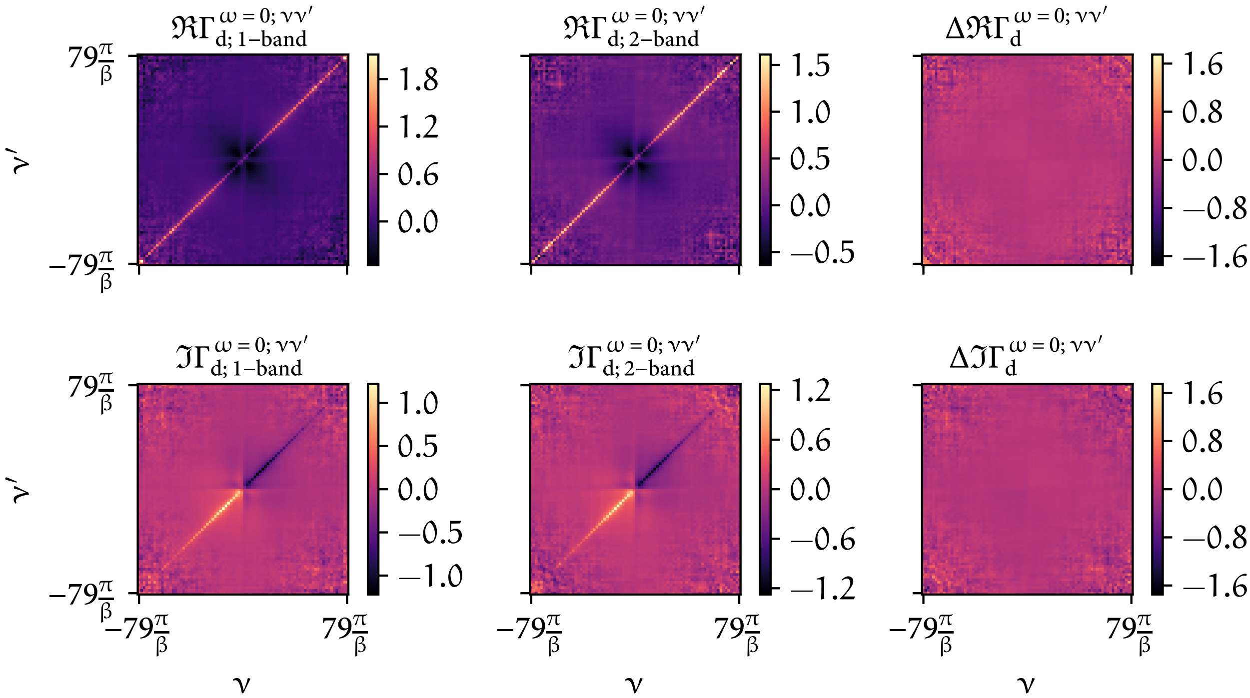


Figure 5.9 – Similar to Fig. 5.6 but this time the number of QMC measurements for the one- and two-particle calculations in w2dynamics were increased by a factor of four.

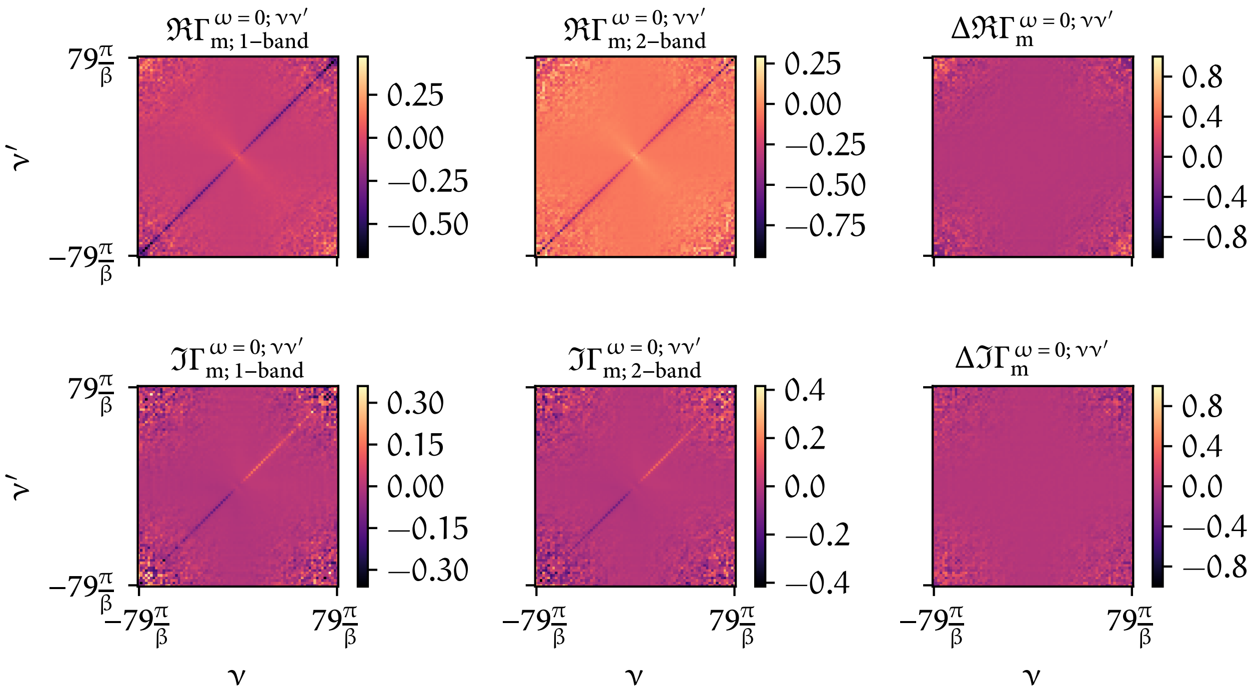
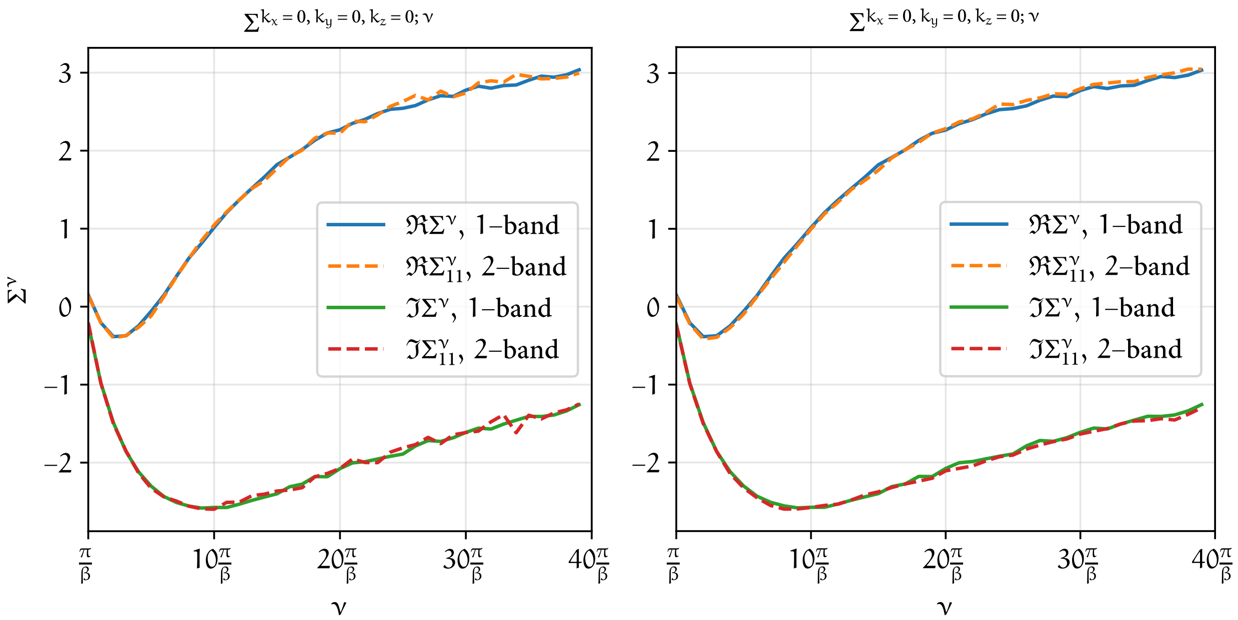


FIGURE 5.10 – Similar to Fig. 5.7 but this time the number of QMC measurements for the one- and two-particle calculations in w2dynamics were increased by a factor of four.

subsequently rotated back by $-\vartheta$. We compared the resulting self-energy to a reference calculation performed without any rotations and found the individual elements to agree up to floating-point precision, with $\sum_{k;12} |\Sigma_{12}^{k;\vartheta\neq 0} - \Sigma_{12}^{k;\vartheta=0}| \lesssim 10^{-9}$ for $\vartheta = \frac{\pi}{2}, \pi$ and $\sum_{k;12} |\Sigma_{12}^{k;\vartheta\neq 0} - \Sigma_{12}^{k;\vartheta=0}| \lesssim 10^{-3}$ for other values of ϑ. We tried this with different numbers of Matsubara frequencies n_ω and n_ν and varying **k**-grid sizes — all yielding indistinguishable results.
Typical results of the self-energy can be seen in Fig. 5.12 and for the irreducible vertex $\Gamma_{d/m}^{\omega=0;\nu\nu'}$ in Fig. 5.13 and Fig. 5.14. This calculation was done for the artificial two-band model with a **k**-grid of $24 \times 24 \times 1$, $n_\omega = 60$, $n_{\nu;\text{core}} = 40$ and $n_{\nu;\text{asympt}} = 0$. Here, we show the (11) orbital of the self-energy and the (1111) orbital component of the irreducible vertex; the (22) and (2222) components of both quantities look very similar.



(A) (B)

FIGURE 5.11 – Frequency dependence of the real and imaginary part of the self-energy for the single-band and the artificial two-band dataset, and for the $\Gamma = (0, 0, 0)$ point. (A) shows the comparison for a lower QMC measurement statistics run for the two-band dataset, whereas (B) for a four-times higher number of QMC measurements. The improved behavior for large frequencies is not surprising, as the increased number of QMC measurements reduces noise.

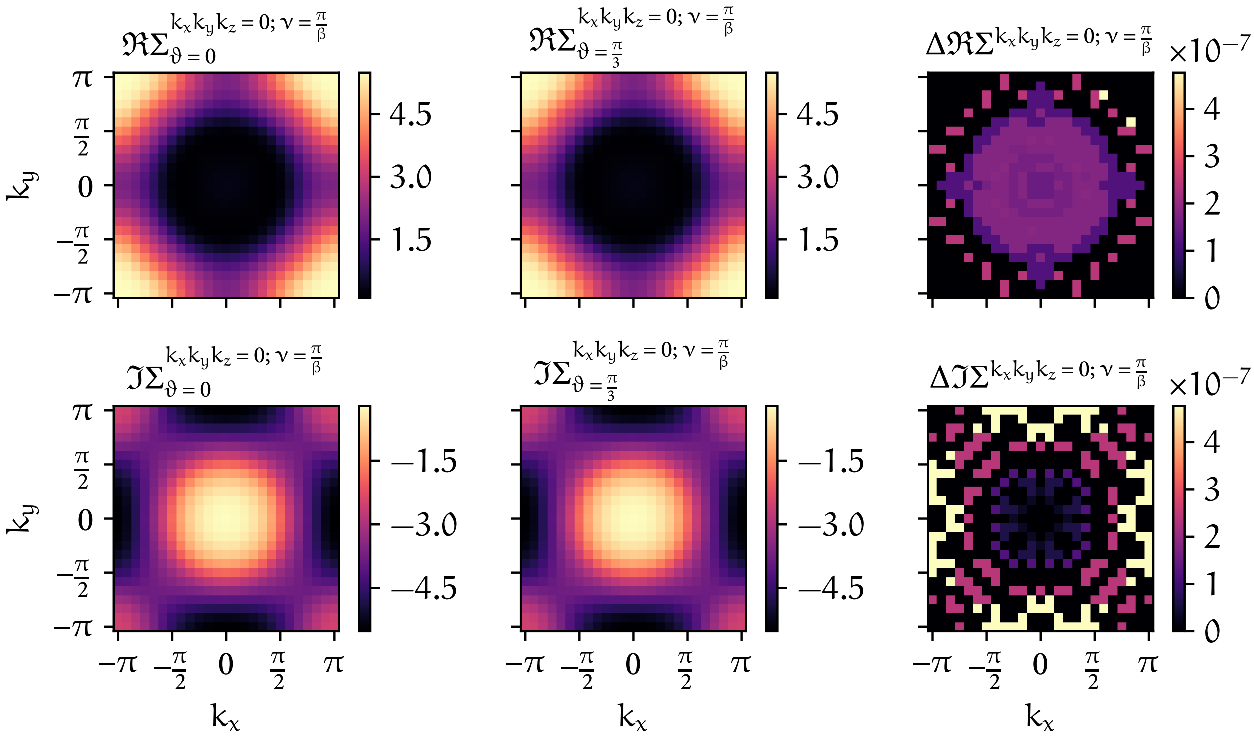


Figure 5.12 – Real and imaginary part of the self-energy for a two-band calculation without a rotation ($\vartheta = 0$, first column) and a calculation, where the input data was rotated before the DΓA calculation and subsequently rotated back by a factor of $\vartheta = \frac{\pi}{3}$ (second column) for both the real and imaginary part of the self-energy (first and second row, respectively). Their differences are shown in the third column.

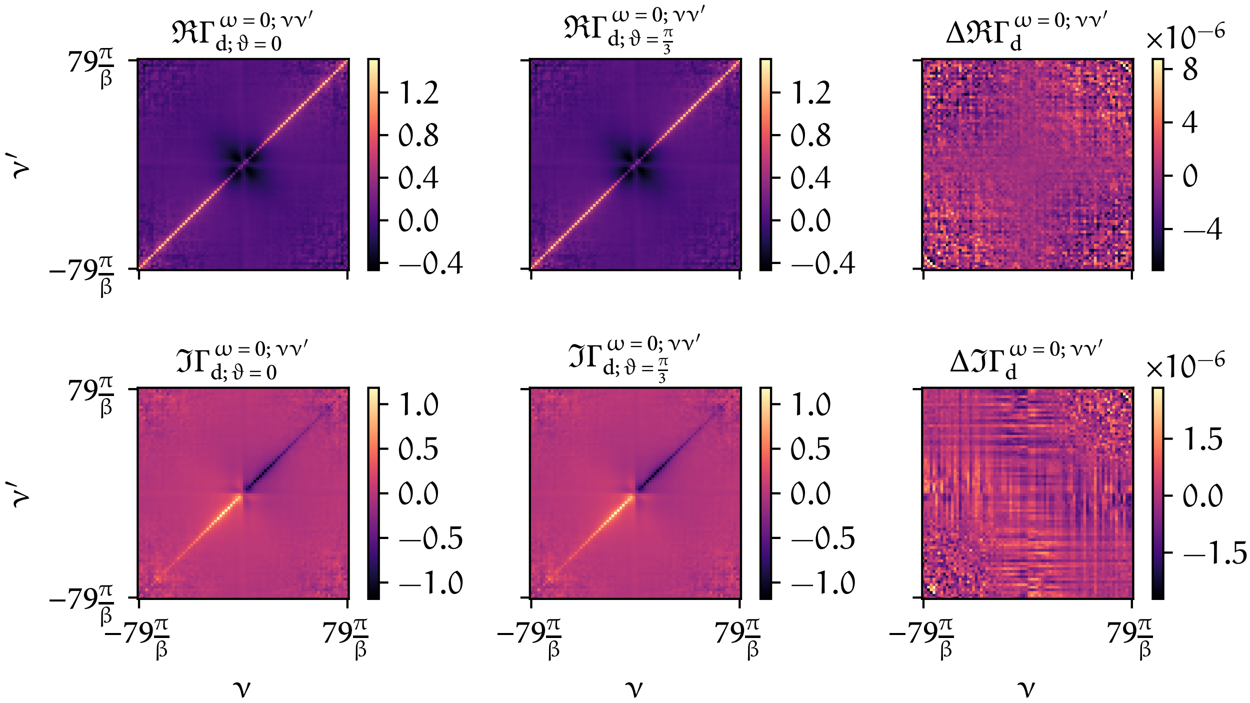


FIGURE 5.13 – Real and imaginary part of the irreducible vertex in the density channel: two-band calculation without a rotation ($\vartheta = 0$, first column) and a calculation, where the input data was rotated before the DΓA calculation and subsequently rotated back by a factor of $\vartheta = \frac{\pi}{3}$ (second column) for both the real and imaginary part of the *local* irreducible vertex $\Gamma_{\mathrm{d}}^{\omega=0;\nu\nu'}$ (first and second row, respectively). Their differences are shown in the third column.

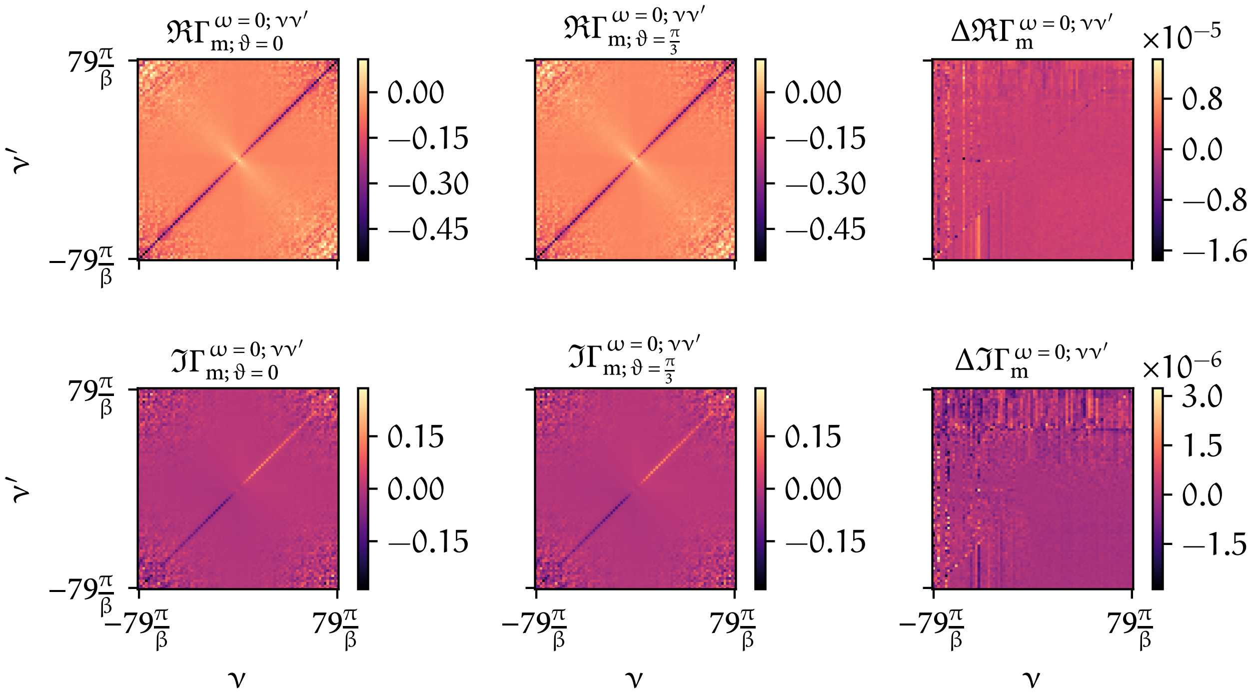


FIGURE 5.14 – Analogous to Fig. 5.13 but for the magnetic spin combination.

Chapter 6
Conclusions and Outlook

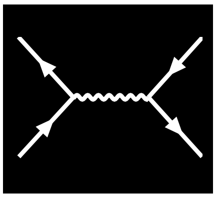
In this thesis, we have developed and implemented a computational framework for the multi-orbital ladder-DΓA, enabling the study of two-particle correlation functions, single-particle quantities and superconducting transition temperatures in strongly correlated electron systems. A central focus was placed on the efficient treatment of vertex asymptotics, which significantly reduces numerical complexity for large frequency boxes and enables calculations down to very low temperatures. Combined with an optimized self-consistency cycle, where two-particle quantities feed back into single-particle properties via the Bethe-Salpeter, Schwinger-Dyson and Dyson equations, the implementation provides a robust and scalable approach to capture both local and non-local correlations beyond dynamical mean-field theory. The code is written in Python, exploits NumPy-based optimizations and uses mpi4py for distributed-memory parallelization, enabling large-scale multi-orbital simulations.

As with most scientific software, there are several promising directions for improving and extending this work:

First, while the implementation of the multi-orbital Eliashberg equation has been completed, its functionality has not yet been fully tested. A thorough validation of this routine would be an important next step, as it would allow for a direct and reliable treatment of superconducting instabilities and frequency-dependent pairing interactions. This, in turn, would open the door to a detailed investigation of the mechanisms driving superconductivity in strongly correlated systems with pronounced multi-orbital character.

Second, to make the framework applicable to modern research, it is essential to perform realistic multi-orbital calculations. By interfacing with ab-initio methods and incorporating material-specific input (e.g. from DFT + Wannier projections and a subsequent DMFT calculation), the code can be applied to experimentally relevant compounds such as cuprates, nickelates and iron-based superconductors. The input data format of the moLDGA code directly allows for such realistic materials calculations.

Overall, the developed framework provides a solid foundation for investigating correlation effects and unconventional superconductivity in multi-orbital systems. By combining vertex asymptotics, an efficient self-consistency cycle and a scalable implementation, this thesis establishes a flexible basis for future research on strongly correlated materials.



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